MTH645 SLIDES

LECTURE # 01

Classical Sets

Set

A set is collection of well defined objects

Classical or Crisp Sets

X, U Universe of discourse or Universal set

x $X = \mathbb{R}$ whole set

NI

$A = \{1, 2, 3\}$

$\phi = \{\}$ Null set

$P(A) = \{$ all subsets of $A\}$ $2^{\#A} = 2^{|A|}$

Classical or Crisp Sets

$X = \{1, 2, 3, 4\}$ $\qquad n_X = 4$

$$P(X) = \{\underline{\phi}, \{1\}, \{2\}, \{3\}, \{4\}, \{1,2\}, \{1,3\}, \{1,4\}, \{2,3\}, \{2,4\}, \{3,4\},$$
$$\{1,2,3\}, \{1,2,4\}, \{1,3,4\}, \{2,3,4\}, \underline{\{1,2,3,4\}}\}$$
$$\qquad\qquad\qquad

Classical or Crisp Sets

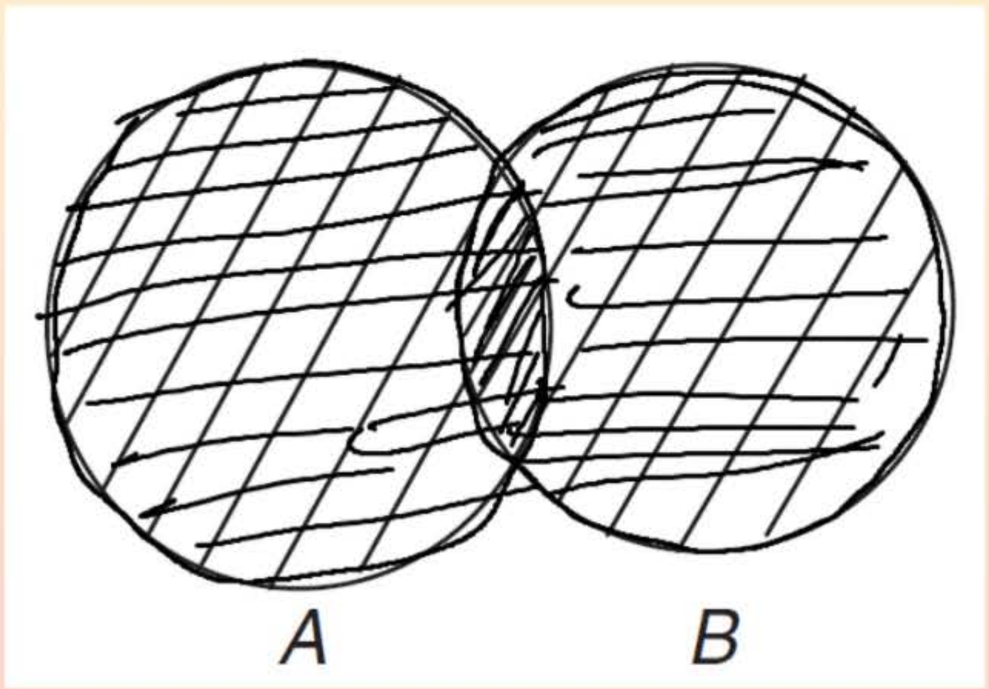
A, B

$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$

$X = N$

$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}$

$A \cup B = \{1, 2, 3, 4, 5\}$



Classical or Crisp Sets

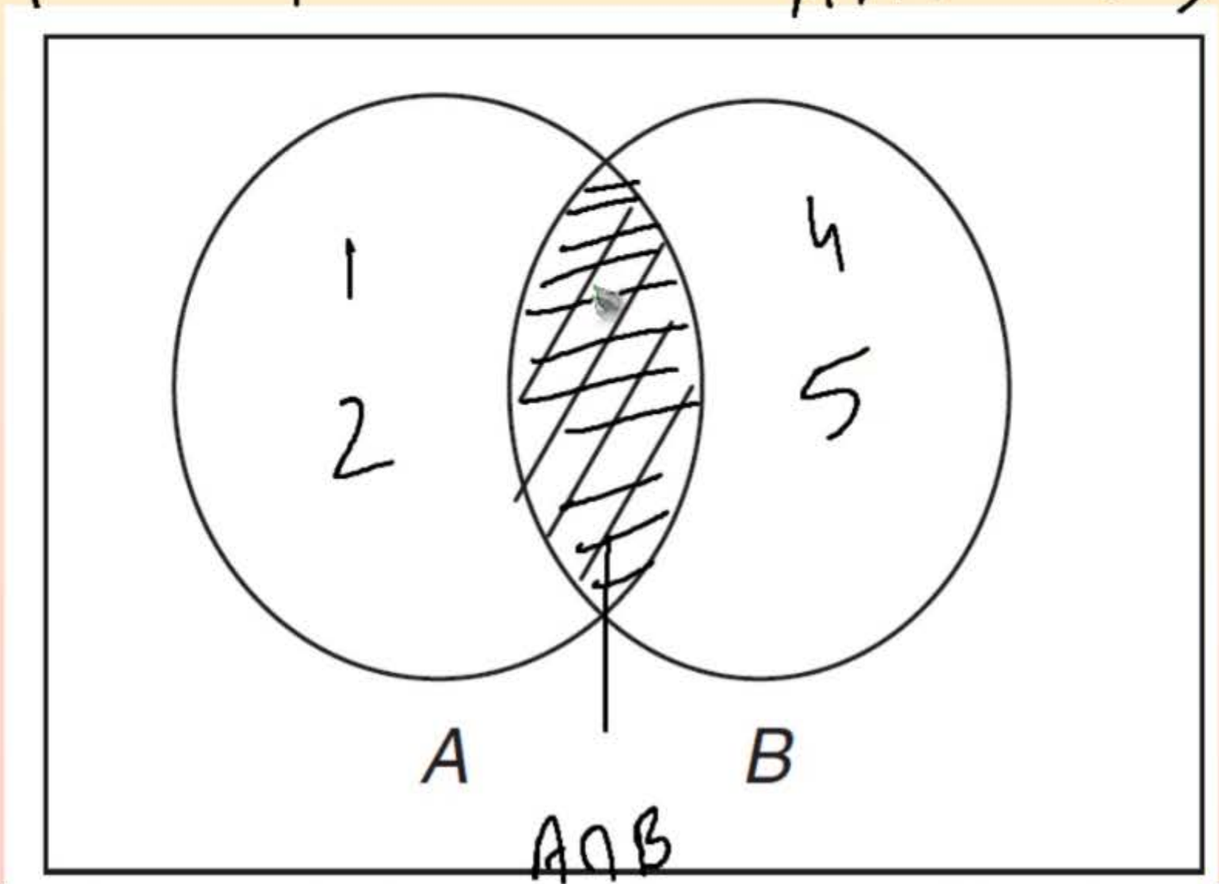
A, B

$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$

$X = N$

$A = \{1, 2, 3\}, \quad B = \{3, 4, 5\}$

$A \cap B = \{3\}$



Classical or Crisp Sets

A

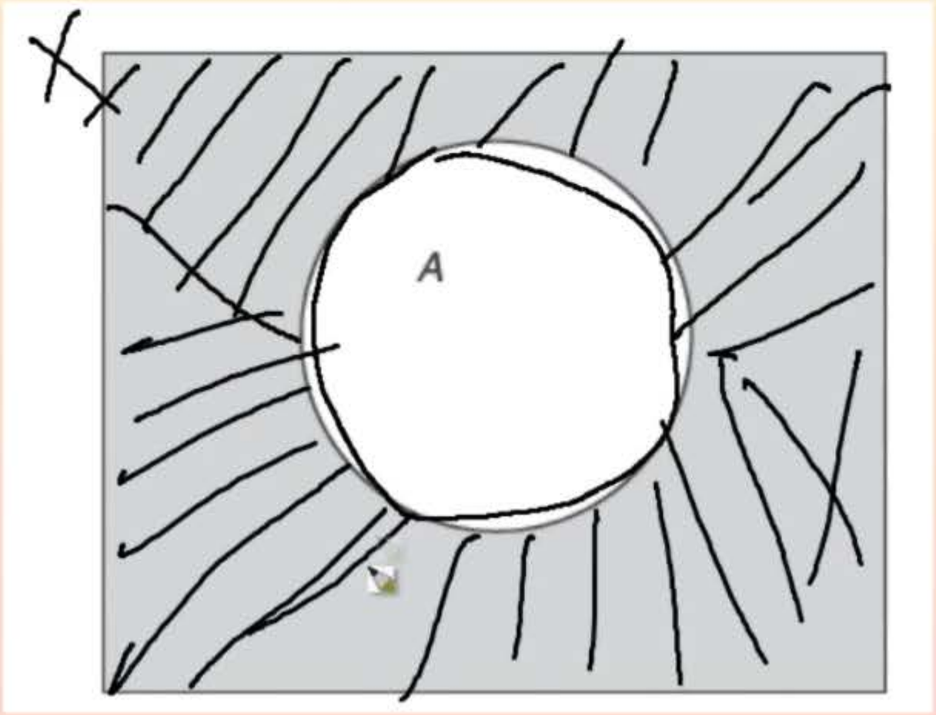
$\bar{A} = \{x \mid x \notin A, x \in X\}$

$X = N$; $A = \{1, 2, 3\}$

$= \{1, 2, 3, 4\}$

$\bar{A} = \{4, 5, 6, \cdots\}$

$\bar{A} = \{4\}$

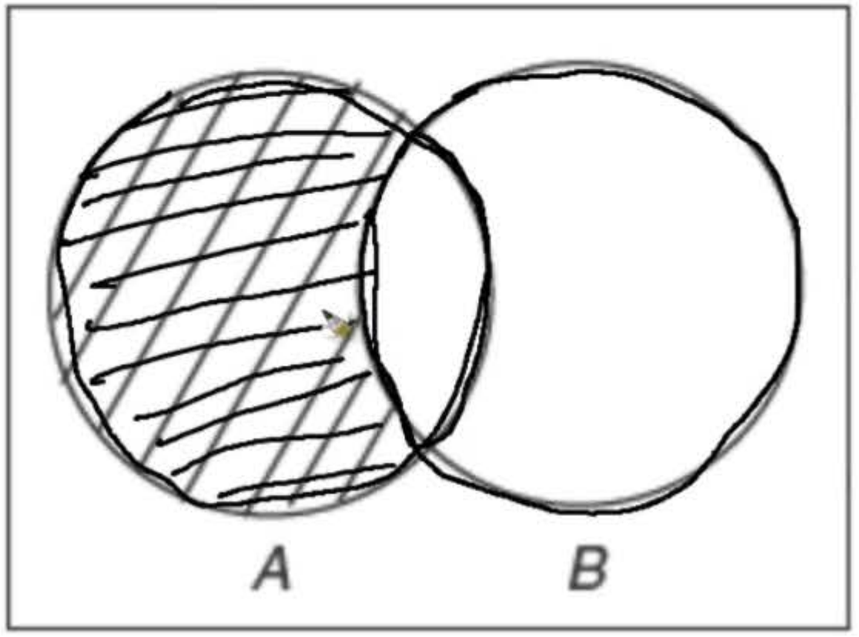


Classical or Crisp Sets

A, B

$$A \setminus B = A \mid B = A - B = \{x \mid x \in A \text{ and } x \notin B\}$$

$A = \{1, 2, 3\}$, $B = \{1, 3, 5, 7\}$, $A \setminus B = \{2\}$



Classical or Crisp Sets

$A \cup B = B \cup A$
$A \cap B = B \cap A$
} Commutative

$A \cup (B \cup C) = (A \cup B) \cup C$
$A \cap (B \cap C) = (A \cap B) \cap C$
} Associative

$A \cup A = A$
$A \cap A = A$
} Idempotency

$A \cup \phi = A$
$A \cap X = A$
$A \cap \phi = \phi$
$A \cup X = X$
} Identiy

$A \subseteq B \subseteq C \Rightarrow A \subseteq C$

Transitivity

$\bar{\bar{A}} = A$ Involution

$A \cup \bar{A} = X$

$A \cap \bar{A} = \phi$

De Morgan's Laws

$\overline{A \cap B} = \bar{A} \cup \bar{B}$

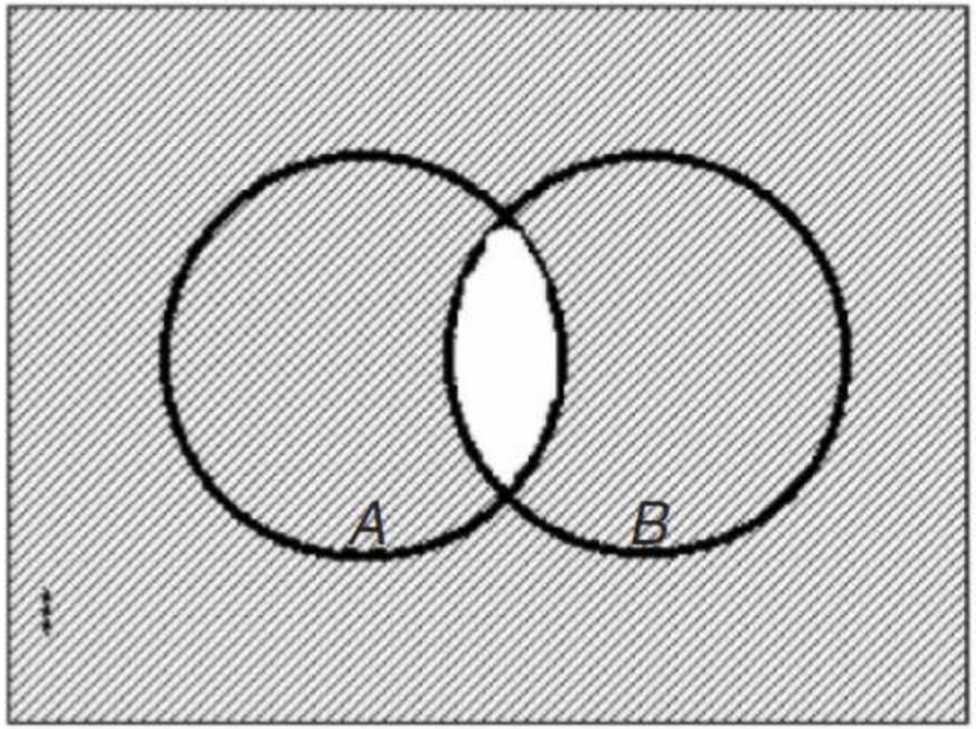
$\overline{A \cup B} = \bar{A} \cap \bar{B}$

Classical or Crisp Sets

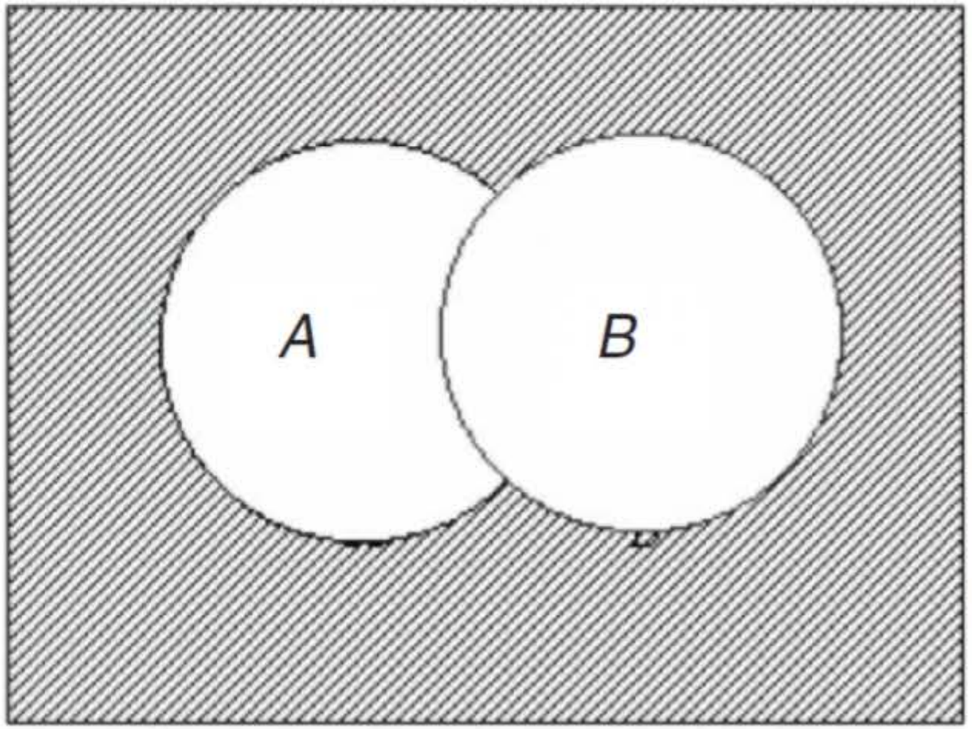
$\overline{A \cup B} = \bar{A} \cap \bar{B}$

$\overline{A \cap B} = \bar{A} \cup \bar{B}$

De Morgan's Laws



$\overline{A \cap B}$



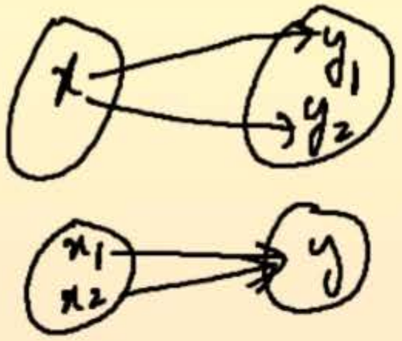
$\overline{A \cup B}$

Classical or Crisp Sets

X, Y

$f : X \longrightarrow Y$

$f(x) = y$



$(x, y_1), (x, y_2)$

$x \in X$

$y \in Y$

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

LECTURE # 02

Characteristic function

Let X be a set and A be a subset of X. Then the function

$$\chi_A(x) = \begin{cases} 1 & \text{if } x \in A \\ 0 & \text{if } x \notin A \end{cases}$$

is called the characteristic function of the set A in X.

$4 \notin A$

$\chi_A(4) = 0$

$A = \{1,2,3\}$

$1 \in A$

$\chi_A(1) = 1$

$U = \{1,2,3,4,5\}$

$2 \in A$

$\chi_A(2) = 1$

$3 \in A$

$\chi_A(3) = 1$

Classical or Crisp Sets

$$U = \{x_1, x_2, x_3, x_4, x_5, x_6\}$$

$$A = \{x_2, x_3, x_5\}$$

$\mu_A(x_1) = 0 \qquad x_1 \notin A$

$\mu_A(x_2) = 1 \qquad x_2 \in A$

$\mu_A(x_3) = 1 \qquad x_3 \in A$

$\mu_A(x_4) = 0 \qquad x_4 \notin A$

$\mu_A(x_5) = 1 \qquad x_5 \in A$

$\mu_A(x_6) = 0 \qquad x_6 \notin A$

$$\mu_A(x) = \begin{cases} 1 & ; \quad x \in A \\ 0 & ; \quad x \notin A \end{cases}$$

$$\mu_A(x) = \begin{cases} 1 & ; \quad x = x_2, x_3, x_5 \\ 0 & ; \quad x = x_1, x_4, x_6 \end{cases}$$

$$A = \{(x_1, 0), (x_2, 1), (x_3, 1), (x_4, 0), (x_5, 1), (x_6, 0)\}$$

$$A = \{(x, \mu_A(x)) \mid x \in A\}$$

Classical or Crisp Sets

$$U = \{x_1, x_2, x_3, x_4, x_5, x_6\}$$

$$A = \{x_2, x_3, x_5\}$$

$$\mu_A(x_1) = 0 \qquad x_1 \notin A$$

$$\mu_A(x_2) = 1 \qquad x_2 \in A$$

$$\mu_A(x_3) = 1 \qquad x_3 \in A$$

$$\mu_A(x_4) = 0 \qquad x_4 \notin A$$

$$\mu_A(x_5) = 1 \qquad x_5 \in A$$

$$\mu_A(x_6) = 0 \qquad x_6 \notin A$$

$$\mu_A(x) = \begin{cases} 1 & ; \ x \in A \\ 0 & ; \ x \notin A \end{cases}$$

$$\mu_A(x) = \begin{cases} 1 & ; \ x = x_2, x_3, x_5 \\ 0 & ; \ x = x_1, x_4, x_6 \end{cases}$$

$$A = \left\{ \frac{x_1}{0}, \frac{x_2}{1}, \frac{x_3}{1}, \frac{x_4}{0}, \frac{x_5}{1}, \frac{x_6}{0} \right\}$$

$$A = \left\{ \frac{0}{x_1}, \frac{1}{x_2}, \frac{1}{x_3}, \frac{0}{x_4}, \frac{1}{x_5}, \frac{0}{x_6} \right\}$$

$$A = \{(x, \mu_A(x)) \mid x \in A\}$$

Classical Sets

Characteristic function

Let X be a set and A be a subset of X. Then the function

$$\chi_A(x) = \begin{cases} 1 & if \quad x \in A \\ 0 & if \quad x \notin A \end{cases}$$

is called the characteristic function of the set A in X.

Classical or Crisp Sets

$$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cup B \longrightarrow \{0,1\}$$

$$A \cup B \longrightarrow \chi_{A \cup B}(x)$$

$$\chi_A(x) \vee \chi_B(x) \longrightarrow \text{maximum operator}$$

$$\max \{ \chi_A(x), \chi_B(x) \}$$

$A = \{1,2\}$, $B = \{2,3\}$

$A \cup B = \{1,2,3\}$

$$\chi_{A \cup B}(1) = \max(\chi_A(1), \chi_B(1))$$
$$= \max(1, 0) = 1$$

Classical or Crisp Sets

$$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cup B \longrightarrow \{0,1\}$$

$$A \cup B \longrightarrow \chi_{A \cup B}(x)$$

$$\chi_A(x) \vee \chi_B(x) \longrightarrow \text{maximum operator}$$

$$\max \{ \chi_A(x), \chi_B(x) \}$$

$A = \{1,2\}$, $B = \{2,3\}$

$A \cup B = \{1,2,3\}$

$$\chi_{A \cup B}(2) = \max(\chi_A(2), \chi_B(2))$$

$$= \max(1,1) = 1$$

Classical or Crisp Sets

$$A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cup B \longrightarrow \{0,1\}$$

$$A \cup B \longrightarrow \chi_{A \cup B}(x)$$

$$\chi_A(x) \vee \chi_B(x) \longrightarrow \text{maximum operator}$$

$$\max \{ \chi_A(x), \chi_B(x) \}$$

$A = \{1,2\}$, $B = \{2,3\}$

$A \cup B = \{1,2,3\}$

$$\chi_{A \cup B}(3) = \max(\chi_A(3), \chi_B(3))$$

$$= \max(0,1) = 1$$

Classical or Crisp Sets

$$A \cap B = \{ x \mid x \in A \text{ and } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cap B \longrightarrow \{0,1\}$$

$$A \cap B \longrightarrow \chi_{A \cap B}(x)$$

$$\chi_A(x) \wedge \chi_B(x) \xrightarrow{\text{minimum operator}} \min\{\chi_A(x), \chi_B(x)\}$$

$A = \{1,2\}$, $B = \{2,3\}$

$A \cap B = \{2\}$

$$\chi_{A \cap B}(1) = \min(\chi_A(1), \chi_B(1)) = \min(1,0) = 0$$

Classical or Crisp Sets

$$A \cap B = \{ x \mid x \in A \text{ and } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cap B \longrightarrow \{0,1\}$$

$$A \cap B \longrightarrow \chi_{A \cap B}(x)$$

$$\chi_A(x) \wedge \chi_B(x) \xrightarrow{\text{minimum operator}} \min\{\chi_A(x), \chi_B(x)\}$$

$A = \{1,2\}, \quad B = \{2,3\}$

$A \cap B = \{2\}$

$$\chi_{A \cap B}(2) = \min(\chi_A(2), \chi_B(2)) = \min(1,1) = 1$$

Classical or Crisp Sets

$$A \cap B = \{ x \mid x \in A \text{ and } x \in B \}$$

$$\chi : A \longrightarrow \{0,1\}$$

$$\chi : A \cap B \longrightarrow \{0,1\}$$

$$A \cap B \longrightarrow \chi_{A \cap B}(x)$$

$$\chi_A(x) \wedge \chi_B(x) \longrightarrow \text{minimum operator}$$

$$\min\{ \chi_A(x), \chi_B(x) \}$$

$A = \{1,2\}, \quad B = \{2,3\}$

$A \cap B = \{2\}$

$$\chi_{A \cap B}(3) = \min(\chi_A(3), \chi_B(3))$$
$$= \min(0, 1) = 0$$

Classical or Crisp Sets

$$X = \{a, b, c, d\}$$

A

$\bar{A} \longrightarrow \chi_{\bar{A}}(x)$

$$1 - \chi_A(x)$$

$A = \{a, b, c\}$

$\chi_A(a) = 1$

$\chi_A(b) = 1$

$\chi_A(c) = 1$

$\chi_A(d) = 0$

$\bar{A}$

$1 - \chi_A(a) = 1 - 1 = 0$

$1 - \chi_A(b) = 1 - 1 = 0$

$1 - \chi_A(c) = 1 - 1 = 0$

$1 - \chi_A(d) = 1 - 0 = 1$

Classical or Crisp Sets

$X = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10, a, b, c\}$

$A = \{1, 3, 5, 7, 9, b\}$

$B = \{1, 2, 4, 5, 7, b, c\}$

$C = \{5, 7, a, b, c\}$

$$A \cup (B \cup C) = (A \cup B) \cup C$$

$$A \cap (B \cap C) = (A \cap B) \cap C$$

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Classical or Crisp Sets

$$X = \{1, 2, 3, 4, 5, a, b, c\}$$

$$A = \{1, 3, 5, b, c\}$$

$$A \cup \phi = A$$

$$A \cap \phi = \phi$$

$$A \cap X = A$$

$$A \cup X = A$$

Classical or Crisp Sets

X, A

$$A \longrightarrow \chi_A(x)$$

$$\chi_A(x) = \begin{cases} 1 & x \in A \quad \text{member} \\ 0 & x \notin A \quad \text{not member} \end{cases}$$

$$A \longrightarrow \{0, 1\}$$

$$A \longrightarrow [0, 1] \quad \checkmark \qquad x \in \%$$

$$(x, \underline{0.7})$$

Classical or Crisp Sets

$X = \{1, 2, 3, 4, 5, 6\}$

$A = \{2, 4, 6\}$

$$\chi_A(x) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

$\chi_A(1) = 0$

$\chi_A(2) = 1$

$\chi_A(3) = 0$

$\chi_A(4) = 1$

$\chi_A(5) = 0$

$\chi_A(6) = 1$

$$A = \{(x, \chi_A(x)) \mid x \in A, \chi_A(x) \in \{0, 1\}\}$$

$$= \{(2,1), (4,1), (6,1)\}$$

$$= \{(1,0), (2,1), (3,0), (4,1), (5,0), (6,1)\}$$

Classical or Crisp Sets

Tall men

5 ft, 6 ft, 6.3 ft

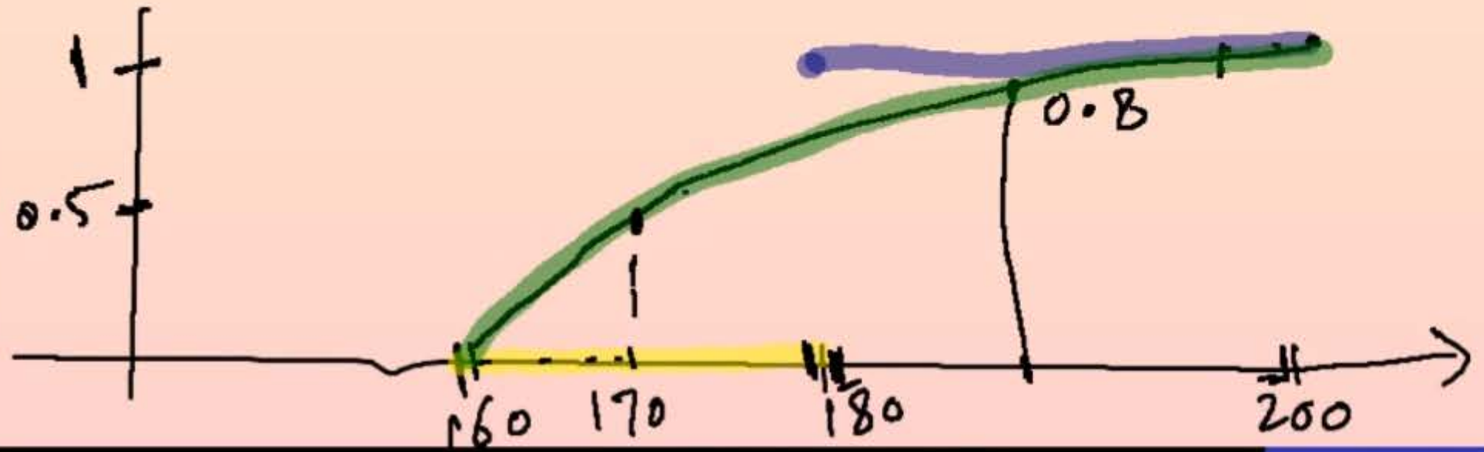
$A = \{\text{Tall men}\}$

✓

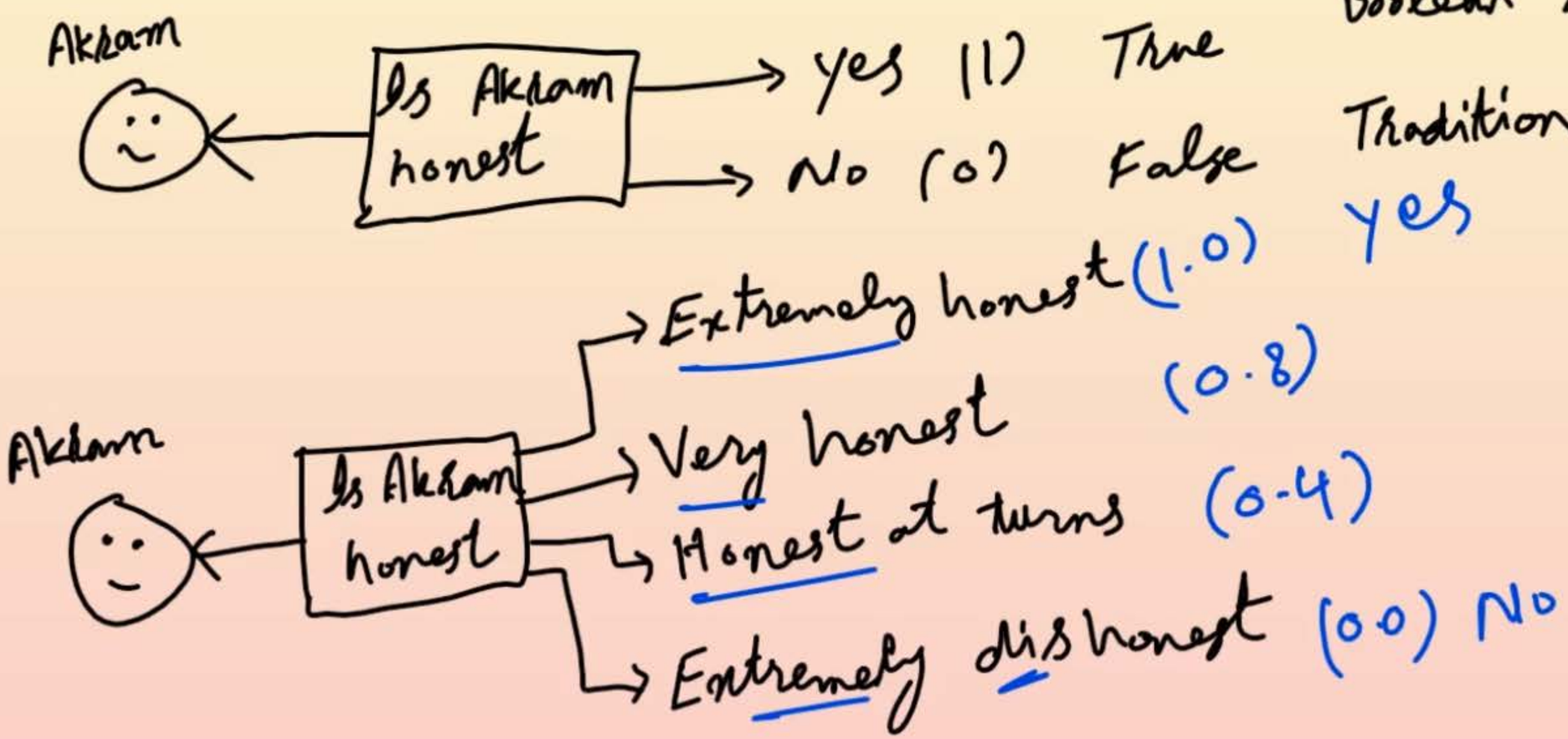
$180 \leq x$ cm

$160 \leq x < 180$

$$\mu_A(x) = \begin{cases} 1 \\ 0 \end{cases}$$



Fuzzy logic is an approach to computing based on degree of truth rather than the usual truth or false (0 or 1) Boolean logic on which the modern computer is based.



Boolean logic

Traditional Binary logic

yes

LECTURE # 03

Fuzzy Set

February 4, 1921, Baku, Azerbaijan – September 6, 2017, Berkeley, California, United States.
Nationality: Azerbaijani, American, Iranian.
Children: Norm Zada, Stella Zadeh

Mathematician, Computer scientist, Electrical engineer, Artificial intelligence researcher, and Professor emeritus of computer science at the University of California, Berkeley.
Zadeh was best known for proposing fuzzy mathematics consisting of these fuzzy-related concepts: fuzzy sets, fuzzy logic, fuzzy algorithms, fuzzy semantics, fuzzy languages, fuzzy control, fuzzy systems, fuzzy probabilities, fuzzy events, and fuzzy information.

Fuzzy Set

A fuzzy set consists of two components: a regular set and a membership function associated with it.

$$A = \{(x, \mu_A(x)) | x \in U\}$$

For an ordinary set, A,

$$\mu_A(x) = \begin{cases} 1 & iif \quad x \in A \\ 0 & iif \quad x \notin A \end{cases}$$

Fuzzy Set

When U is a finite set $\{x_1, x_2, x_3, ..., x_n\}$, the fuzzy set on U may be represented as

$$A = \sum_{i=1}^{n} \frac{x_i}{\mu_A(x_i)}$$

When U is an infinite set, the fuzzy set may be represented as

$$A = \int_x \frac{x}{\mu_A(x)}$$

Fuzzy Set

Example

Let $U = \{Akram, Aslam, Amir, Asif\}$. Evaluated by a girl, the fuzzy set "handsome boys", may be characterized as

$A = \{(Akram, 0.3), (Aslam, 0.5), (Amir, 0.7), (Asif, 0.9)\}$

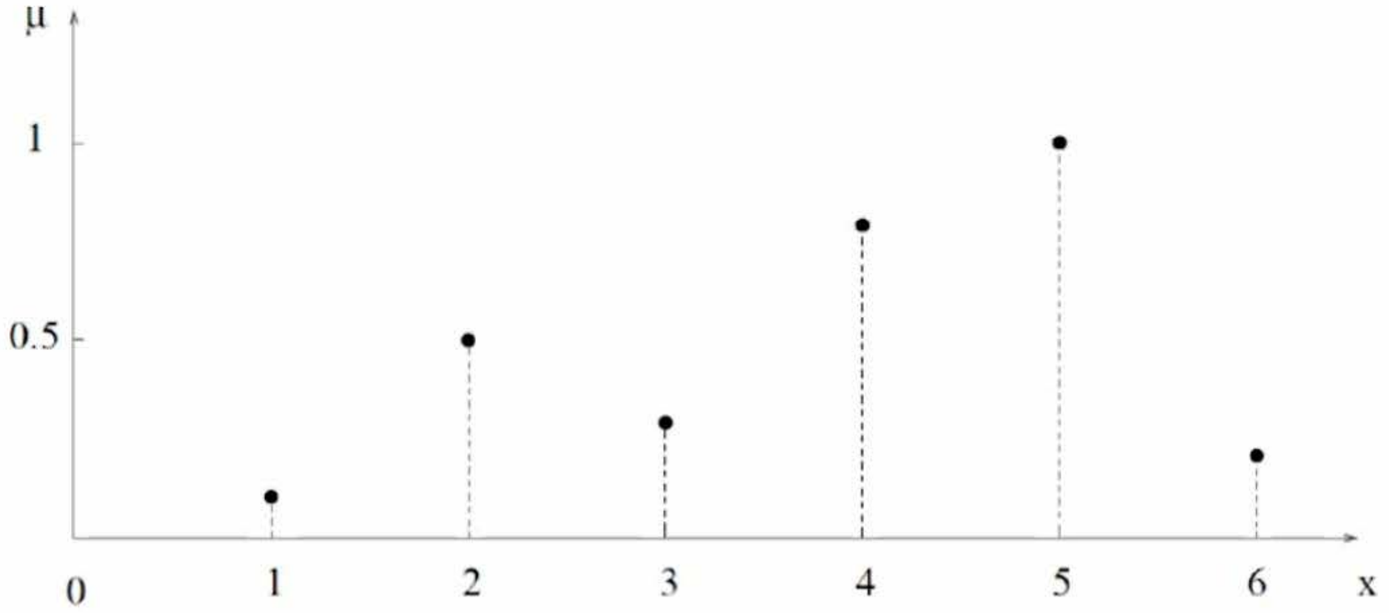
or

$$A = \frac{Akram}{0.3} + \frac{Aslam}{0.5} + \frac{Amir}{0.7} + \frac{Asif}{0.9}$$

Example

$A = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.3), (x_4, 0.8), (x_5, 1), (x_6, 0.2)\}$

$A = \{(1, 0.1), (2, 0.5), (3, 0.3), (4, 0.8), (5, 1), (6, 0.2)\}$



Fuzzy set $\mathcal{A} = \{(1, 0.1), (2, 0.5), (3, 0.3), (4, 0.8), (5, 1), (6, 0.2)\}$

Example

Fuzzy set of numbers close to 10 is

$$A = \{(x, \mu_A(x)) | x \in [5, 15], \mu_A(x) = \frac{1}{1 + (x - 10)^2}\}$$

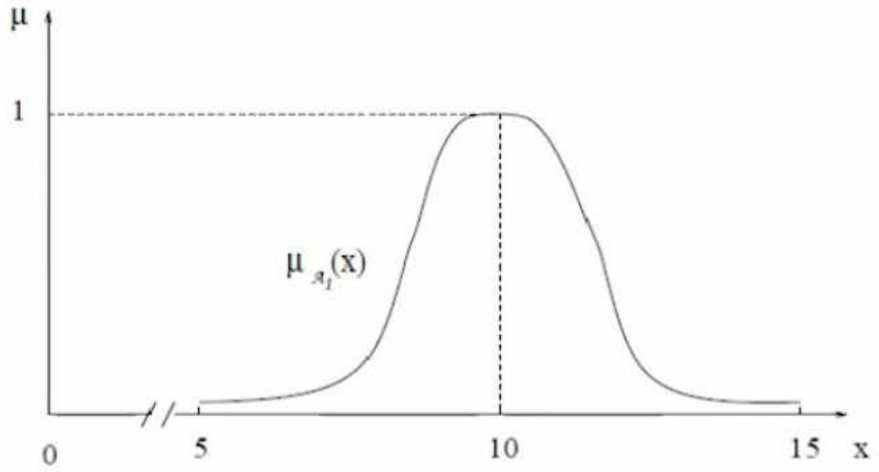


Figure: Real numbers close to 10

Example

Fuzzy set of integers close to 10 is

$$A = \{(7, 0.1), (8, 0.3), (9, 0.8), (10, 1), (11, 0.8), (12, 0.3), (13, 0.1)\}$$

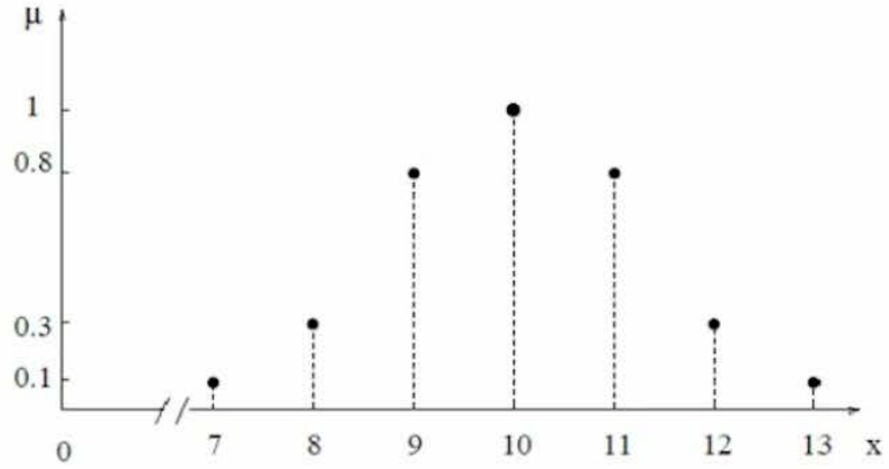
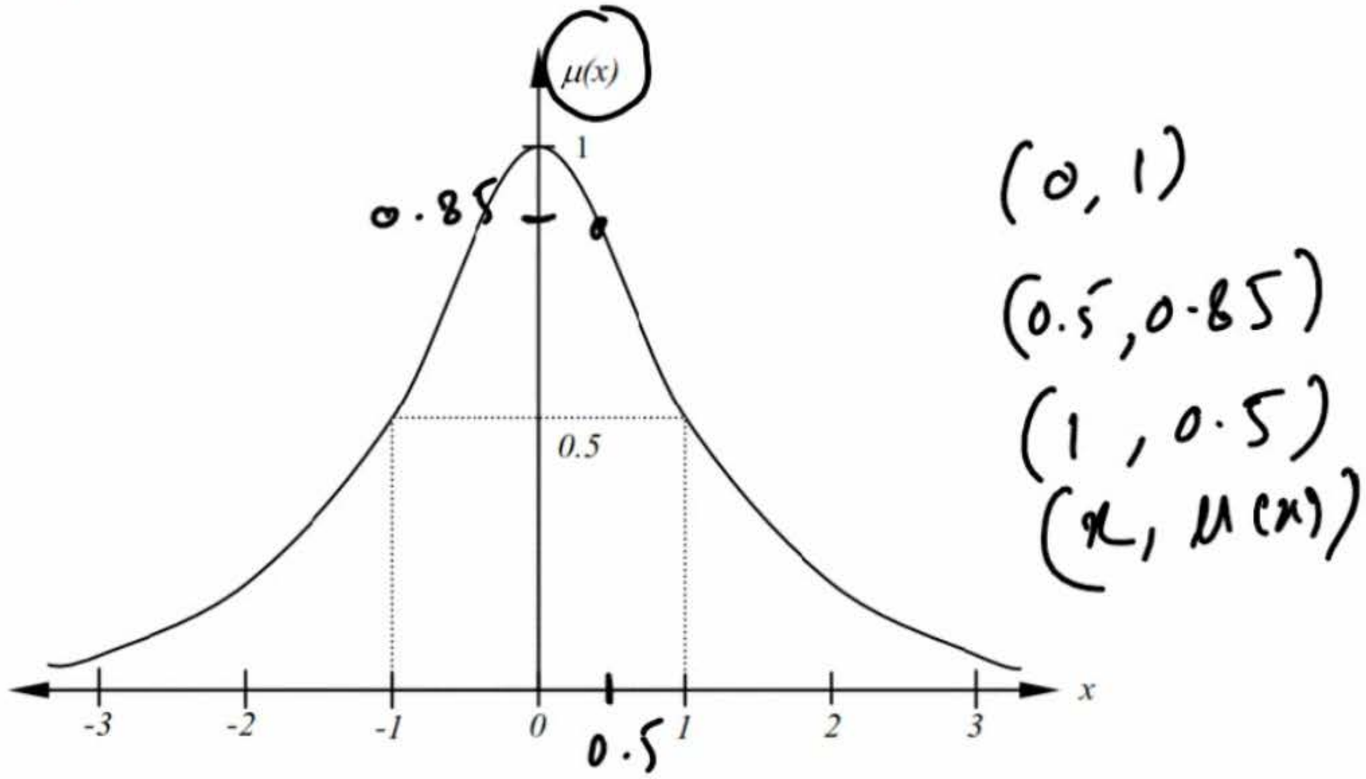


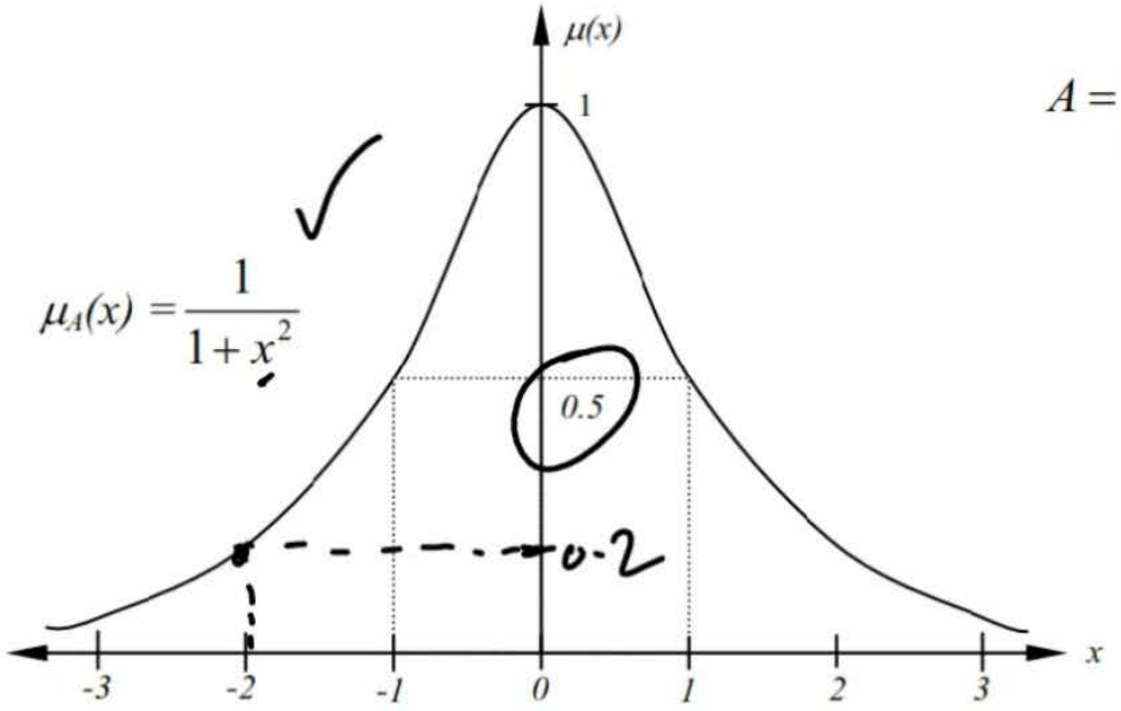
Figure: Integers close to 10

Fuzzy Set

Example Let's define a fuzzy set A ={real number near 0}. The boundary for set "real number near 0" is pretty ambiguous.



Example Let's define a fuzzy set A ={real number near 0}. The boundary for set "real number near 0" is pretty ambiguous.



$$\mu_A(x) = \frac{1}{1+x^2}$$

$$A = \int \mu_A(x)/x \qquad \text{where } \mu_A(x) = \frac{1}{1+x^2}$$

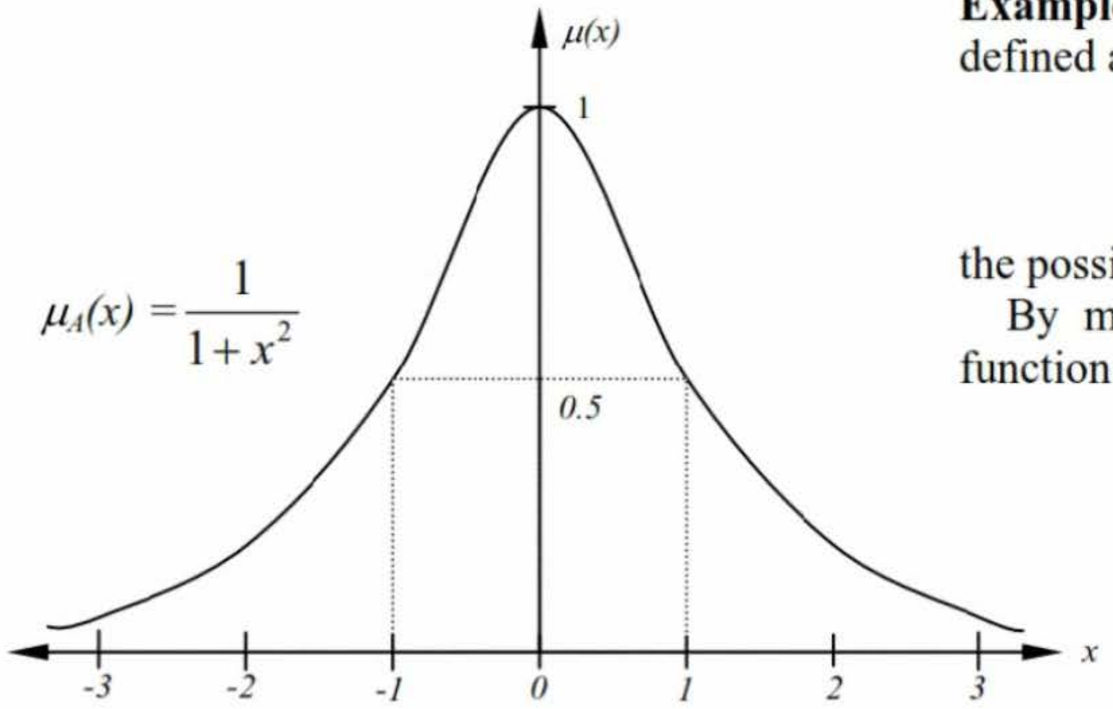
$$\frac{1}{1+(-2)^2} = \frac{1}{5} = 0.2$$

$x = 1$

$$\frac{1}{1+1^2} = \frac{1}{2} = 0.5$$

Fuzzy Set

Example Let's define a fuzzy set A ={real number near 0}. The boundary for set "real number near 0" is pretty ambiguous.



$$\mu_A(x) = \frac{1}{1+x^2}$$

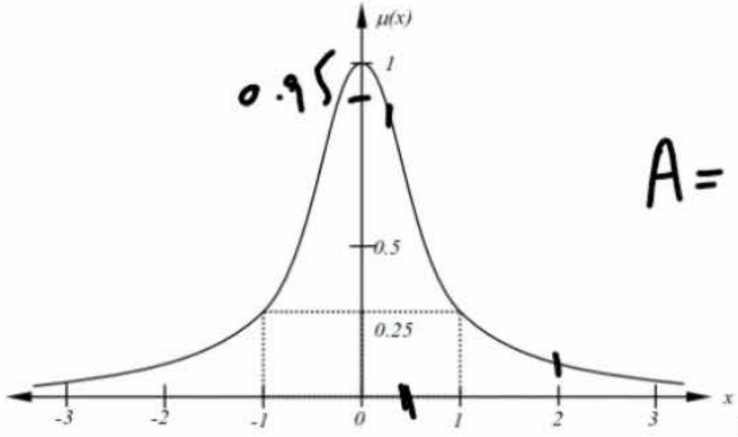
Example Another fuzzy set A ={real number very near 0} can be defined and its membership function is

$$\mu_B(x) = \left(\frac{1}{1+x^2}\right)^2$$

the possibility of 1 is 0.25, that of 2 is 0.04 and of 3 is 0.01 .

By modifying the above function, it is able to denote membership function of fuzzy set A = {real number near a} as,

$$\mu_A(x) = \frac{1}{1+(x-a)^2}$$



$$A = \int \frac{\mu_A(x)}{x}$$

$$= \int \frac{\frac{1}{1+(x-a)^2}}{x}$$

Expression of Fuzzy Set

Expression

- $A = \{(x, \mu_A(x))\}$
- $A = \frac{\mu_A(x_1)}{x_1} + \frac{\mu_A(x_2)}{x_2} + \cdots + \frac{\mu_A(x_n)}{x_n} + \cdots$
- $A = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$ ✓
- $A = \int \frac{\mu_A(x)}{x}$ ✓

Expression of Fuzzy Set

Example

$A = \sum_{i=1}^{n} \frac{\mu_A(x_i)}{x_i}$ is an other form of representation of fuzzy set. Represent the following fuzzy sets by this form.

(1) $A = \{(3, 0.9), (2, 0.3), (4, 1.0), (7, 0.7)\}$

(2) $B = \{(a, \mu_B(a)), (b, \mu_B(b)), (c, \mu_B(c)), (d, \mu_B(d)), (e, \mu_B(e))\}$

(1) $\frac{0.9}{3} + \frac{0.3}{2} + \frac{1.0}{4} + \frac{0.7}{7}$

(2) $\frac{\mu_B(a)}{a} + \frac{\mu_B(b)}{b} + \frac{\mu_B(c)}{c} + \frac{\mu_B(d)}{d} + \frac{\mu_B(e)}{e}$

Prove that $(\tilde{A} \cup \tilde{A}')(x) \geq 0.5$

Membership value of $\tilde{A} \cup \tilde{A}'$ is always greater then or equal to 0.5

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x in $\tilde{A}$.

So, by definition of compliment we have

$$\tilde{A}'(x) = 1 - \tilde{A}(x) \longrightarrow (1)$$

$$(\tilde{A} \cup \tilde{A}')(x) = \max\left(\tilde{A}(x), \tilde{A}'(x)\right)$$

$$= \max\left(\tilde{A}(x), 1 - \tilde{A}(x)\right) \text{ max by equation (1)}$$

Case I: If $\tilde{A}(x) \geq 0.5$, then obviously $1 - \tilde{A}(x) \leq 0.5$

$\Rightarrow \tilde{A}'(x) \leq 0.5$

$$\Rightarrow \max\left(\tilde{A}(x), \tilde{A}'(x)\right) \geq 0.5$$

Prove that $(\tilde{A} \cup \tilde{A}')(x) \geq 0.5$

Membership value of $\tilde{A} \cup \tilde{A}'$ is always greater then or equal to 0.5

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x in $\tilde{A}$.

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$$\tilde{A}'(x) = 1 - \tilde{A}(x) \longrightarrow (1)$$

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$$= \max\left(\tilde{A}(x), 1 - \tilde{A}(x)\right) \text{ max by equation (1)}$$

Case II: If $\tilde{A}(x) \leq 0.5$, then obviously $1 - \tilde{A}(x) \geq 0.5$

$$\Rightarrow \max\left(\tilde{A}(x), \tilde{A}'(x)\right) \geq 0.5$$

Hence, $(\tilde{A} \cup \tilde{A}')(x) \geq 0.5$

Prove that $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Membership value of $\tilde{A} \cap \tilde{A}'$ is always ~~greater~~ less then or equal to 0.5

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x. So, by definition of compliment we have $\tilde{A}'(x) = 1 - \tilde{A}(x)$

$$(\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x))$$

$$= \min(\tilde{A}(x), 1 - \tilde{A}(x))$$

Case I: If $\tilde{A}(x) \geq 0.5$, then obviously $\tilde{A}'(x) = 1 - \tilde{A}(x) \leq 0.5$

$$\Rightarrow \min(\tilde{A}(x), 1 - \tilde{A}(x)) \leq 0.5$$

Prove that $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

Membership value of $\tilde{A} \cap \tilde{A}'$ is always ~~greater~~ less then or equal to 0.5

Consider $\tilde{A}$ be a fuzzy set and $\tilde{A}(x)$ be the membership value of x. So, by definition of compliment we have $\tilde{A}'(x) = 1 - \tilde{A}(x)$

$$(\tilde{A} \cap \tilde{A}')(x) = \min(\tilde{A}(x), \tilde{A}'(x))$$

$$= \min(\tilde{A}(x), 1 - \tilde{A}(x))$$

Case II: If $\tilde{A}(x) \leq 0.5$, then obviously $\tilde{A}'(x) = 1 - \tilde{A}(x) \geq 0.5$

So, $\min(\tilde{A}(x), 1 - \tilde{A}(x)) \leq 0.5$

Hence, $(\tilde{A} \cap \tilde{A}')(x) \leq 0.5$

LECTURE #04

Support of a fuzzy set $\tilde{A}$

Support

A set is said to be the support of a fuzzy set $\tilde{A}$ if the membership values of elements are greater then zero.

$$\text{Support}(\tilde{A}) = \{x : x \in X \text{ and } \tilde{A}(x) > 0\}$$

e.g. $X = \{x_1, x_2, x_3, x_4, x_5\}$

$$\tilde{A} = \{(x_1, 0), (x_2, 0.2), (x_3, 0.9), (x_4, 1), (x_5, 0)\}$$

$$\text{Support}(\tilde{A}) = \{x_2, x_3, x_4\}$$

Normal fuzzy set

Normal fuzzy set

A fuzzy set is said to be normal fuzzy set if there exist atleast one element whose membership value is one.

$$X = \{x_1, x_2, x_3, x_4, x_5\}$$

$$\tilde{A} = \{(x_1, 0), (x_2, 0.2), (x_3, 1), (x_4, 0.5), (x_5, 0.9)\}$$

Normal fuzzy set

Normal fuzzy set

A fuzzy set is said to be normal fuzzy set if there exist atleast one element whose membership value is one.

Is $\tilde{A}: \mathbb{R} \longrightarrow [0,1]$, defined by $\tilde{A}(x) = e^{-x^2}$ a normal fuzzy set. Yes

$\because \quad \tilde{A}(x) = e^{-x^2} \quad x = 0 \quad \tilde{A}(0) = e^{-(0)^2} = e^0 = 1$

Empty fuzzy set

Empty fuzzy set

A fuzzy set is said to be empty set if all elements of universal set maps on zero i.e. the membership value of all elements is zero is called empty fuzzy set.

$$X = \{x_1, x_2, x_3, x_4\}$$

$$\tilde{A} = \{(x_1, 0), (x_2, 0), (x_3, 0), (x_4, 0)\}$$

$\tilde{A}$ = empty fuzzy set.

Cardinality of fuzzy set

Cardinality of fuzzy set

Cardinality of a fuzzy set is the sum of all membership values of the elements of the set.

$X = \{x_1, x_2, x_3, x_4, x_5\}$

$A = \{x_1, x_2, x_3\} \quad |A| = 3$

$A = \{(x_1, 1), (x_2, 1), (x_3, 1), (x_4, 0), (x_5, 0)\}$

$|A| = 1 + 1 + 1 = 3$

$\tilde{F} = \{(x_1, 0.1), (x_2, 0.5), (x_3, 0.7)\}$

$|\tilde{F}| = 0.1 + 0.5 + 0.7 = 1.3$

Degree of subsethood of $\tilde{A}$ in $\tilde{B}$

Degree of subsethood

Degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is defined as $S\left(\tilde{A}, \tilde{B}\right) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|}$

e.g. $\tilde{A} = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.5)\}$

$\tilde{B} = \{x_1, 0.5), (x_2, 0.4), (x_3, 0.2)\}$

$\tilde{A} \cap \tilde{B} = \{(x_1, 0.1), (x_2, 0.3), (x_3, 0.2)\}$

$|\tilde{A} \cap \tilde{B}| = 0.1 + 0.3 + 0.2 = 0.6$

$|\tilde{A}| = 0.1 + 0.3 + 0.5 = 0.9$

So, degree of subsethood of $\tilde{A}$ in $\tilde{B}$ is

$$S(\tilde{A}, \tilde{B}) = \frac{|\tilde{A} \cap \tilde{B}|}{|\tilde{A}|} = \frac{0.6}{0.9} = 0.67$$

Note:
$S(\tilde{A}, \tilde{B}) \in [0, 1]$

Union

Union

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cup \tilde{B}$.

Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

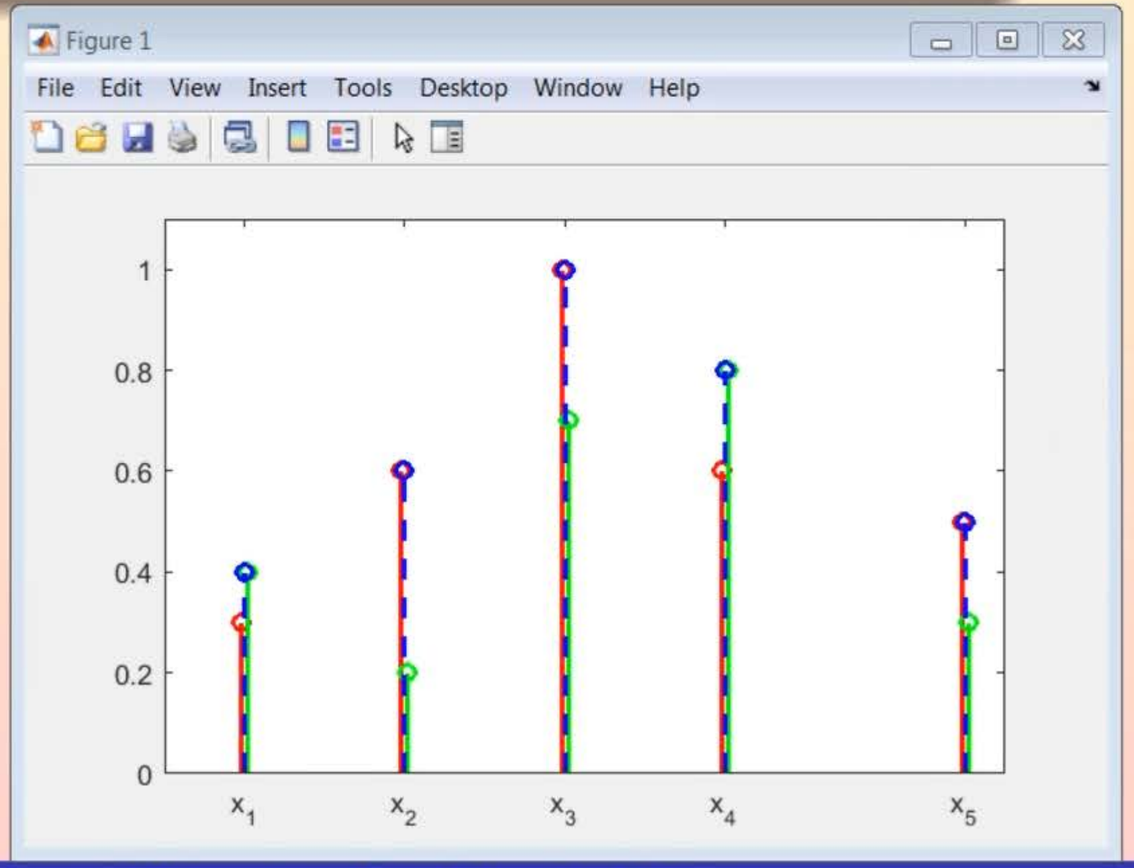
$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\tilde{A} \cup \tilde{B} = \{(x_1, 0.4), (x_2, 0.6), (x_3, 1), (x_4, 0.8), (x_5, 0.5)\}$$

Union

Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$
$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$
$$\tilde{A} \cup \tilde{B} = \{(x_1, 0.4), (x_2, 0.6), (x_3, 1), (x_4, 0.8), (x_5, 0.5)\}$$



Intersection

Intersection

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The intersection of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cap \tilde{B}$.

Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

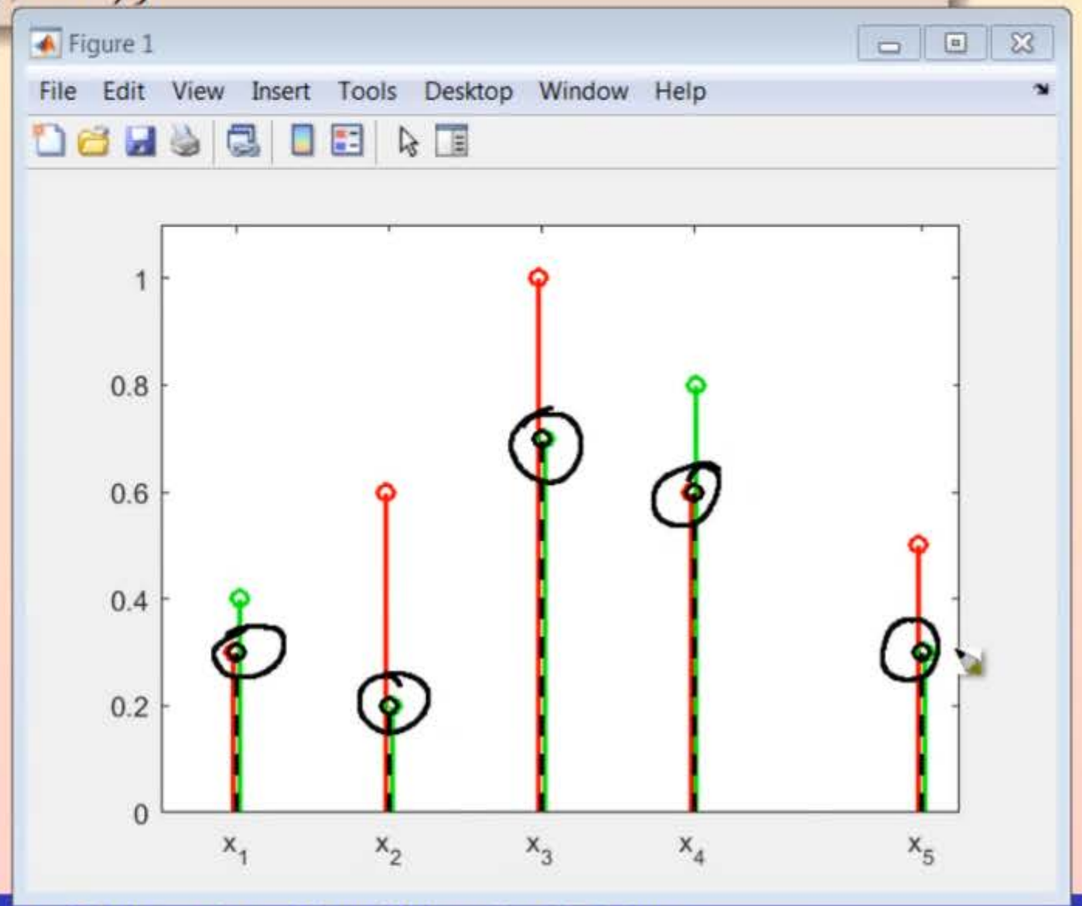
$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\tilde{A} \cap \tilde{B} = \{(x_1, 0.3), (x_2, 0.2), (x_3, 0.7), (x_4, 0.6), (x_5, 0.3)\}$$

Intersection

Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$
$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$
$$\tilde{A} \cap \tilde{B} = \{(x_1, 0.3), (x_2, 0.2), (x_3, 0.7), (x_4, 0.6), (x_5, 0.3)\}$$



LECTURE # 05

Complement

Complement

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.
Let $\tilde{A} \in \tilde{F}(X)$. The Complement of $\tilde{A}$ is the fuzzy set $\tilde{B}$ with

$$\mu_{\tilde{B}}(x) = 1 - \mu_{\tilde{A}}(x), \quad \forall x \in X.$$

We denote $\tilde{B} = \bar{\tilde{A}}$.

Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\bar{\tilde{A}} = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0), (x_4, 0.4), (x_5, 0.5)\}$$

$$\bar{\tilde{B}} = \{(x_1, 0.6), (x_2, 0.8), (x_3, 0.3), (x_4, 0.2), (x_5, 0.7)\}$$

Complement

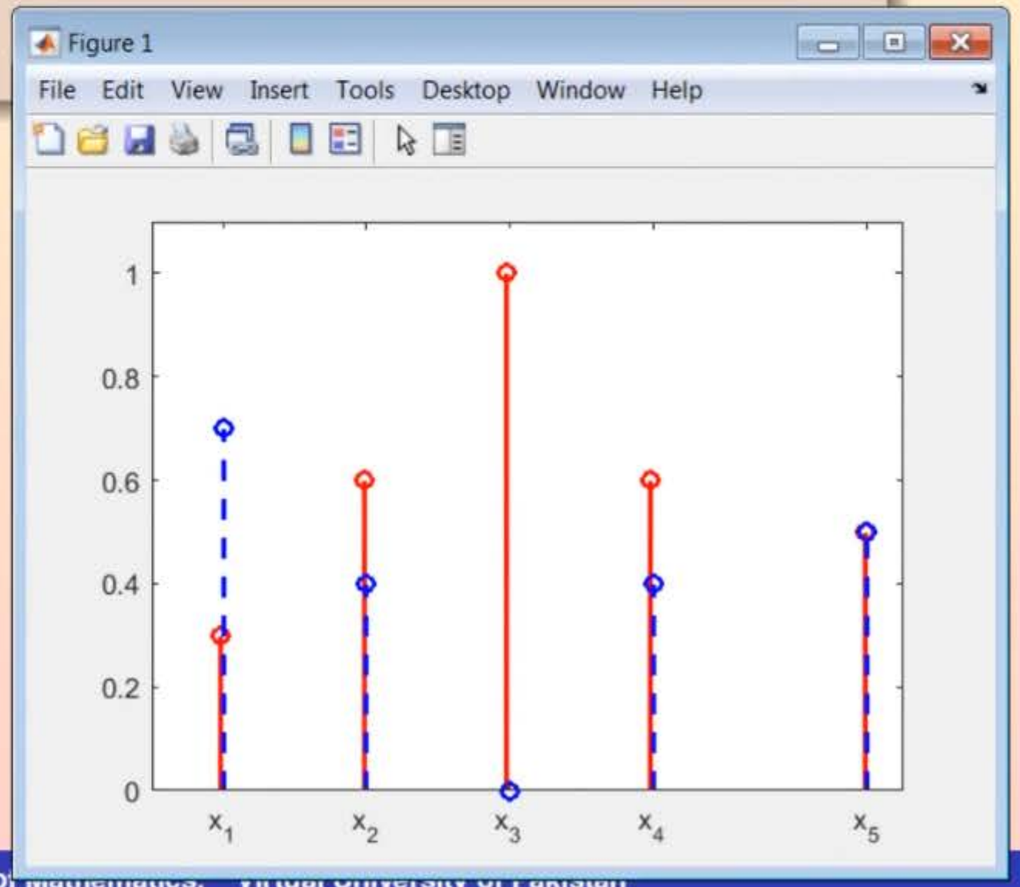
Example

$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\bar{\tilde{A}} = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0), (x_4, 0.4), (x_5, 0.5)\}$$

$$\bar{\tilde{B}} = \{(x_1, 0.6), (x_2, 0.8), (x_3, 0.3), (x_4, 0.2), (x_5, 0.7)\}$$



Complement

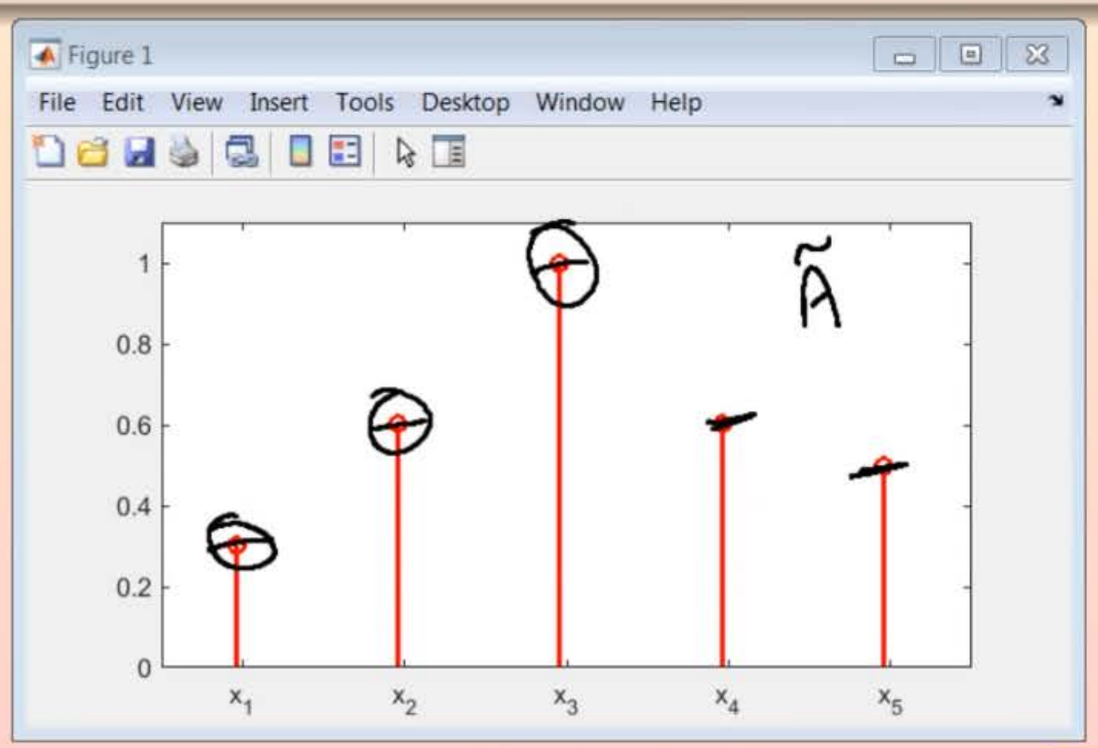
Example

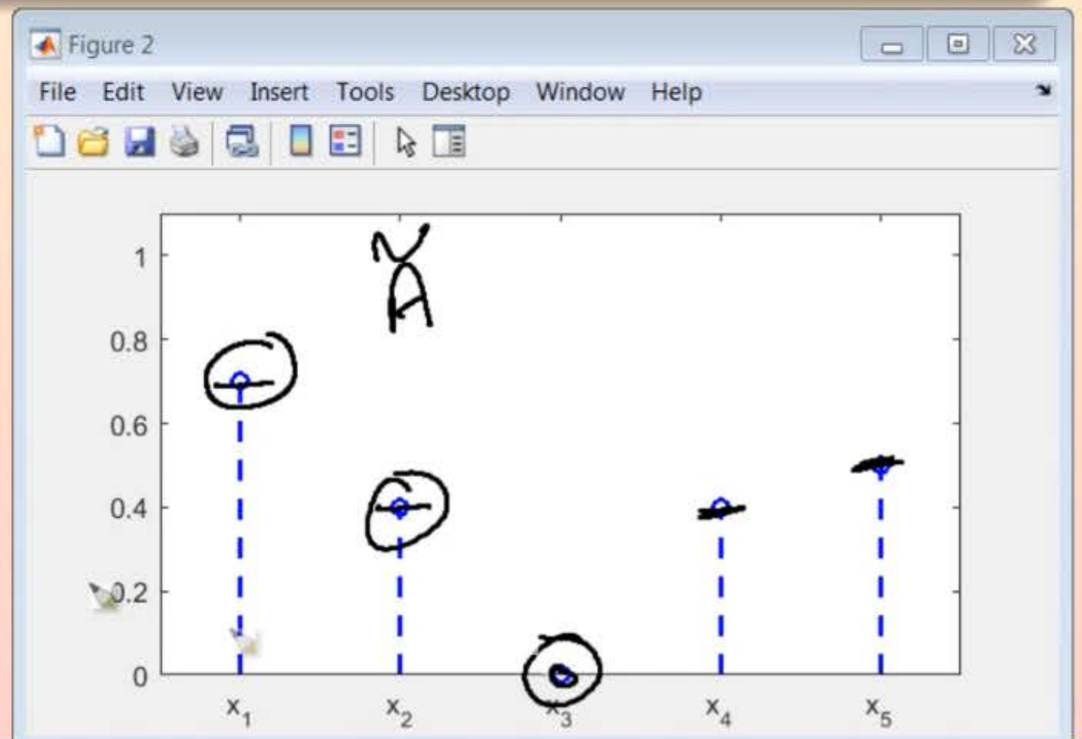
$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\bar{\tilde{A}} = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0), (x_4, 0.4), (x_5, 0.5)\}$$

$$\bar{\tilde{B}} = \{(x_1, 0.6), (x_2, 0.8), (x_3, 0.3), (x_4, 0.2), (x_5, 0.7)\}$$





Complement

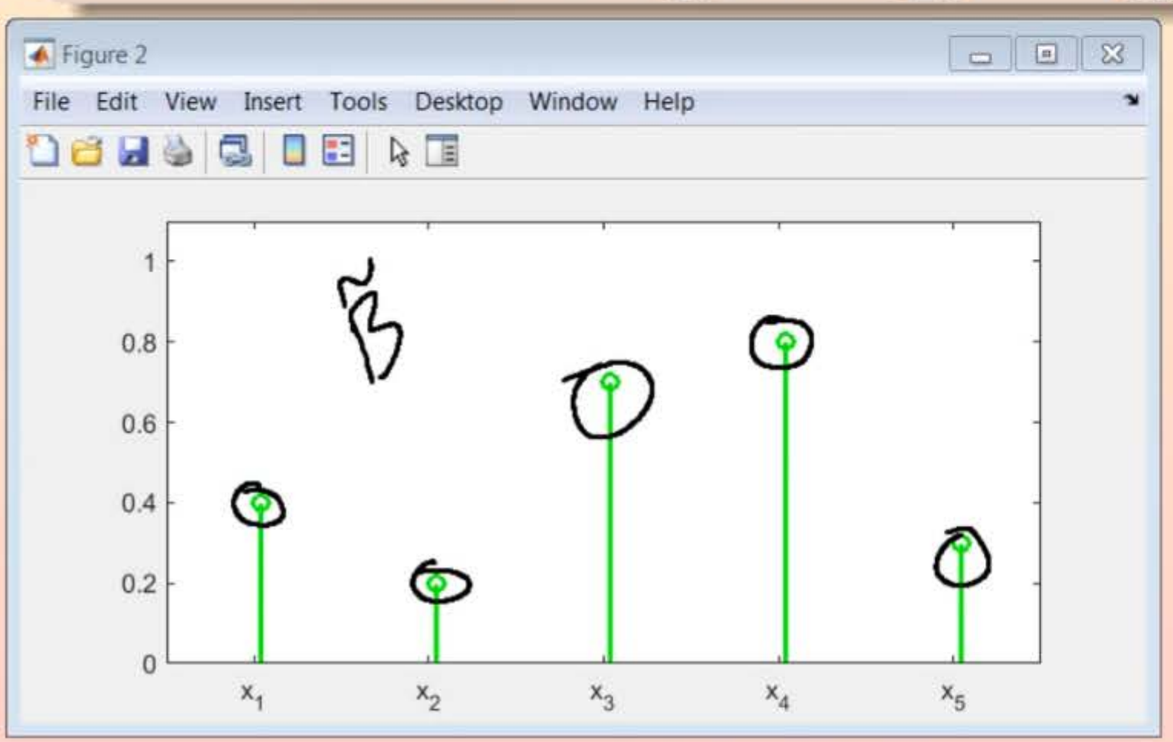
Example

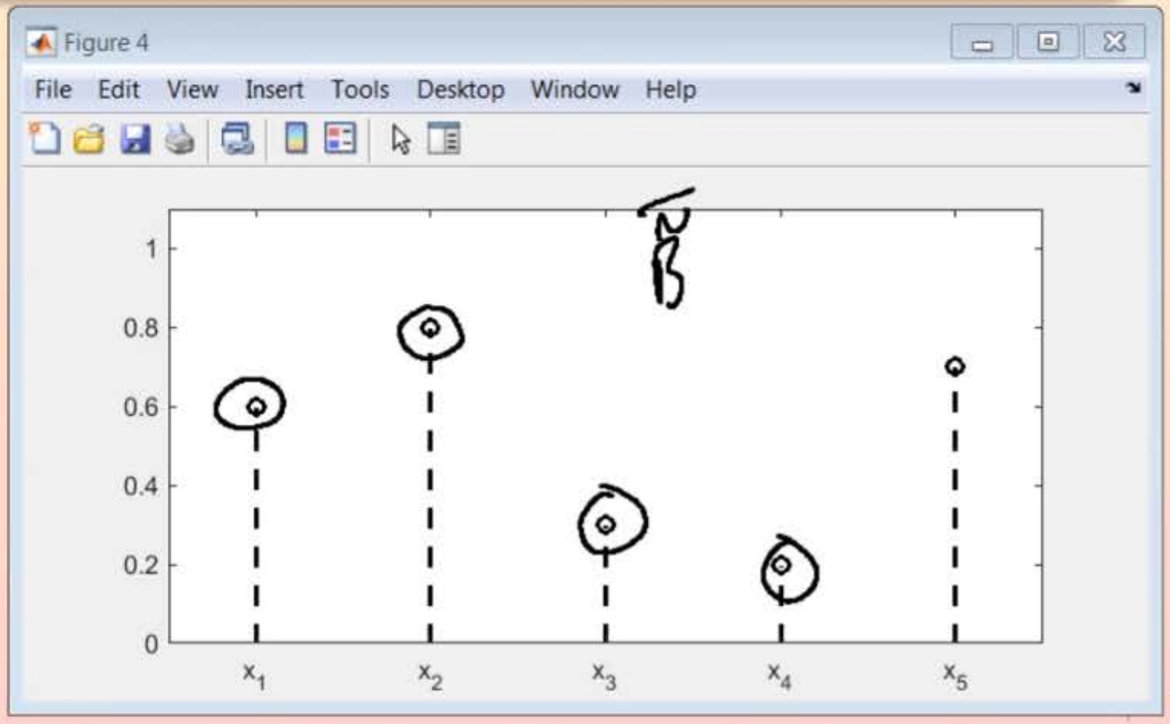
$$\tilde{A} = \{(x_1, 0.3), (x_2, 0.6), (x_3, 1), (x_4, 0.6), (x_5, 0.5)\}$$

$$\tilde{B} = \{(x_1, 0.4), (x_2, 0.2), (x_3, 0.7), (x_4, 0.8), (x_5, 0.3)\}$$

$$\bar{\tilde{A}} = \{(x_1, 0.7), (x_2, 0.4), (x_3, 0), (x_4, 0.4), (x_5, 0.5)\}$$

$$\bar{\tilde{B}} = \{(x_1, 0.6), (x_2, 0.8), (x_3, 0.3), (x_4, 0.2), (x_5, 0.7)\}$$





Let $X = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$, Let two fuzzy sets on X are

$\tilde{A} = \{(0, 0), (1, 0), (2, 0.1), (3, 0.2), (4, 0.3), (5, 0.8), (6, 0.9), (7, 1), (8, 1), (9, 1)\}$

$\tilde{B} = \{(0, 1), (1, 1), (2, 0.9), (3, 0.8), (4, 0.7), (5, 0.5), (6, 0.4), (7, 0.2), (8, 0.2), (9, 0)\}$

Determine $\tilde{A} \vee \tilde{B}$, $\tilde{A} \wedge \tilde{B}$, $\tilde{A}'$

$(\tilde{A} \vee \tilde{B})(x) = \max\{\tilde{A}(x), \tilde{B}(x)\}$.

$\tilde{A} \vee \tilde{B} = \{(0, 1), (1, 1), (2, 0.9), (3, 0.8), (4, 0.7), (5, 0.8), (6, 0.9), (7, 1), (8, 1), (9, 1)\}$

$\tilde{A} \wedge \tilde{B} = \min\{\tilde{A}(x), \tilde{B}(x)\}$

$= \{(0, 0), (1, 0), (2, 0.1), (3, 0.2), (4, 0.3), (5, 0.5), (6, 0.4), (7, 0.2), (8, 0.2), (9, 0)\}$

Let $X = \{0,1,2,3,4,5,6,7,8,9\}$, Let two fuzzy sets on X are

$\tilde{A} = \{(0,0), (1,0), (2,0.1), (3,0.2), (4,0.3), (5,0.8), (6,0.9), (7,1), (8,1), (9,1)\}$

$\tilde{B} = \{(0,1), (1,1), (2,0.9), (3,0.8), (4,0.7), (5,0.5), (6,0.4), (7,0.2), (8,0.2), (9,0)\}$

Determine $\tilde{A} \vee \tilde{B}$, $\tilde{A} \wedge \tilde{B}$, $\tilde{A}'$

$$\tilde{A}'(x) = 1 - \tilde{A}(x)$$

$\tilde{A}' = \{(0,1), (1,1), (2,0.9), (3,0.8), (4,0.7), (5,0.2), (6,0.1), (7,0), (8,0), (9,0)\}$

(i) $(A \cup B)^c = A^c \cap B^c$ — $(\tilde{A} \vee \tilde{B})' = \tilde{A}' \wedge \tilde{B}'$

(ii) $(A \cap B)^c = A^c \cup B^c$ — $(\tilde{A} \wedge \tilde{B})' = \tilde{A}' \vee \tilde{B}'$

De Morgan's Laws

$$\tilde{A} = \{(1, 0.2), (3, 0.5), (5, 1), (7, 0.6), (10, 0.4)\}$$

$$\tilde{B} = \{(1, 0.5), (3, 0.2), (5, 0.8), (7, 0.7), (10, 0.3)\}$$

$$\tilde{A} \cup \tilde{B} = \{(1, 0.5), (3, 0.5), (5, 1), (7, 0.7), (10, 0.4)\}$$

$$\overline{\tilde{A} \cup \tilde{B}} = \{(1, 0.5), (3, 0.5), (5, 0), (7, 0.3), (10, 0.6)\}$$

$$\bar{\tilde{A}} = \{(1, 0.8), (3, 0.5), (5, 0), (7, 0.4), (10, 0.6)\}$$

$$\bar{\tilde{B}} = \{(1, 0.5), (3, 0.8), (5, 0.2), (7, 0.3), (10, 0.7)\}$$

$$\bar{\tilde{A}} \cap \bar{\tilde{B}} = \{(1, 0.5), (3, 0.5), (5, 0), (7, 0.3), (10, 0.6)\}$$

$$\overline{\tilde{A} \cup \tilde{B}} = \bar{\tilde{A}} \cap \bar{\tilde{B}}$$

$$\overline{\tilde{A} \cap \tilde{B}} = \bar{\tilde{A}} \cup \bar{\tilde{B}}$$

Union

Union

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

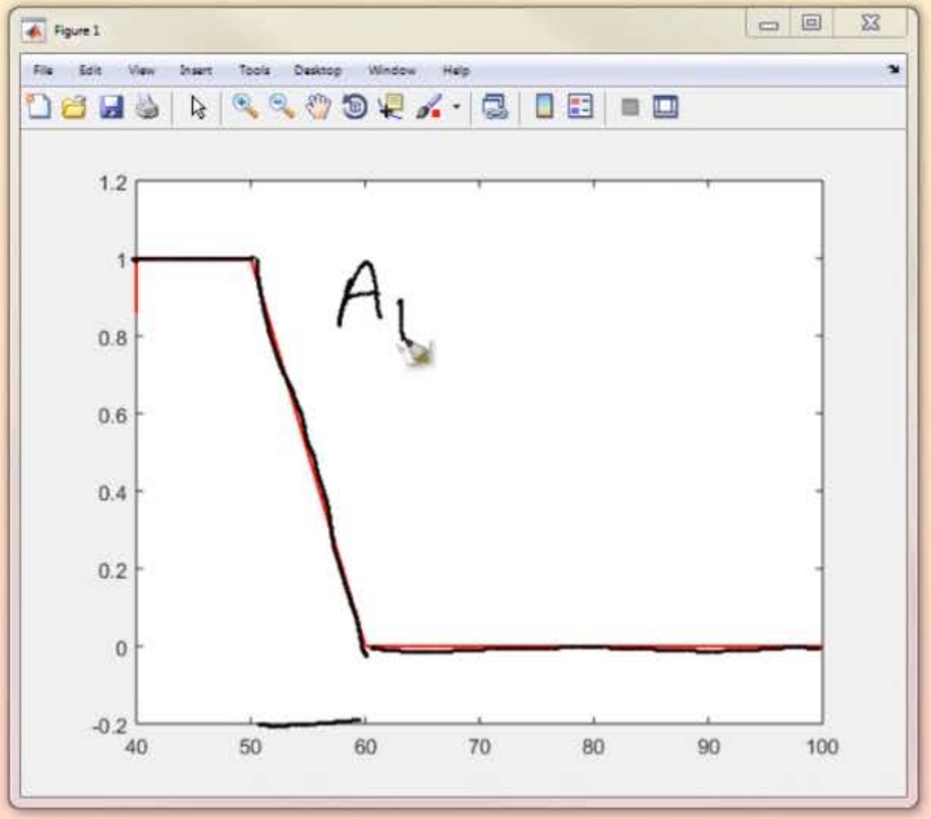
Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cup \tilde{B}$.

Example

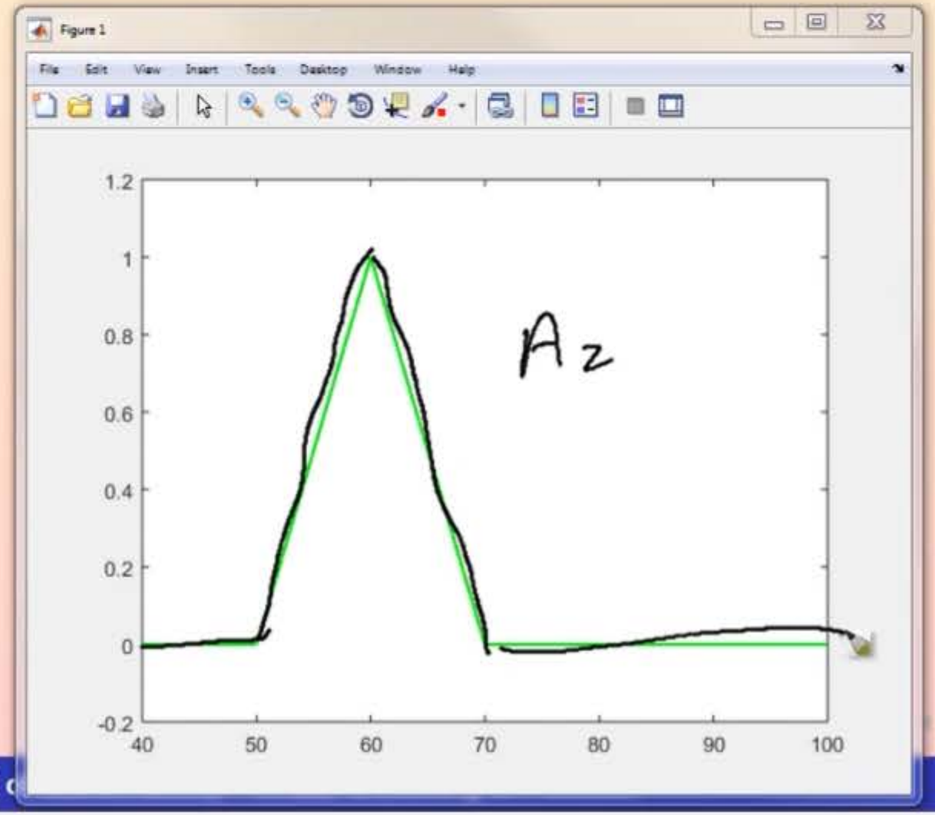
$$A_1(x) = \begin{cases} 1 & \text{if } 40 \leq x < 50 \\ 1 - \frac{x-50}{10} \ \checkmark & \text{if } 50 \leq x < 60 \\ 0 & \text{if } 60 \leq x \leq 100 \end{cases}$$



Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} \checkmark & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} \checkmark & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$



Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

$$A_1(x) \vee A_2(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 55 \\ \frac{x-50}{10} & \text{if } 55 \le x \le 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

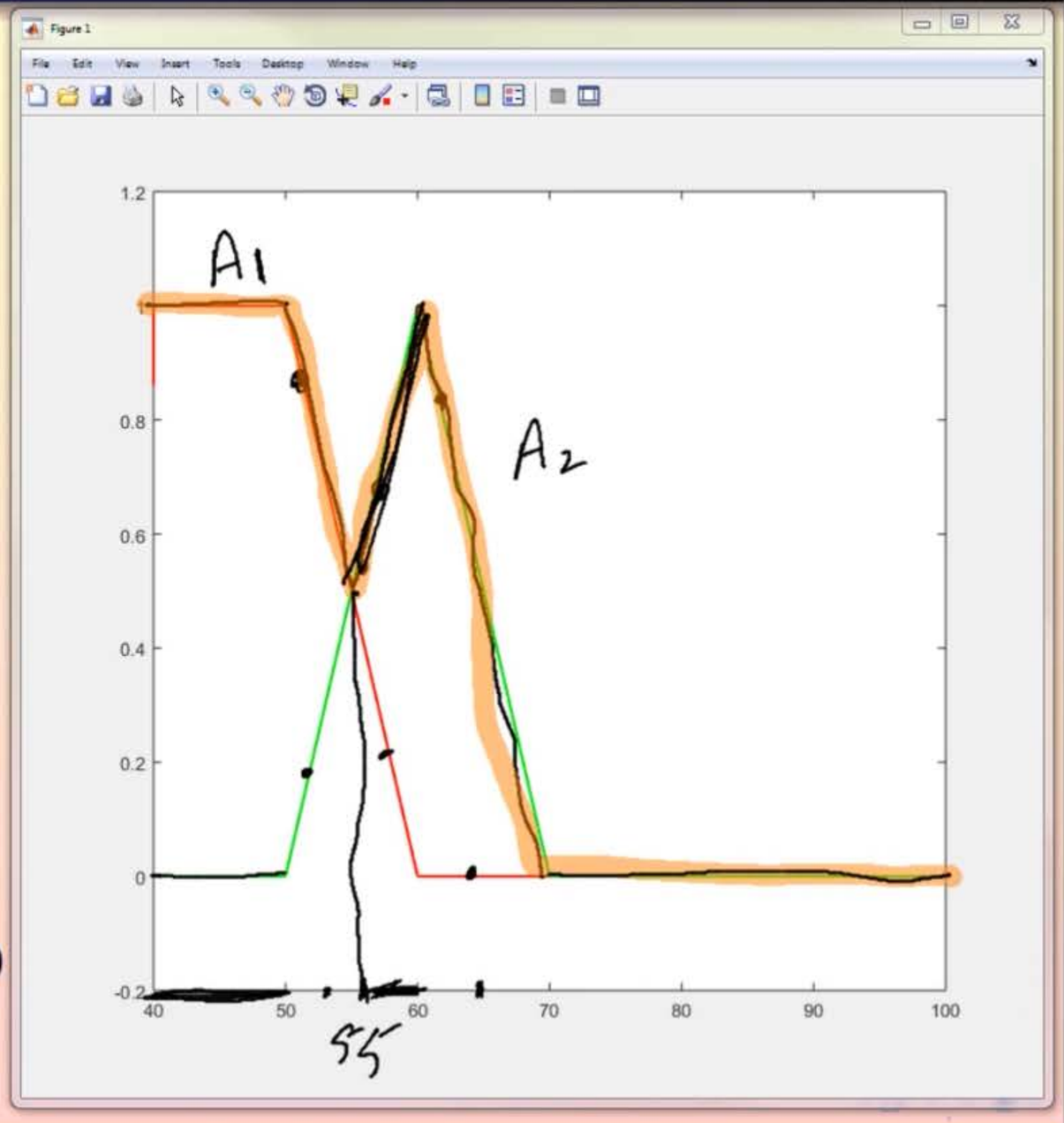
max

Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

$$A_1(x) \vee A_2(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 55 \\ \frac{x-50}{10} & \text{if } 55 \le x \le 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

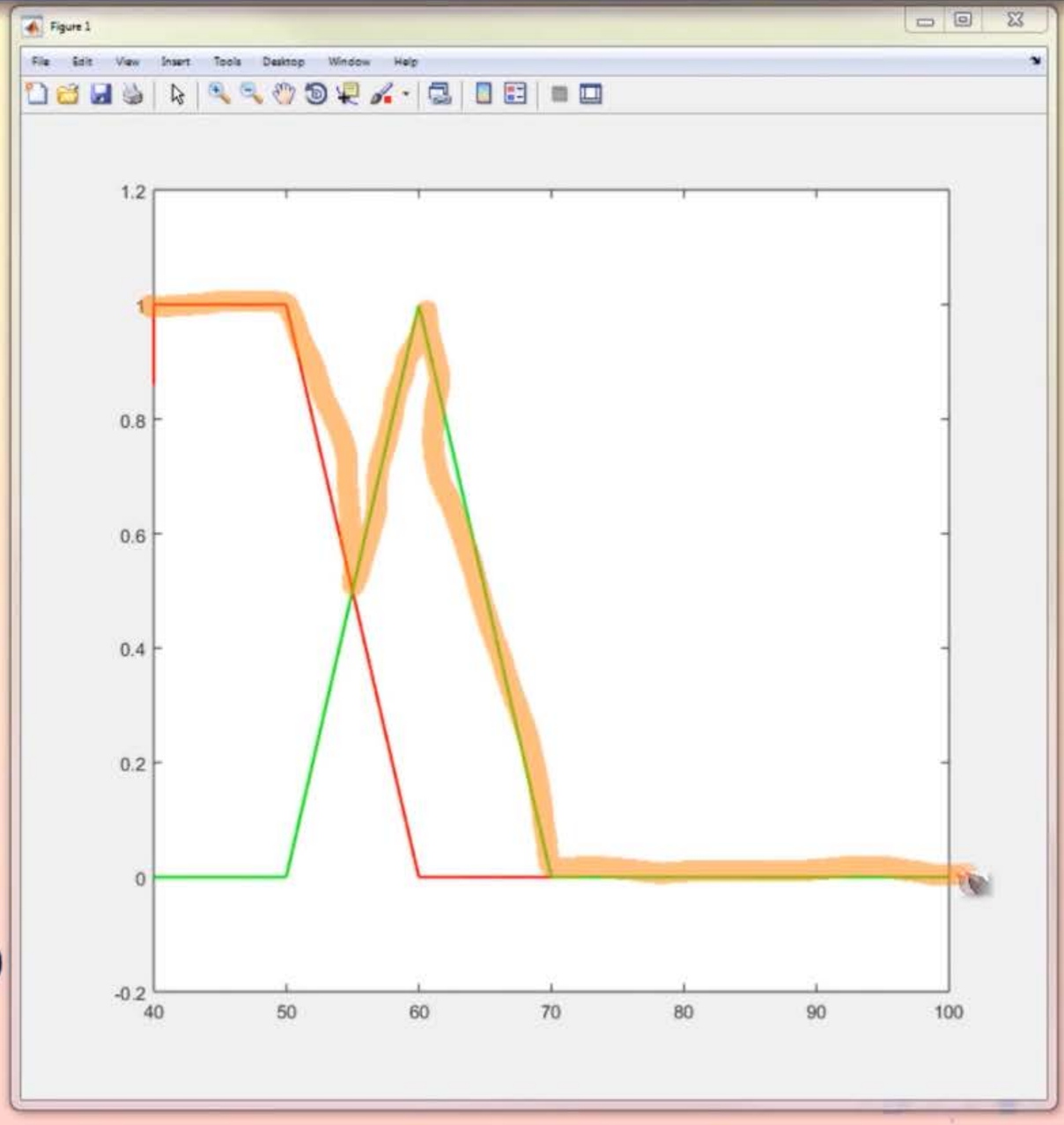


Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \leq x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \leq x < 60 \\ 0 & \text{if } 60 \leq x \leq 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \leq x < 50 \\ \frac{x-50}{10} & \text{if } 50 \leq x < 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \leq x < 70 \\ 0 & \text{if } 70 \leq x \leq 100 \end{cases}$$

$$A_1(x) \vee A_2(x) = \begin{cases} 1 & \text{if } 40 \leq x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \leq x < 55 \\ \frac{x-50}{10} & \text{if } 55 \leq x \leq 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \leq x < 70 \\ 0 & \text{if } 70 \leq x \leq 100 \end{cases}$$



LECTURE # 06

Intersection

Intersection

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

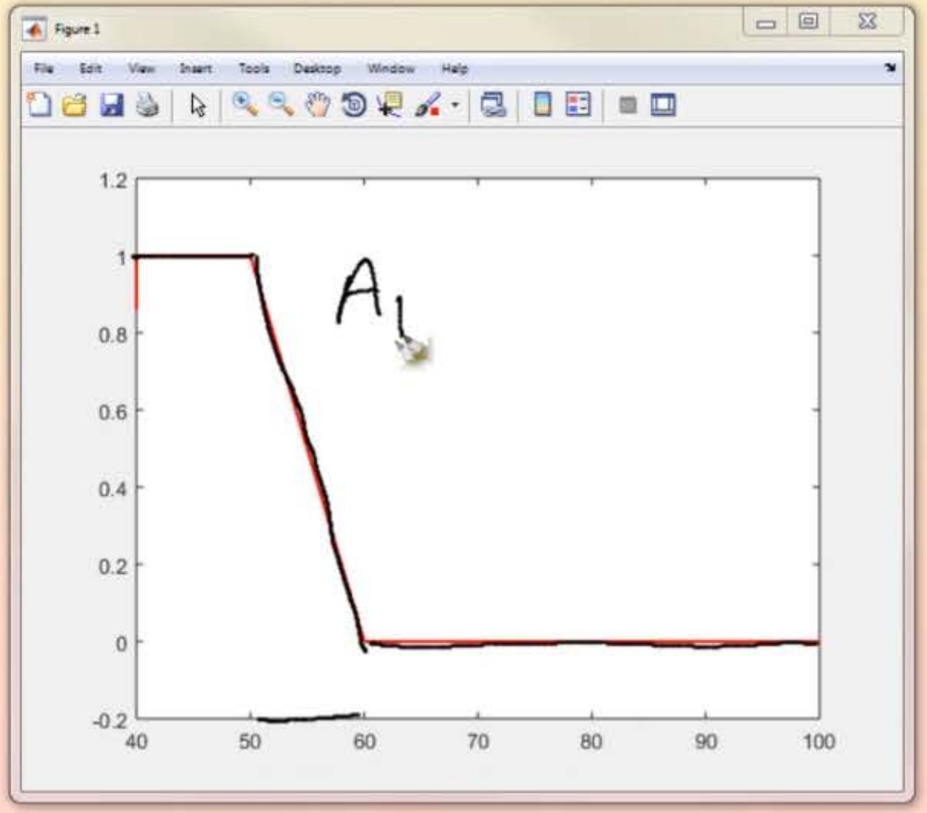
Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The intersection of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cap \tilde{B}$.

Example

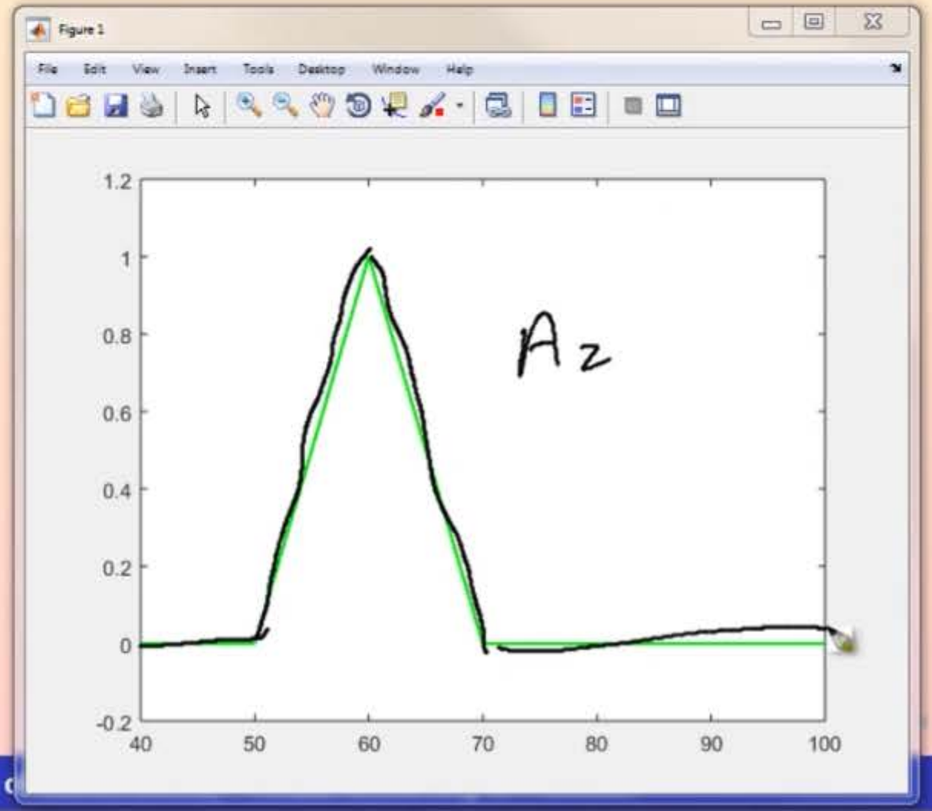
$$A_1(x) = \begin{cases} 1 & \text{if } 40 \leq x < 50 \\ 1 - \frac{x-50}{10} \checkmark & \text{if } 50 \leq x < 60 \\ 0 & \text{if } 60 \leq x \leq 100 \end{cases}$$



Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

$$A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} \checkmark & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} \checkmark & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$



Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases} \qquad A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 1 - \frac{x-60}{10} & \text{if } 60 \le x < 70 \\ 0 & \text{if } 70 \le x \le 100 \end{cases}$$

$$A_1(x) \wedge A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 55 \\ 1 - \frac{x-50}{10} & \text{if } 55 \le x \le 60 \\ 0 & \text{if } 60 < x \le 100 \end{cases}$$

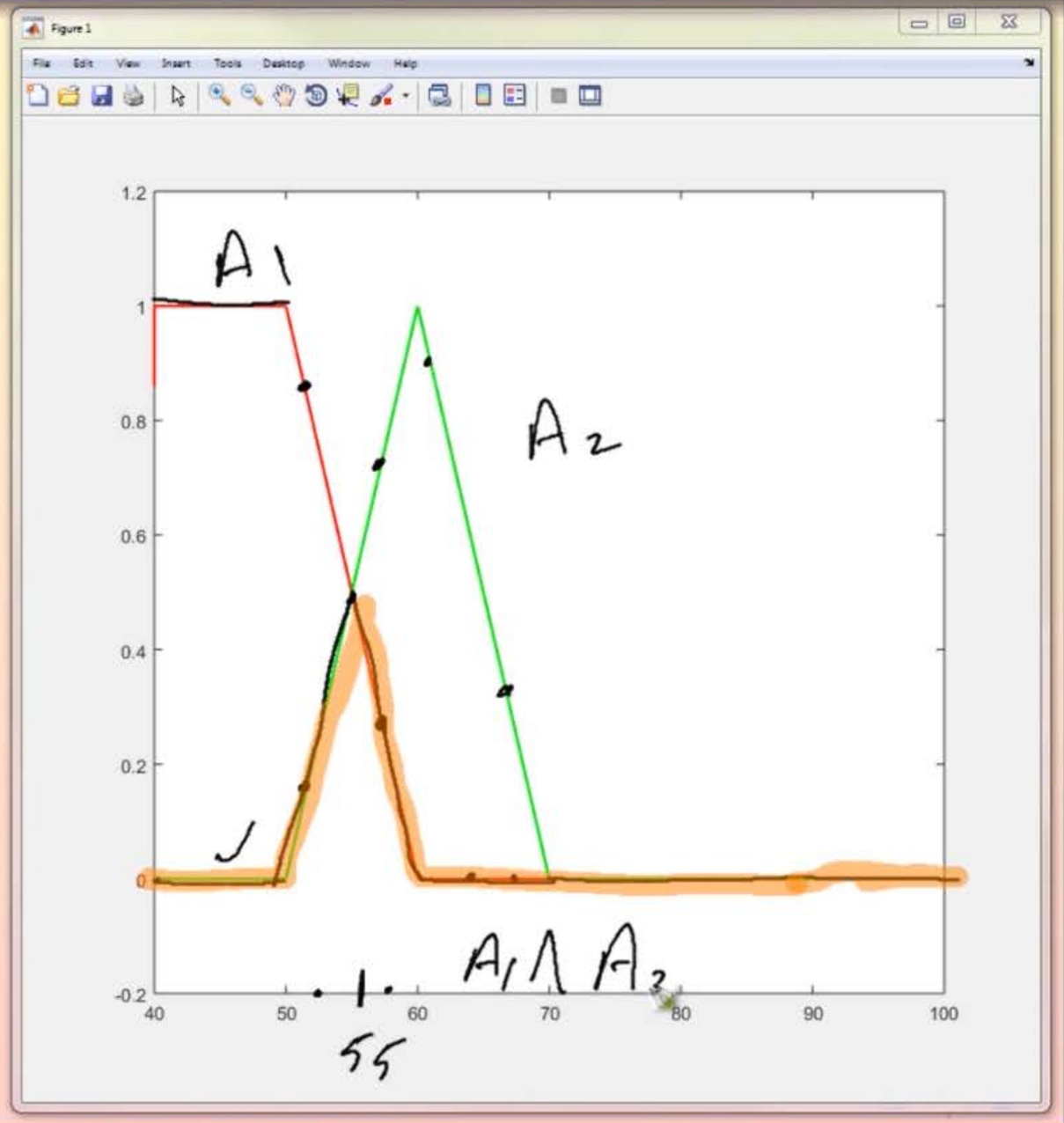
min

Example

$$A_1(x) = \begin{cases} 1 & \text{if } 40 \le x < 50 \\ 1 - \frac{x-50}{10} & \text{if } 50 \le x < 60 \\ 0 & \text{if } 60 \le x \le 100 \end{cases}$$

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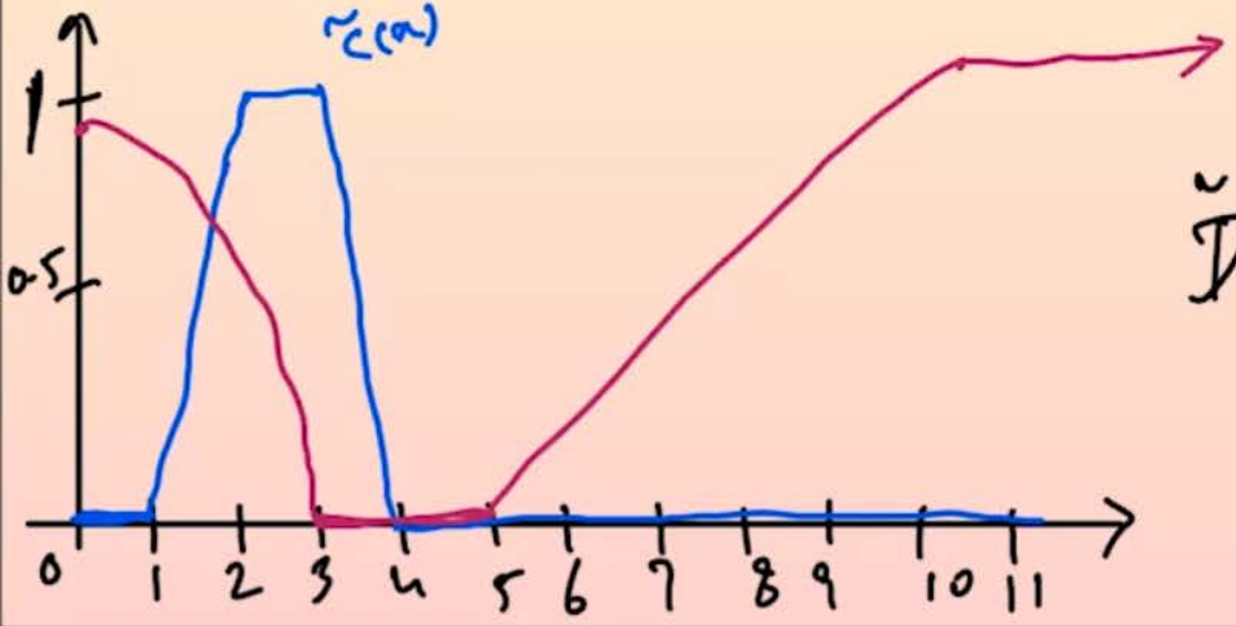
$$A_1(x) \wedge A_2(x) = \begin{cases} 0 & \text{if } 40 \le x < 50 \\ \frac{x-50}{10} & \text{if } 50 \le x < 55 \\ 1 - \frac{x-50}{10} & \text{if } 55 \le x \le 60 \\ 0 & \text{if } 60 < x \le 100 \end{cases}$$



Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where $X =$ non negative real numbers.

Draw the graph of $C \wedge D'$ when

$$\tilde{C}(x) = \begin{cases} 0 & ; \ 0 < x < 1 \\ x-1 & ; \ 1 \le x < 2 \\ 1 & \ \ 2 \le x < 3 \\ 4-x & ; \ 3 \le x < 4 \\ 0 & ; \ 4 \le x \end{cases} \quad ; \quad \tilde{D}(x) = \begin{cases} e^{x-3} & ; \ 0 \le x < 3 \\ 1 & ; \ 3 \le x < 5 \\ 1 - \frac{x-5}{2} & ; \ 5 \le x < 10 \\ 0 & ; \ 10 \le x \end{cases}$$



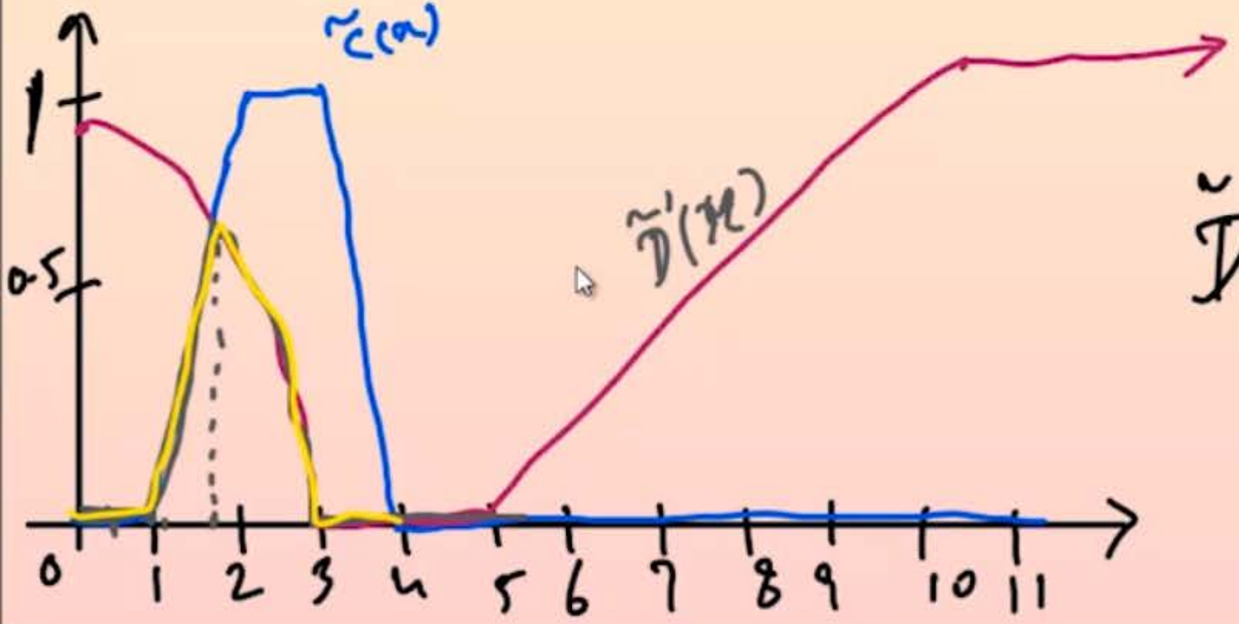
$$\tilde{D}'(x) = \begin{cases} 1 - e^{x-3} & ; \ 0 \le x < 3 \\ 0 & ; \ 3 < x < 5 \\ \frac{x-5}{2} & ; \ 5 \le x < 10 \\ 1 & ; \ 10 \le x \end{cases}$$

Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where X = non negative real numbers.

Draw the graph of $\tilde{C} \wedge \tilde{D}'$ when

$$(\tilde{C} \wedge \tilde{D}')(x) = \min\{\tilde{C}(x), \tilde{D}'(x)\}$$

$$\tilde{C}(x) = \begin{cases} 0 & ; \ 0 < x < 1 \\ x-1 & ; \ 1 \le x < 2 \\ 1 & \ \ 2 \le x < 3 \\ 4-x & ; \ 3 \le x < 4 \\ 0 & ; \ 4 \le x \end{cases}$$



$$\tilde{D}'(x) = \begin{cases} 1 - e^{x-3} & ; \ 0 \le x < 3 \\ 0 & ; \ 3 < x < 5 \\ \frac{x-5}{2} & ; \ 5 \le x < 10 \\ 1 & ; \ 10 \le x \end{cases}$$

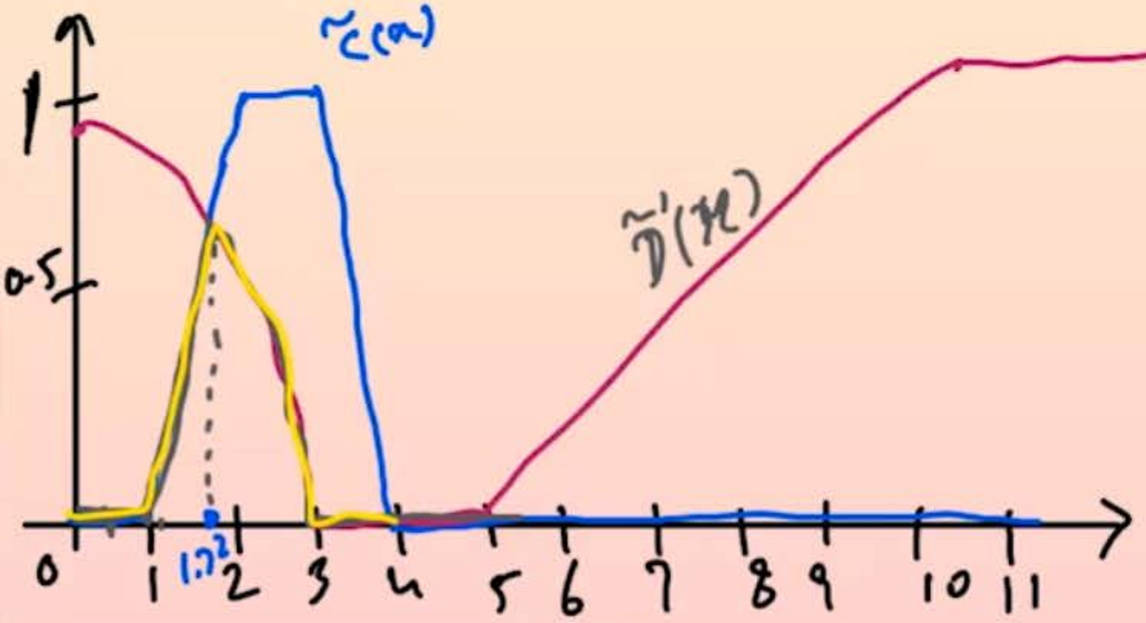
Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where X = non negative real numbers.

Draw the graph of $\tilde{C} \wedge \tilde{D}'$ when

$$\tilde{C}(x) = \begin{cases} 0 & ; \ 0 < x < 1 \\ x-1 & ; \ 1 \le x < 2 \\ 1 & ; \ 2 \le x < 3 \\ 4-x & ; \ 3 \le x < 4 \\ 0 & ; \ 4 \le x \end{cases}$$

$\tilde{C}(x)$

$$(\tilde{C} \wedge \tilde{D}')(x) = \begin{cases} 0 & 0 \le x < 1 \\ x-1 & 1 \le x < 1.72 \\ 1-e^{x-3} & 1.72 \le x < 3 \\ 0 & 3 \le x \end{cases}$$



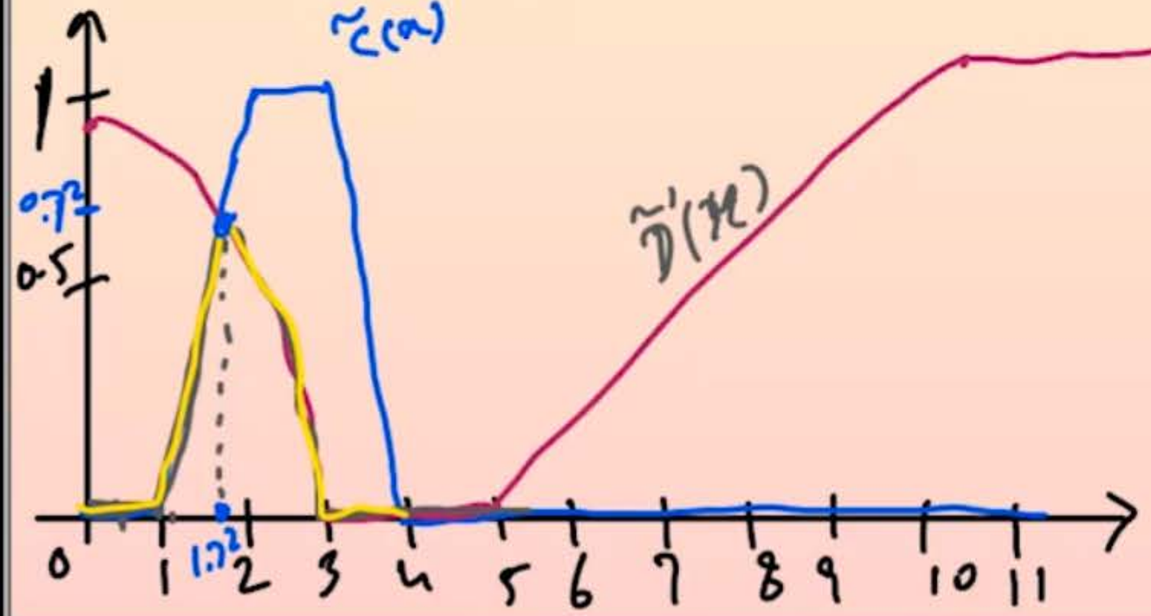
$$\tilde{D}'(x) = \begin{cases} 1-e^{x-3} & ; \ 0 \le x < 3 \\ 0 & ; \ 3 < x < 5 \\ \frac{x-5}{2} & ; \ 5 \le x < 10 \\ 1 & ; \ 10 \le x \end{cases}$$

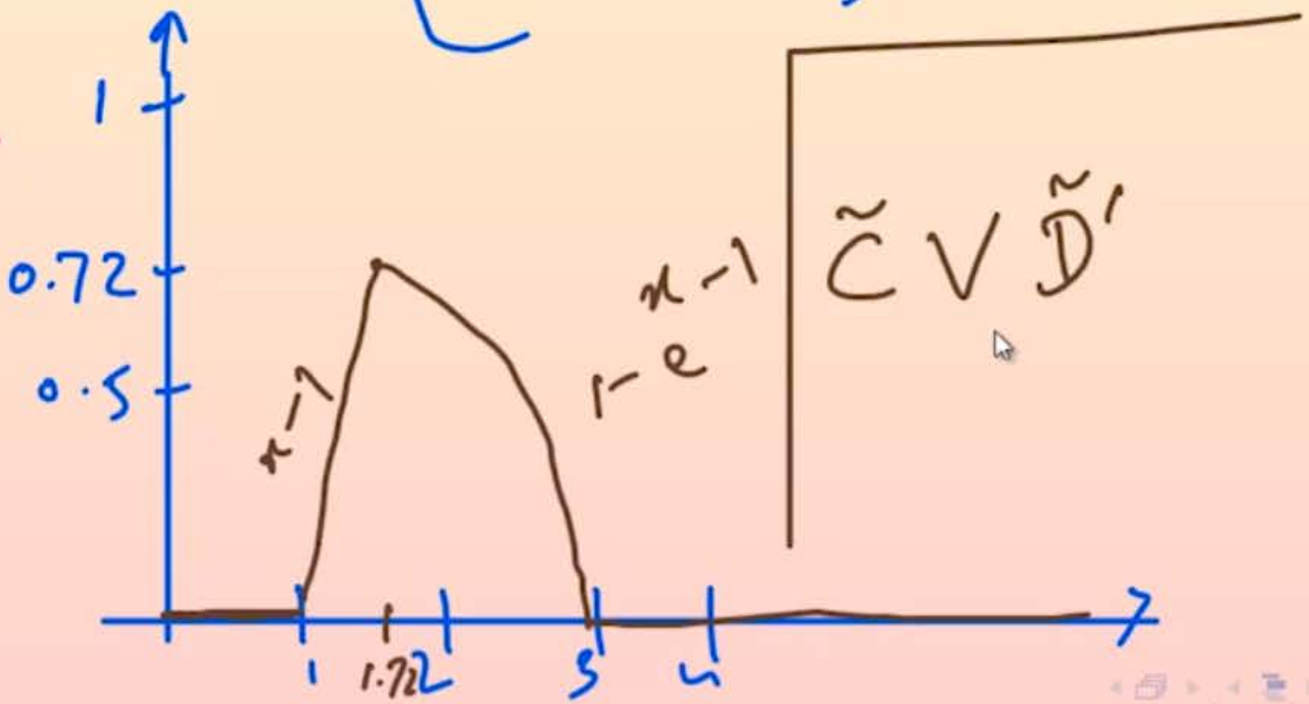
Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where X = non negative real numbers.

Draw the graph of $\tilde{C} \wedge \tilde{D}'$ when

$$\tilde{C}(x) = \begin{cases} 0 & ; \ 0 < x < 1 \\ x-1 & ; \ 1 \le x < 2 \\ 1 & ; \ 2 \le x < 3 \\ 4-x & ; \ 3 \le x < 4 \\ 0 & ; \ 4 \le x \end{cases}$$

$$(\tilde{C} \wedge \tilde{D}')(x) = \begin{cases} 0 & 0 \le x < 1 \\ x-1 & 1 \le x < 1.72 \\ 1-e^{x-3} & 1.72 \le x < 3 \\ 0 & 3 \le x \end{cases}$$





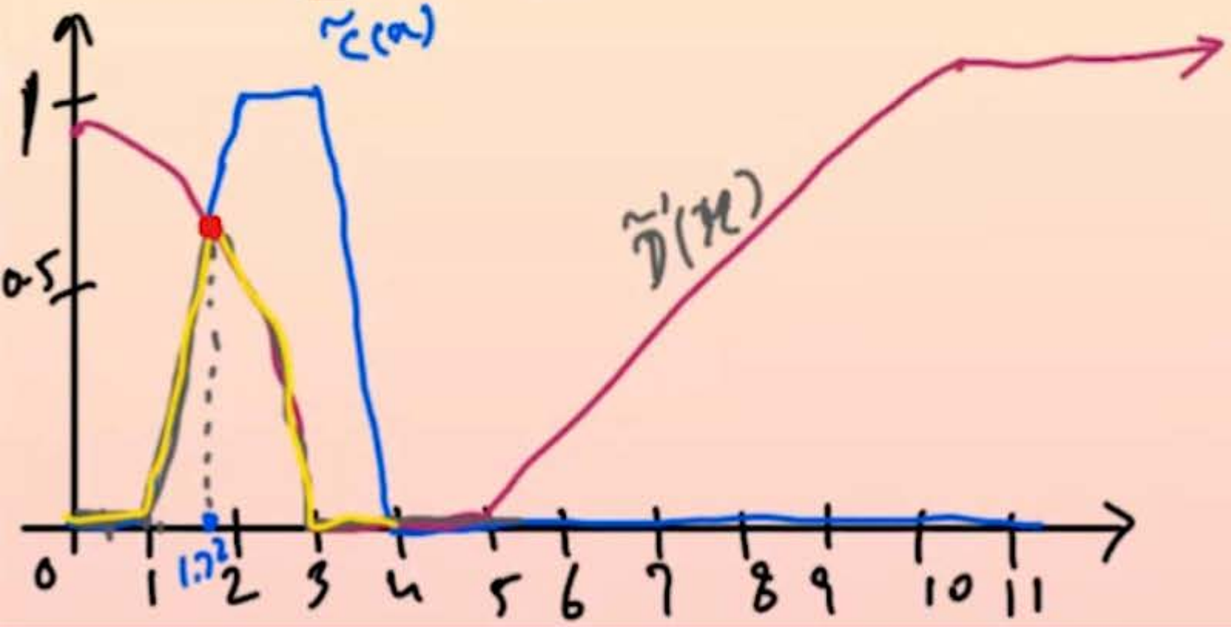
Let $\tilde{C}$ and $\tilde{D}$ be fuzzy sets where X = non negative real numbers.

Draw the graph of $\tilde{C} \wedge \tilde{D}'$ when

$$\tilde{C}(x) = \begin{cases} 0 & ; \ 0 < x < 1 \\ x-1 & ; \ 1 \le x < 2 \\ 1 & ; \ 2 \le x < 3 \\ 4-x & ; \ 3 \le x < 4 \\ 0 & ; \ 4 \le x \end{cases}$$

$\tilde{C}(x)$

$$(\tilde{C} \wedge \tilde{D}')(x) = \begin{cases} 0 & 0 \le x < 1 \\ x-1 & 1 \le x < 1.72 \\ 1-e^{x-3} & 1.72 \le x < 3 \\ 0 & 3 \le x \end{cases}$$



$$x-1 = 1-e^{x-3}$$

$$1.72-1 = 1-e^{1.72-3}$$

$$0.72 = 0.7219\ldots$$

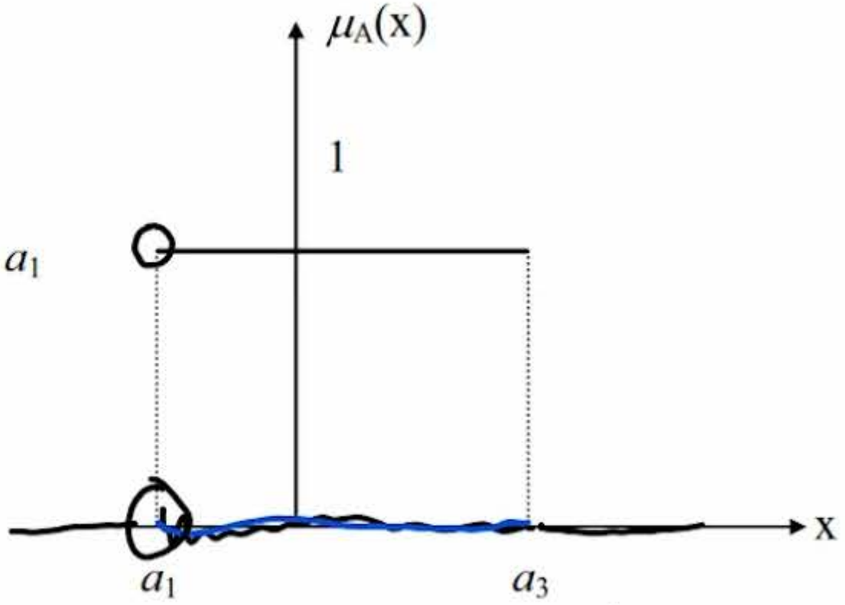
Interval as Fuzzy Set

$[a,b) \subset R$

When interval is defined on real number $\Re$, this interval is said to be a subset of $\Re$. For instance, if interval is denoted as $A = [a_1, a_3]$ $a_1, a_3 \in \Re$, $a_1 < a_3$, we may regard this as one kind of sets. Expressing the interval as membership function is shown in the following :

$$\mu_A(x) = \begin{cases} 0, & x < a_1 \\ 1, & a_1 \le x \le a_3 \\ 0, & x > a_3 \end{cases}$$

If $a_1 = a_3$, this interval indicates a point. That is, $[a_1, a_1] = a_1$



Interval $A = [a_1, a_3]$

Union

Definition

The union of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

$$\mathbb{I}_1 \cup \mathbb{I}_2 = [a_1, b_1] \cup [a_2, b_2] = [\min\{a_1, a_2\}, \max\{b_1, b_2\}]$$

provided that $[a_1, b_1] \cap [a_2, b_2] \neq \phi$. Otherwise, it is undefined (since the result is not an interval).

$I_1 = [-1, 0]$, $I_2 = [2, 10]$

$I_1 \cup I_2$ = undefined



Union

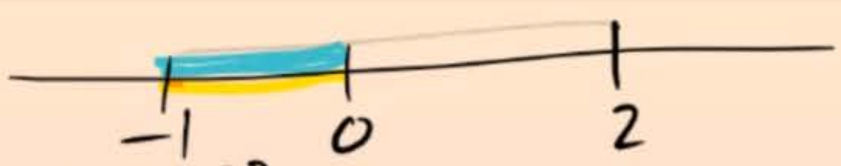
Definition

The union of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

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provided that $[a_1, b_1] \cap [a_2, b_2] \neq \phi$. Otherwise, it is undefined (since the result is not an interval).

$I_1 = [-1, 0]$, $I_2 = [-1, 2]$



$I_1 \cup I_2 = [\min\{-1, -1\}, \max\{0, 2\}]$

$= [-1, 2]$

Union

Definition

The union of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

$$\mathbb{I}_1 \cup \mathbb{I}_2 = [a_1, b_1] \cup [a_2, b_2] = [\min\{a_1, a_2\}, \max\{b_1, b_2\}]$$

provided that $[a_1, b_1] \cap [a_2, b_2] \neq \phi$. Otherwise, it is undefined (since the result is not an interval).

$I_1 = [-1, 2]$, $I_2 = [2, 10]$

$I_1 \cup I_2 = [\min\{-1, 2\}, \max\{2, 10\}]$

$= [-1, 10]$



LECTURE # 07

Intersection

Definition

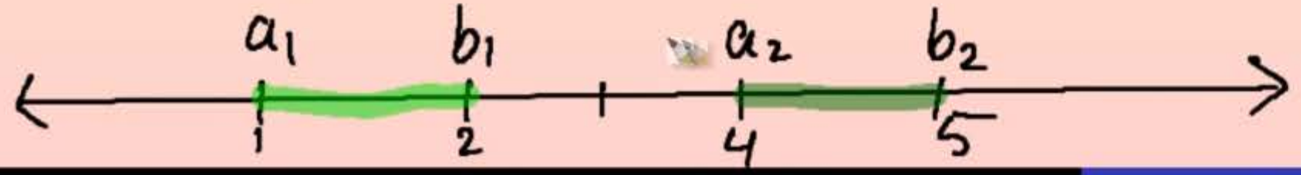
The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [1,2]$; $I_2 = [4,5]$

$$I_1 \cap I_2 = [1,2] \cap [4,5]$$
$$= [\max\{1,4\}, \min\{2,5\}]$$
$$= [4,2] \neq \text{Interval}$$
$$= \phi$$



Intersection

Definition

The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

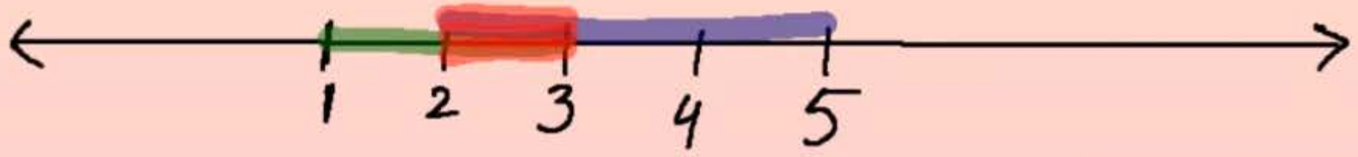
and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [1,3]$; $I_2 = [2,5]$

$I_1 \cap I_2 = [1,3) \cap [2,5]$

$= [\max[1,2), \min(3,5)]$

$= [2,3]$



Intersection

Definition

The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

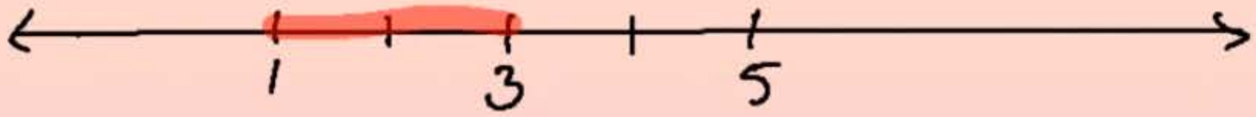
$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [1,3]$; $I_2 = [1,5]$

$I_1 \cap I_2 = [1,3] \cap [1,5]$

$= [1,3]$



Intersection

Definition

The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

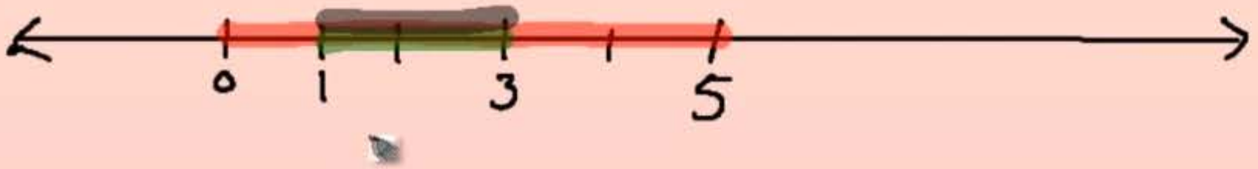
$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [0, 5]$; $I_2 = [1, 3]$

$I_1 \cap I_2 = [0, 5] \cap [1, 3]$

$= [1, 3]$



Intersection

Definition

The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

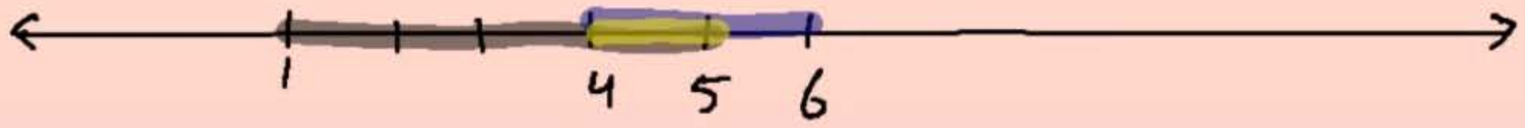
$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [1, 5]$; $I_2 = [4, 6]$

$I_1 \cap I_2 = [1, 5] \cap [4, 6]$

$= [4, 5]$



Intersection

Definition

The intersection of two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ is defined to be

$$\mathbb{I}_1 \cap \mathbb{I}_2 = [a_1, b_1] \cap [a_2, b_2] = [\max\{a_1, a_2\}, \min\{b_1, b_2\}]$$

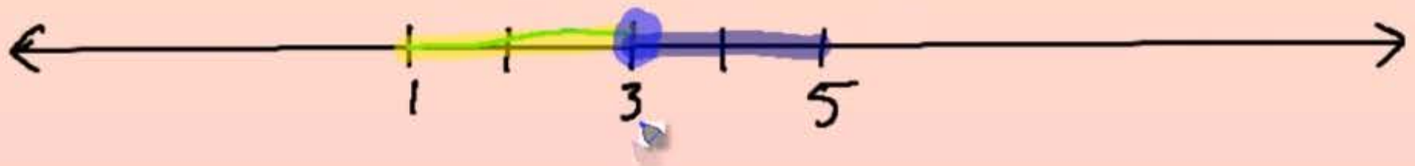
and $[a_1, b_1] \cap [a_2, b_2] = \phi$ if and only if $a_1 > b_2$ or $a_2 > b_1$.

$I_1 = [1,3]$; $I_2 = [3,5]$

$I_1 \cap I_2 = [1,3) \cap [3,5]$

$= [3,3]$

$= 3$



Equality

Definition

Two intervals $\mathbb{I}_1 = [a_1, b_1]$ and $\mathbb{I}_2 = [a_2, b_2]$ are said to be equal, denoted by $[a_1, b_1] = [a_2, b_2]$, if and only if $a_1 = a_2$ and $b_1 = b_2$.

$I_1 = [1, 3]$; $I_2 = [1, 2+1]$

$1 = 1$; $3 = 2+1$



Inequality

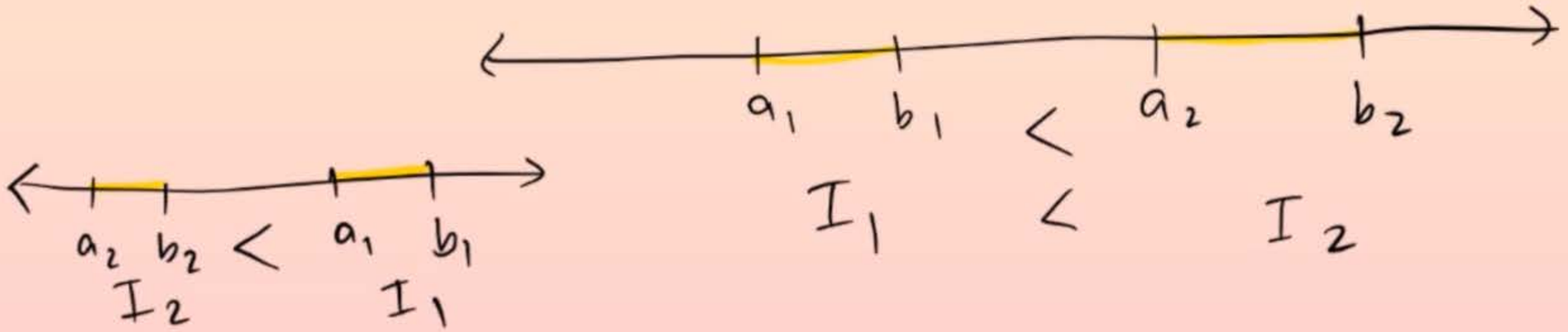
$2 < 3$

$2 > 1$

$I_1 \leqq \geqq I_2$

Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.



Inequality

Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

$I_1 = (1,3]$, $I_2 = (2,5]$



can not be compared

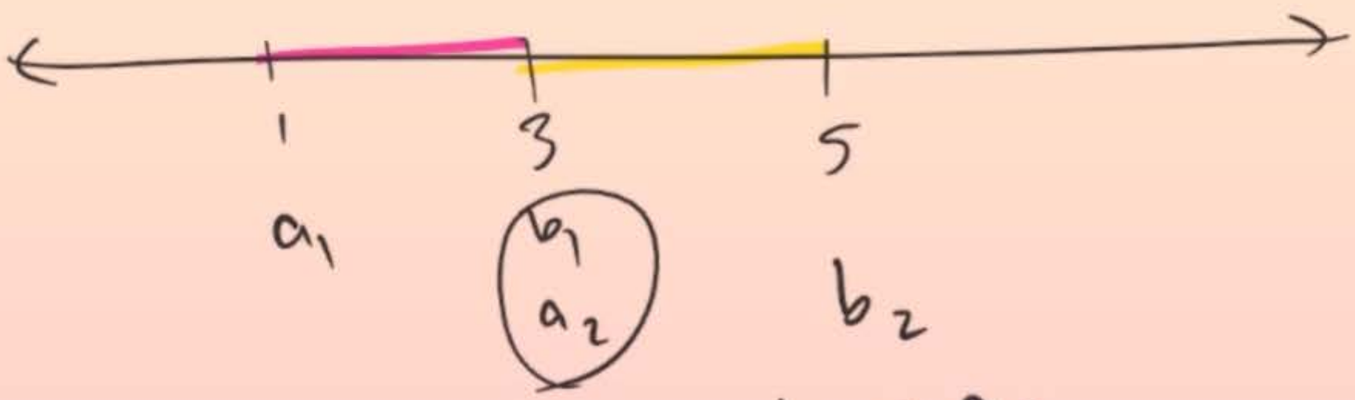
Inequality

Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

$I_1 = [1, 3)$, $I_2 = [3, 5]$

cannot be compared

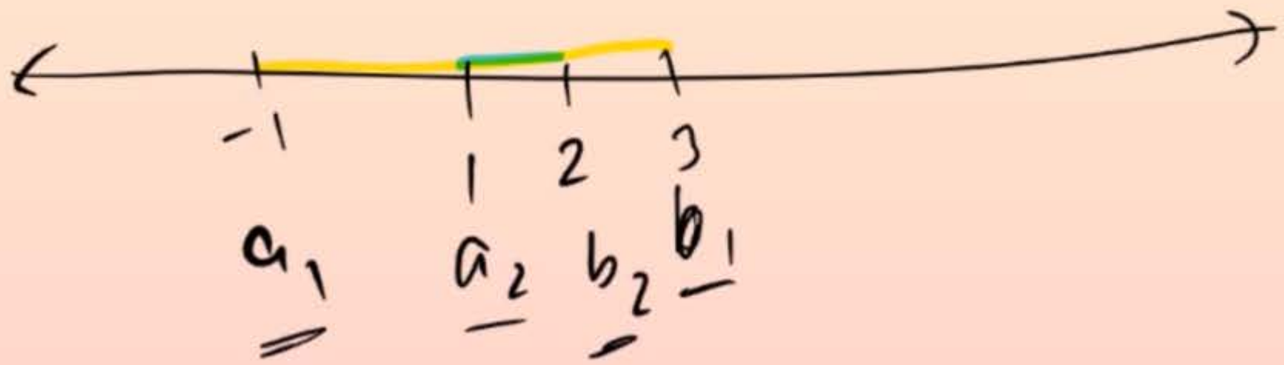


Inequality

Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

$I_1 = [-1, 3]$, $I_2 = [1, 2]$

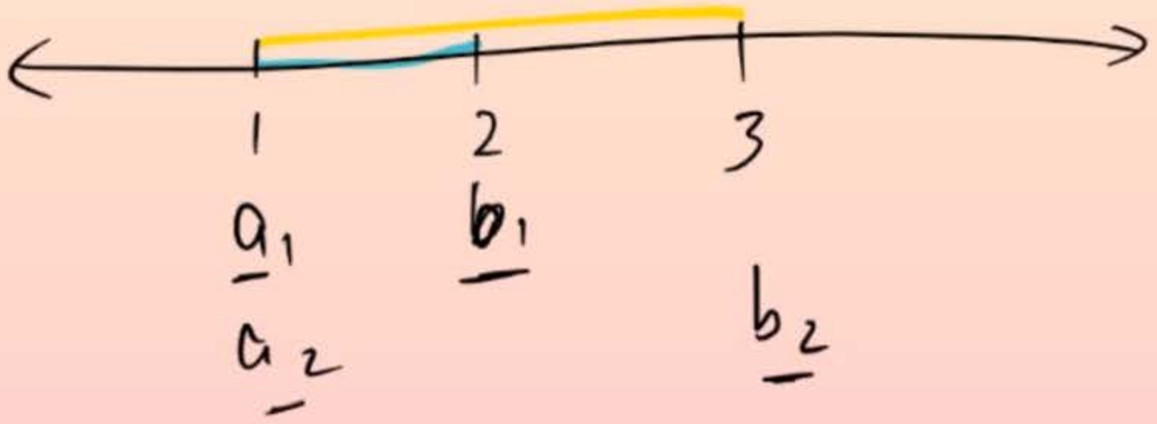


Inequality

Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

$I_1 = [1, 2)$, $I_2 = [1, 3]$

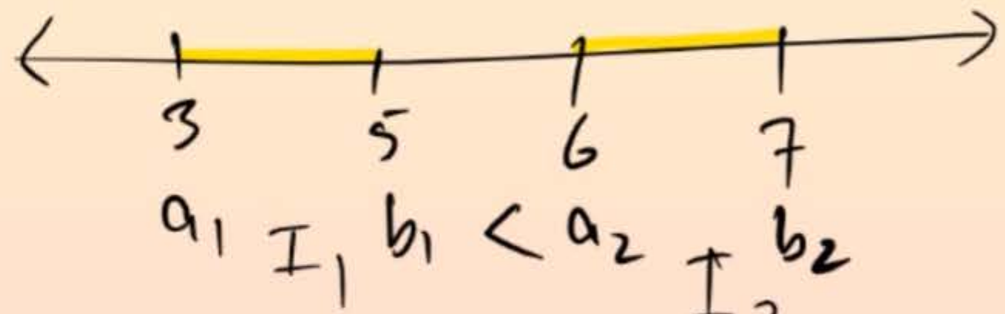


Inequality

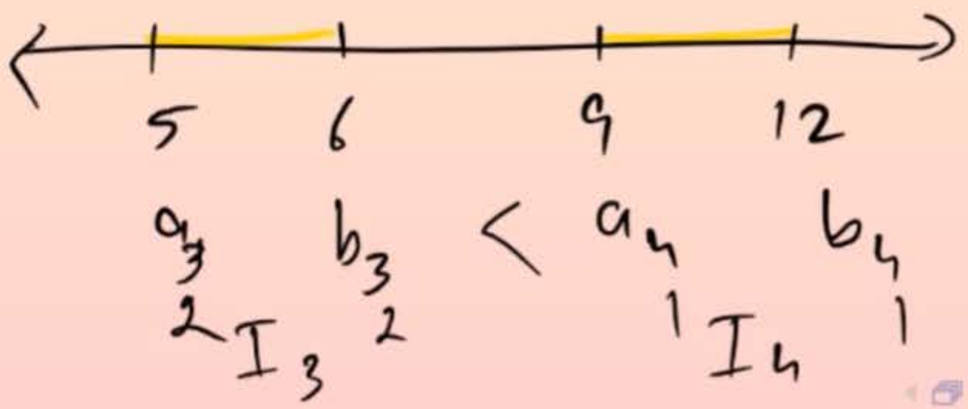
Definition

Interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be less than (resp., greater than) interval $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] < [a_2, b_2]$ (resp., $[a_1, b_1] > [a_2, b_2]$) if and only if $b_1 < a_2$ (resp., $a_1 > b_2$). Otherwise, they cannot be compared. Note that the relations $\leq$ and $\geq$ are not defined for intervals.

$I_1 = [3, 5)$, $I_2 = [6, 7]$



$I_3 = [5, 6]$, $I_4 = [9, 12]$



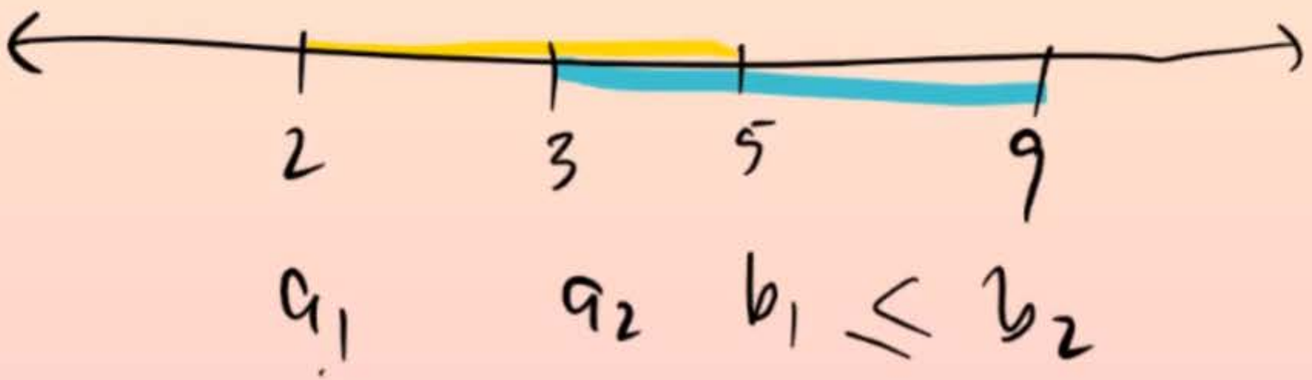
$I_2 < I_1$

Inclusion

Definition

The interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be included in $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] \subseteq [a_2, b_2]$ if and only if $a_2 \leqslant a_1$ and $b_1 \leqslant b_2$. This is equivalent to saying that the interval $\mathbb{I}_1 = [a_1, b_1]$ is a subset or subinterval of $\mathbb{I}_2 = [a_2, b_2]$.

$I_1 = [2, 5]$, $I_2 = [3, 9]$



Inclusion

Definition

The interval $\mathbb{I}_1 = [a_1, b_1]$ is said to be included in $\mathbb{I}_2 = [a_2, b_2]$, denoted by $[a_1, b_1] \subseteq [a_2, b_2]$ if and only if $a_2 \leqslant a_1$ and $b_1 \leqslant b_2$. This is equivalent to saying that the interval $\mathbb{I}_1 = [a_1, b_1]$ is a subset or subinterval of $\mathbb{I}_2 = [a_2, b_2]$.

$I_1 = [3, 5], \quad I_2 = [2, 9]$



$a_2 \leq a_1 \qquad b_1 \leq b_2 \qquad I_1 \subseteq I_2$

Theorem

Theorem

The addition and multiplication operations of intervals are commutative and associative but not distributive. More precisely:

(1) $X + Y = Y + X$

(2) $X \times Y = Y \times X$

(3) $X + (Y + Z) = (X + Y) + Z$

(4) $X \times (Y \times Z) = (X \times Y) \times Z$

(5) $X + \mathbf{0} = \mathbf{0} + X = X$; *where* $\mathbf{0} = [0, 0]$

(6) $X \times \mathbf{1} = \mathbf{1} \times X = X$; *where* $\mathbf{1} = [1, 1]$

(7) $Z(X + Y) \neq ZX + ZY$, *except when*

(a) $Z = [z, z]$ *is a point ; or*
(b) $X = Y = \mathbf{0}$ *; or*
(c) $xy \geq 0$ *for all* $x \in X$ *and* $y \in Y$.
In general, we only have the subdistributive law: $Z(X + Y) \subseteq ZX + ZY$

Proof (1)

$$X + Y = \{x + y \mid x \in X;\ y \in Y\} \qquad x+y=z\in R$$
$$= \{y + x \mid y \in Y;\ x \in X\} \qquad y+x$$
$$= Y + X$$

Proof (2)

$$\begin{aligned} X \times Y &= \{x \times y \mid x \in X ; \ y \in Y\} \\ &= \{y \times x \mid y \in Y ; \ x \in X\} \\ &= Y \times X \end{aligned}$$

$X + (Y + Z) = (X + Y) + Z$

Proof (3)

$$
\begin{aligned}
X + (Y + Z) &= \{x + a \mid x \in X;\ a \in (Y + Z)\} \\
&= \{x + (y + z) \mid x \in X;\ y \in Y;\ z \in Z\} \\
&= \{(x + y) + z \mid x \in X;\ y \in Y;\ z \in Z\} \\
&= \{(b + z \mid b \in (X + Y);\ z \in Z\} \\
&= (X + Y) + Z
\end{aligned}
$$

$a \in Y+Z$

$y+z \mid y \in Y, z \in Z$

Proof (4)

$$
\begin{aligned}
X \times (Y \times Z) &= \{x \times a \mid x \in X;\ a \in (Y \times Z)\} \\
&= \{x \times (y \times z) \mid x \in X;\ y \in Y;\ z \in Z\} \\
&= \{(x \times y) \times z \mid x \in X;\ y \in Y;\ z \in Z\} \\
&= \{(b \times z \mid b \in (X \times Y);\ z \in Z\} \\
&= (X \times Y) \times Z
\end{aligned}
$$

$X + \mathbf{0} = \mathbf{0} + X = X$; where $\mathbf{0} = [0, 0]$

Proof (5)

$$
\begin{aligned}
X + \mathbf{0} &= \{x + 0 \mid x \in X ; \ 0 \in \mathbf{0}\} \\
&= \{0 + x \mid 0 \in \mathbf{0} ; \ x \in X\} \qquad 0 + x = x \\
&= \{x \mid x \in X\} \\
&= X
\end{aligned}
$$

$$X + 0 = 0 + X = X$$

$X \times \mathbf{0} = \mathbf{0} \times X = \mathbf{0}$; where $\mathbf{0} = [0, 0]$

Proof (5)

$$
\begin{aligned}
X \times \mathbf{0} &= \{x \times 0 \mid x \in X;\ 0 \in \mathbf{0}\} \\
&= \{0 \times x \mid 0 \in \mathbf{0};\ x \in X\} \qquad 0 \times x = 0 \\
&= \{0 \mid 0 \in \mathbf{0}\} \\
&= \mathbf{0}
\end{aligned}
$$

$x \cdot 0 = 0 \cdot x = 0$

$X \times \mathbf{1} = \mathbf{1} \times X = X$; where $\mathbf{1} = [1, 1]$

Proof (6)

$$
\begin{aligned}
X \times \mathbf{1} &= \{x \times 1 \mid x \in X ; 1 \in \mathbf{1}\} \\
&= \{1 \times x \mid 1 \in \mathbf{1} ; \ x \in X\} \qquad / \ x\,x = x \\
&= \{x \mid x \in X\} \\
&= X
\end{aligned}
$$

$$X \cdot 1 = 1 \cdot X = X$$

$Z(X+Y) \neq ZX+ZY$

Proof (7 (a))

$$
\begin{aligned}
z(X+Y) &= \{z \cdot a \mid a \in (X+Y)\} \\
&= \{z \cdot (x+y) \mid x \in X;\ y \in Y\} \\
&= \{z \cdot x + z \cdot y) \mid x \in X;\ y \in Y\} \\
&= zX + zY
\end{aligned}
$$

$Z(X+Y) \neq ZX+ZY$

Proof (7 (b))

$$\begin{aligned} z(0+0) &= Z\mathbf{0} \\ &= \mathbf{0} \\ &= Z\mathbf{0} + Z\mathbf{0} \end{aligned}$$

LECTURE # 08

Convex Set

Definition (Convex set) The term convex is applicable to a set A in R^n (n-dimensional Euclidian vector space) if the followings are satisfied.

i) Two arbitrary points s and r are defined in A.

$$r = (r_i \mid i \in N_n), \quad s = (s_i \mid i \in N_n).$$

(N is a set of positive integers)

ii) For arbitrary real number λ between 0 and 1, point t is involved in A where t is

$$t = (\lambda r_i + (1-\lambda)s_i \mid i \in N_n)$$

$\lambda r + (1-\lambda) s \in A$

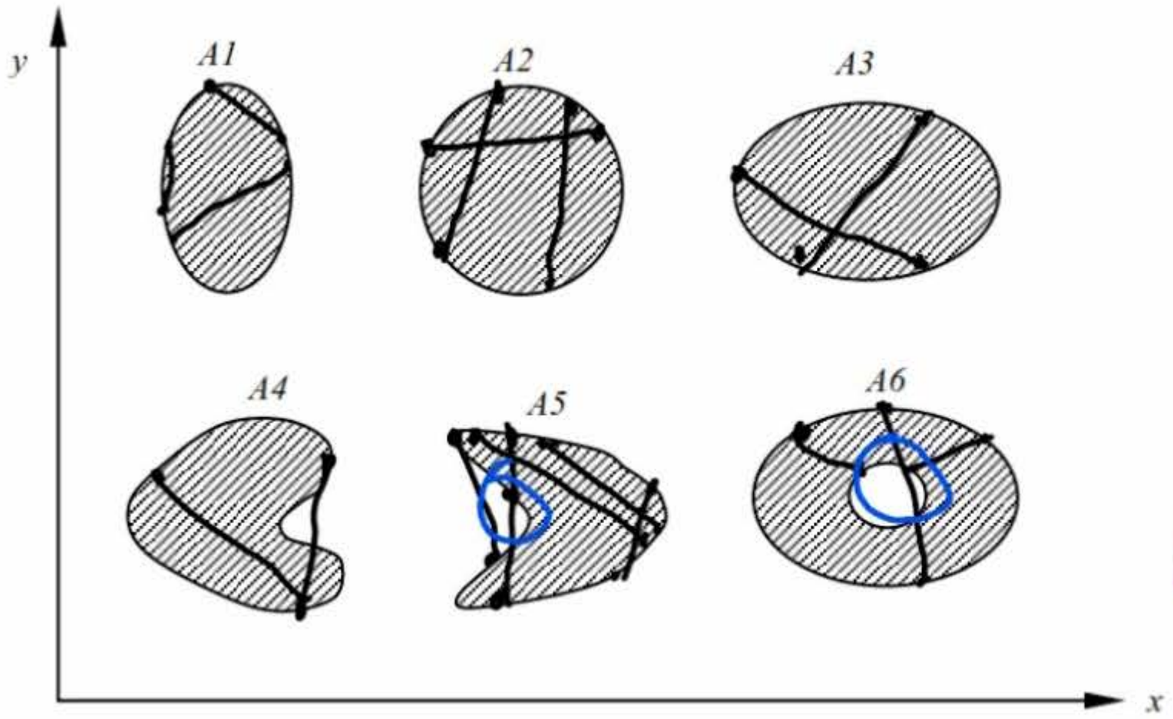
$s, r \in A$

$\lambda \in [0,1]$

$t \in A$

Convex Set

$\lambda x + (1-\lambda) y \in A$



A_1, A_2, A_3 Convex sets

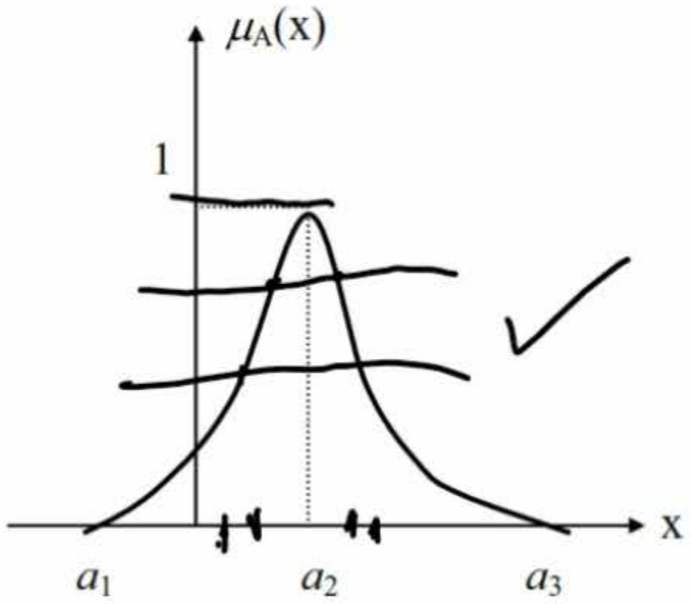
A_4, A_5, A_6

Convex Fuzzy Set

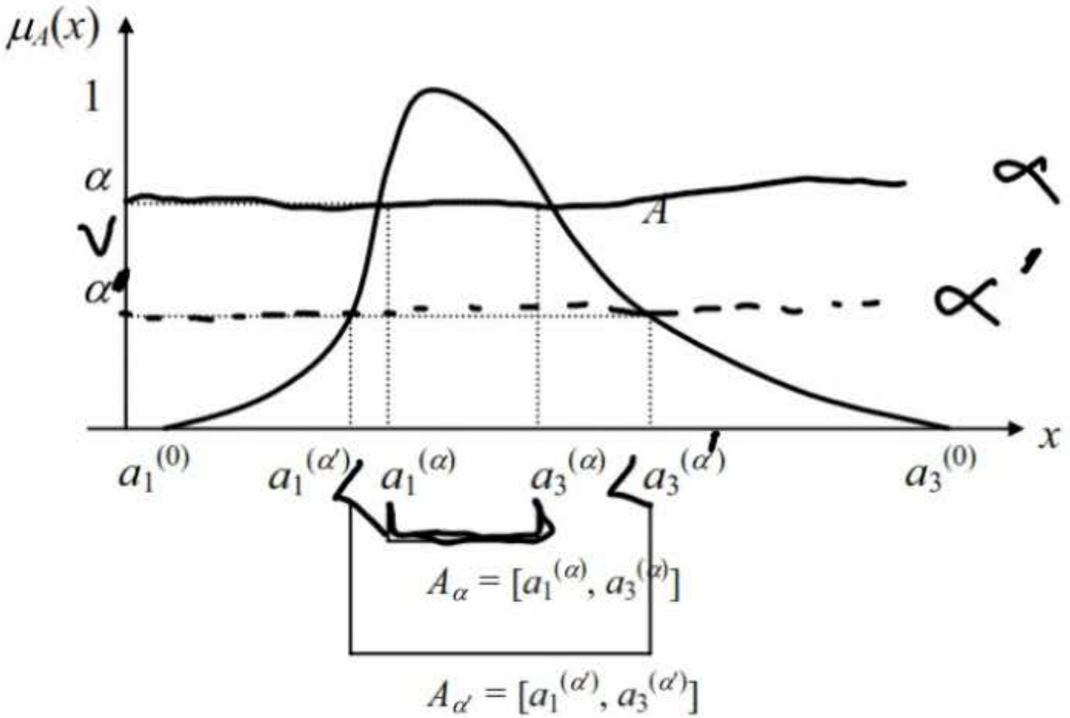
The convex condition is that the line by α-cut is continuous and α-cut interval satisfies the following relation.

$$A_\alpha = [a_1^{(\alpha)}, a_3^{(\alpha)}]$$

$$(\alpha' < \alpha) \Rightarrow (a_1^{(\alpha')} \leq a_1^{(\alpha)}, a_3^{(\alpha')} \geq a_3^{(\alpha)})$$



Fuzzy Number $A = [a_1, a_2, a_3]$



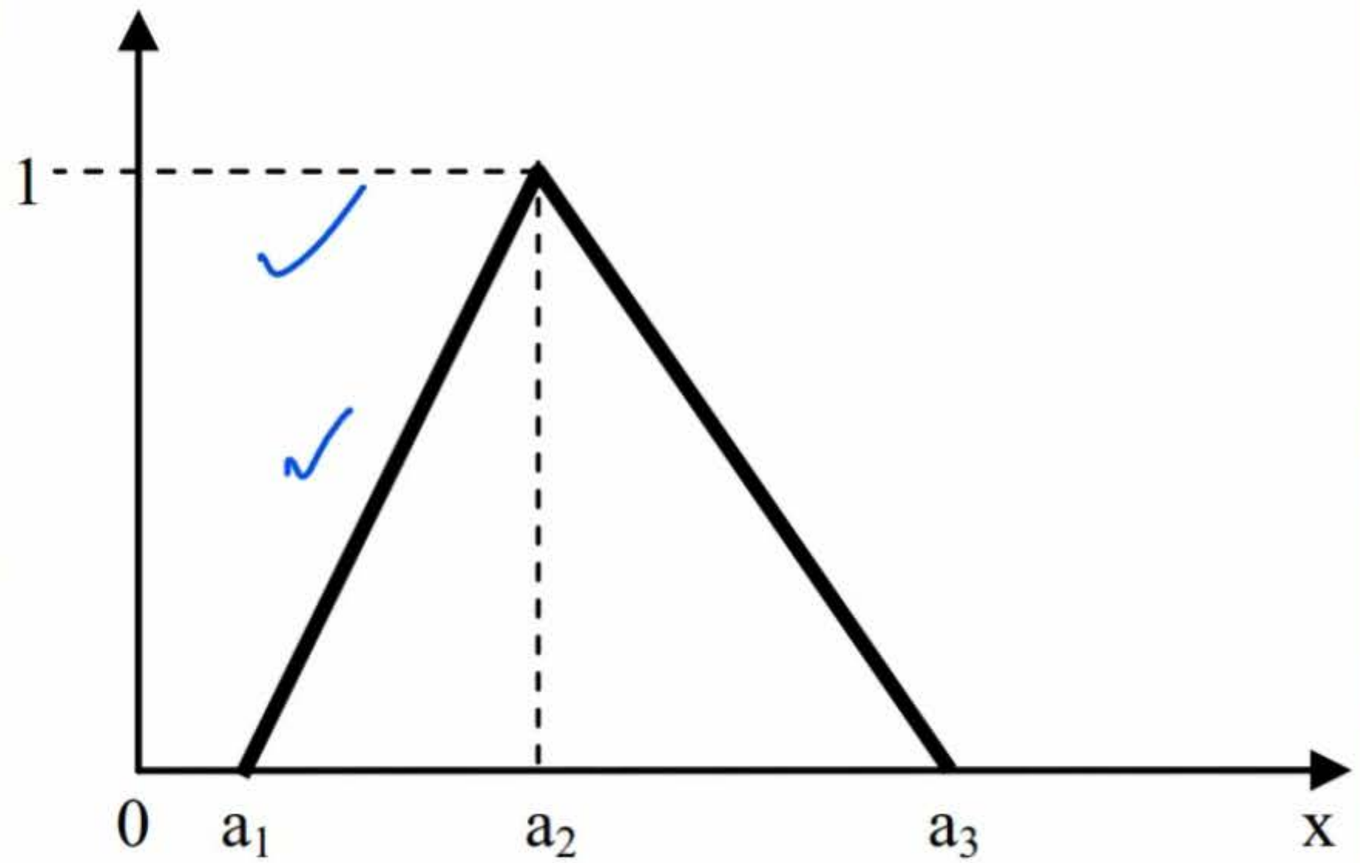
α-cut of fuzzy number $(\alpha' < \alpha) \Rightarrow (A_\alpha \subset A_{\alpha'})$

The convex condition may also be written as, $(\alpha' < \alpha) \Rightarrow (A_\alpha \subset A_{\alpha'})$

Triangular Fuzzy Numbers

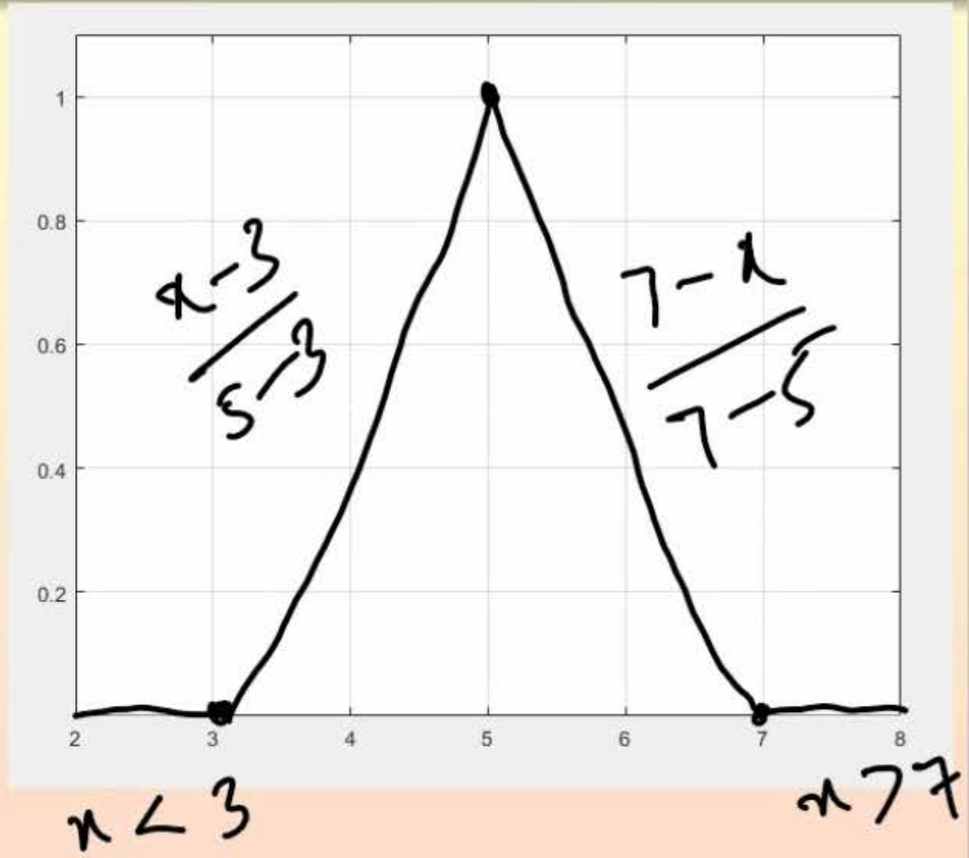
Definition A fuzzy number represented with three points is called triangular fuzzy number. It defined as follows: $\tilde{A} = (a_1, a_2, a_3)$.

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \frac{x-a_1}{a_2-a_1} & a_1 \leq x \leq a_2 \\ \frac{a_3-x}{a_3-a_2} & a_2 \leq x \leq a_3 \\ 0 & x > a_3, \end{cases}$$



Triangular Fuzzy Numbers

$$\mu_{\hat{A}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{5-3} & 3 \le x \le 5 \\ \frac{7-x}{7-5} & 5 \le x \le 7 \\ 0 & x > 7. \end{cases}$$



$A = (3, 5, 7)$

$\frac{3-3}{5-3} = 0$

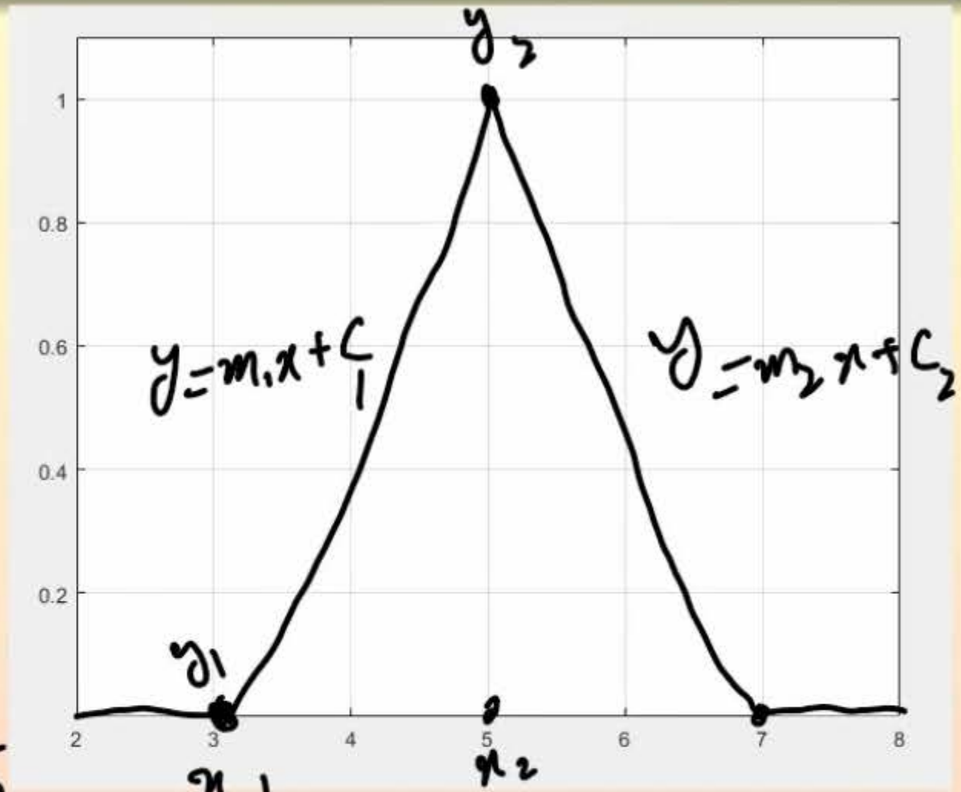
$\frac{5-3}{5-3} = 1$

$\frac{7-5}{7-5} = 1$

$\frac{7-7}{7-5} = 0$

Triangular Fuzzy Numbers

$$\mu_{\hat{A}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{5-3} & 3 \le x \le 5 \\ \frac{7-x}{7-5} & 5 \le x \le 7 \\ 0 & x > 7. \end{cases}$$



$\hat{A} = (3, 5, 7)$

$y = m_1 x + c$

$m_1 = \frac{1-0}{5-3} = \frac{1}{2}$

$m_2 = \frac{0-1}{7-5} = -\frac{1}{2}$

$y = \frac{1}{2} x + c_1$

$0 = \frac{3}{2} + c_1$

$y = \frac{1}{2} x - \frac{3}{2} = \frac{x-3}{2}$

$= \frac{x-3}{5-3}$

Triangular Fuzzy Numbers

$$\mu_{\hat{A}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{5-3} & 3 \le x \le 5 \\ \frac{7-x}{7-5} & 5 \le x \le 7 \\ 0 & x > 7. \end{cases}$$

$A = (3, 5, 7)$

$y = m_1 x + C$

$m_1 = \frac{1-0}{5-3} = \frac{1}{2}$

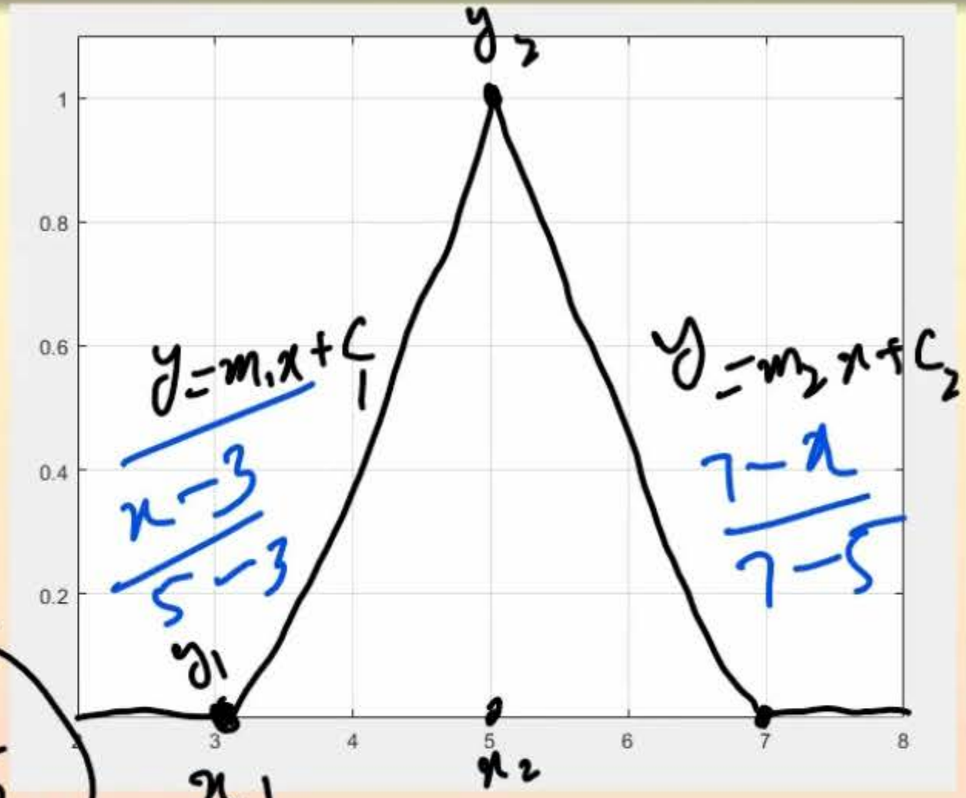
$m_2 = \frac{0-1}{7-5} = \frac{-1}{2}$

$y = \frac{-1}{2}x + C_2$

$0 = \frac{-1}{2}(7) + C_2$

$y = \frac{-1}{2}x + \frac{7}{2}$

$= \frac{7-x}{2} = \frac{7-x}{7-5}$



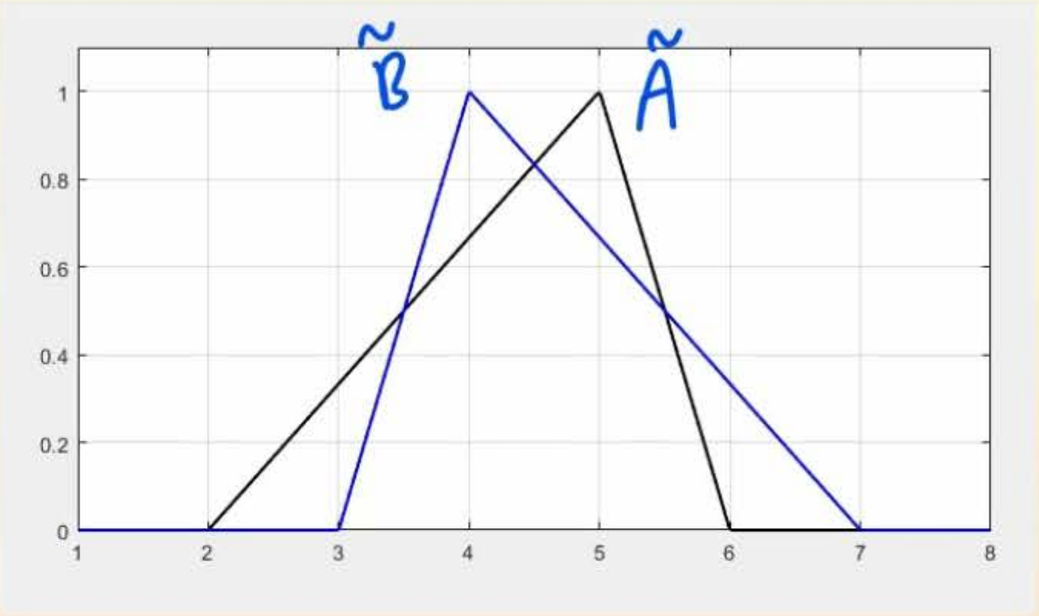
LECTURE # 09

Triangular Fuzzy Numbers

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \frac{x-2}{5-2} & 2 \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{4-3} & 3 \le x \le 4 \\ \frac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$



Union

Union

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The union of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cup \tilde{B}$.

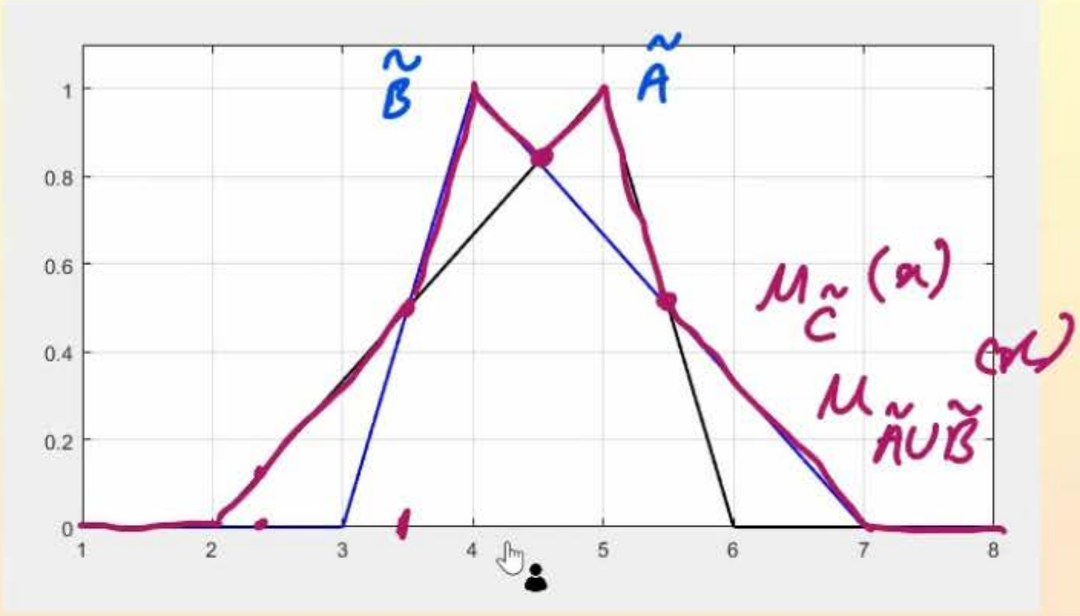
Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \dfrac{x-2}{5-2} & 2 \le x \le 5 \\ \dfrac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \dfrac{x-3}{4-3} & 3 \le x \le 4 \\ \dfrac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$



$0 \qquad x < 2$

$$\frac{x-2}{5-2} = \frac{x-3}{4-3} \quad \Big| \quad x-2 = 3x-9$$

$$3x - x - 9 + 2 = 0$$

$$2x - 7 = 0$$

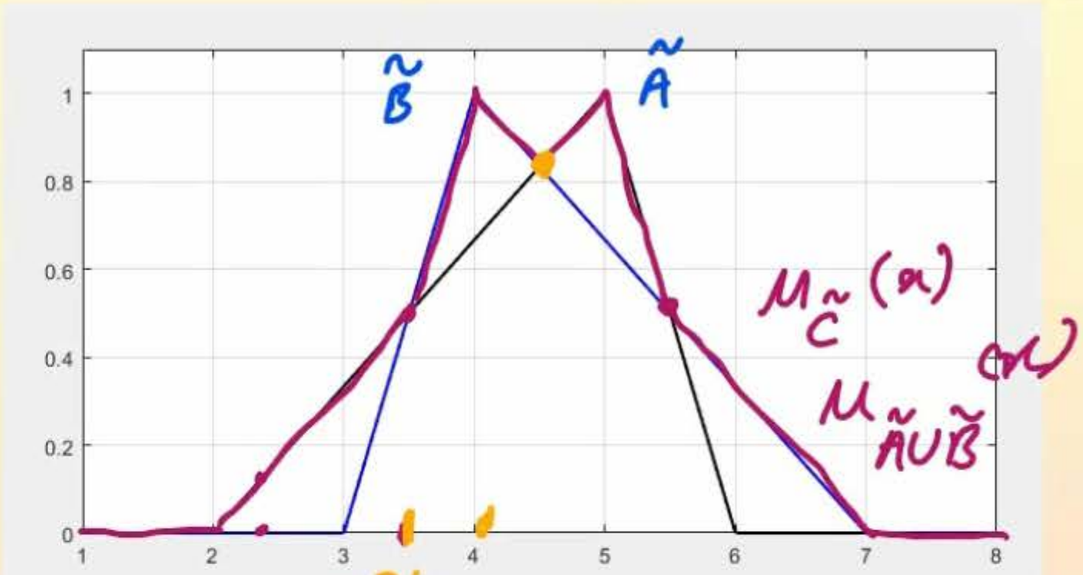
$$x = 7/2$$

Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \dfrac{x-2}{5-2} & 2 \le x \le 5 \\ \dfrac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$



$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \dfrac{x-3}{4-3} & 3 \le x \le 4 \\ \dfrac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$

$$\frac{x-2}{3} = \frac{7-x}{3}$$

$$x - 2 = 7 - x$$

$$2x = 9$$

$$x = \frac{9}{2}$$

$7/2$

$$\begin{cases} 0 & x < 2 \\ \dfrac{x-2}{5-2} & 2 \le x \le 7/2 \\ \dfrac{x-3}{4-3} & 7/2 \le x \le 4 \\ \dfrac{7-x}{7-4} & 4 \le x \le \end{cases}$$

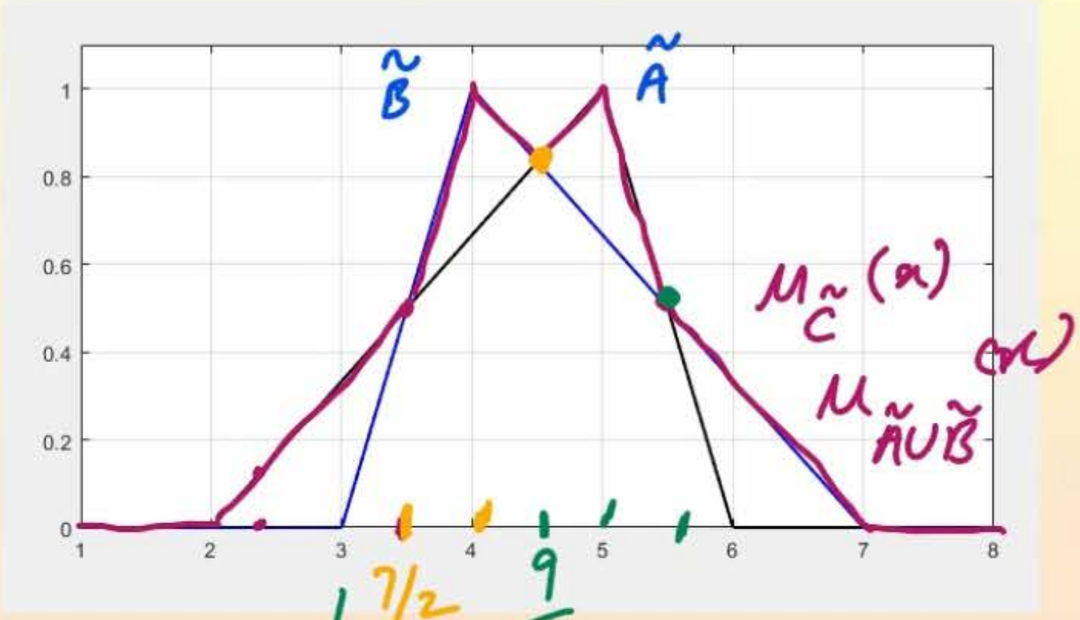
Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \dfrac{x-2}{5-2} & 2 \le x \le 5 \\ \dfrac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \dfrac{x-3}{4-3} & 3 \le x \le 4 \\ \dfrac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$



$$\begin{cases} \dfrac{x-2}{5-2} & \dfrac{9}{2} \le x \le 5 \\ \dfrac{6-x}{6-5} & 5 \le x \le \end{cases}$$

$$\begin{cases} 0 & x < 2 \\ \dfrac{x-2}{5-2} & 2 \le x \le 7/2 \\ \dfrac{x-3}{4-3} & 7/2 \le x \le 4 \\ \dfrac{7-x}{7-4} & 4 \le x \le \dfrac{9}{2} \end{cases}$$

Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \frac{x-2}{5-2} & 2 \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{4-3} & 3 \le x \le 4 \\ \frac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$

$$\frac{6-x}{6-5} = \frac{7-x}{7-4}$$

$$\frac{6-x}{1} = \frac{7-x}{3}$$

$$18 - 3x = 7 - x$$

$$3x - 18 + 7 - x = 0$$

$$2x - 11 = 0$$

$$x = 11/2$$

Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \max\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \vee \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \frac{x-2}{5-2} & 2 \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{4-3} & 3 \le x \le 4 \\ \frac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$

$\mu_{\tilde{A} \cup \tilde{B}}(x)$

$$= \begin{cases} 0 & x < 2 \\ \frac{x-2}{5-2} & 2 \le x \le 7/2 \\ \frac{x-3}{4-3} & 7/2 \le x \le 4 \\ \frac{7-x}{7-4} & 4 \le x \le \frac{9}{2} \\ \frac{x-2}{5-2} & \frac{9}{2} \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le \frac{11}{2} \\ \frac{7-x}{7-4} & 11/2 \le x \le 7 \\ 0 & x > 7 \end{cases}$$

Intersection

Intersection

Let $\tilde{F}(X)$ denote the collection of fuzzy sets on a given universe of discourse X.

Let $\tilde{A}, \tilde{B} \in \tilde{F}(X)$. The intersection of $\tilde{A}$ and $\tilde{B}$ is the fuzzy set $\tilde{C}$ with

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

We denote $\tilde{C} = \tilde{A} \cap \tilde{B}$.

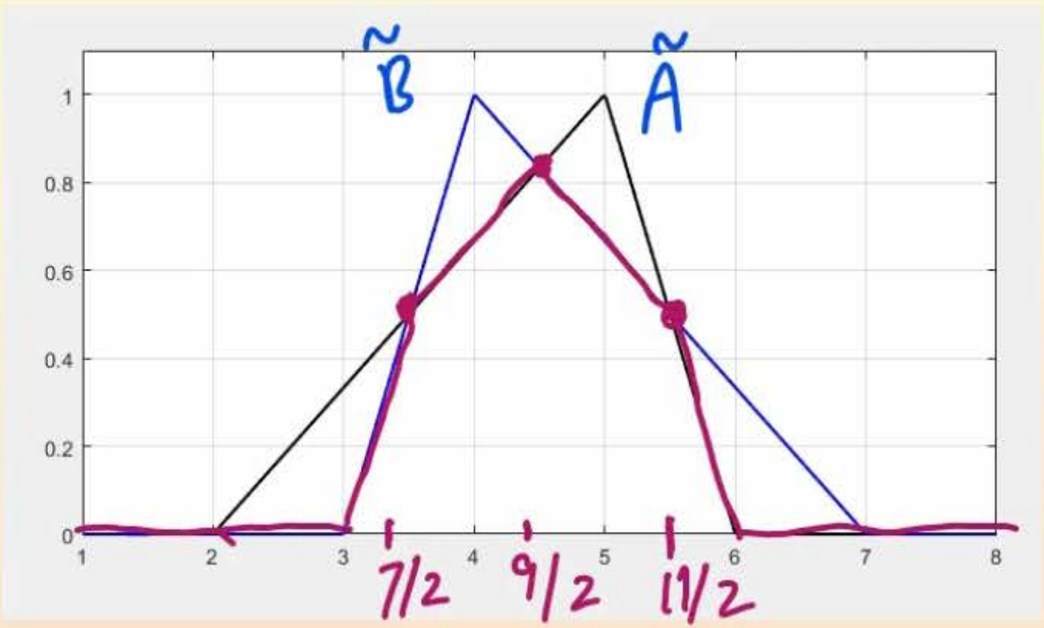
Triangular Fuzzy Numbers

$$\mu_{\tilde{C}}(x) = \min\{\mu_{\tilde{A}}(x), \mu_{\tilde{B}}(x)\} = \mu_{\tilde{A}}(x) \wedge \mu_{\tilde{B}}(x), \quad \forall x \in X.$$

$\tilde{A} = (2,5,6)$, $\tilde{B} = (3,4,7)$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 2 \\ \frac{x-2}{5-2} & 2 \le x \le 5 \\ \frac{6-x}{6-5} & 5 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

$$\mu_{\tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{4-3} & 3 \le x \le 4 \\ \frac{7-x}{7-4} & 4 \le x \le 7 \\ 0 & x > 7 \end{cases}$$



$$\mu_{\tilde{A} \cap \tilde{B}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{4-3} & 3 \le x \le 7/2 \\ \frac{x-2}{5-2} & 7/2 \le x \le 9/2 \\ \frac{7-x}{7-4} & 9/2 \le x \le 11/2 \\ \frac{6-x}{6-5} & 11/2 \le x \le 6 \\ 0 & x > 6 \end{cases}$$

Addition

Definition

Let $\tilde{A} = (a_1, b_1, c_1)$ and $\tilde{B} = (a_2, b_2, c_2)$ be two triangular fuzzy numbers, then their addition is defined as

$$\tilde{A} + \tilde{B} = (a_1, b_1, c_1) + (a_2, b_2, c_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2)$$

$\tilde{A} = (2,3,5)$; $\tilde{B} = (4,7,8)$

$\tilde{A} + \tilde{B} = (2,3,5) + (4,7,8)$

$= (2+4, 3+7, 5+8)$

$= (6, 10, 13)$

EDITOR | PUBLISH | VIEW

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Go To | Find — NAVIGATE
Insert | Comment | Indent — EDIT
Breakpoints — BREAKPOINTS
Run | Run and Advance | Run Section | Advance | Run and Time — RUN

```
1 - x=linspace(0,14,1000);
2 - A=trimf(x,[2 3 5]);
3 - B=trimf(x,[4 7 8]);
4 - TriSum=fuzarith(x,A,B,'sum');
5
6 - plot(x,A,'b--','LineWidth',2)
7 - hold on
8 - plot(x,B,'m:','LineWidth',2)
9 - plot(x,TriSum,'r','LineWidth',2)
10
11 - ylim([-0.05 1.2])
12   ylabel('Membership Value')
```

script | Ln 12 | Col 27

EDITOR | PUBLISH | VIEW

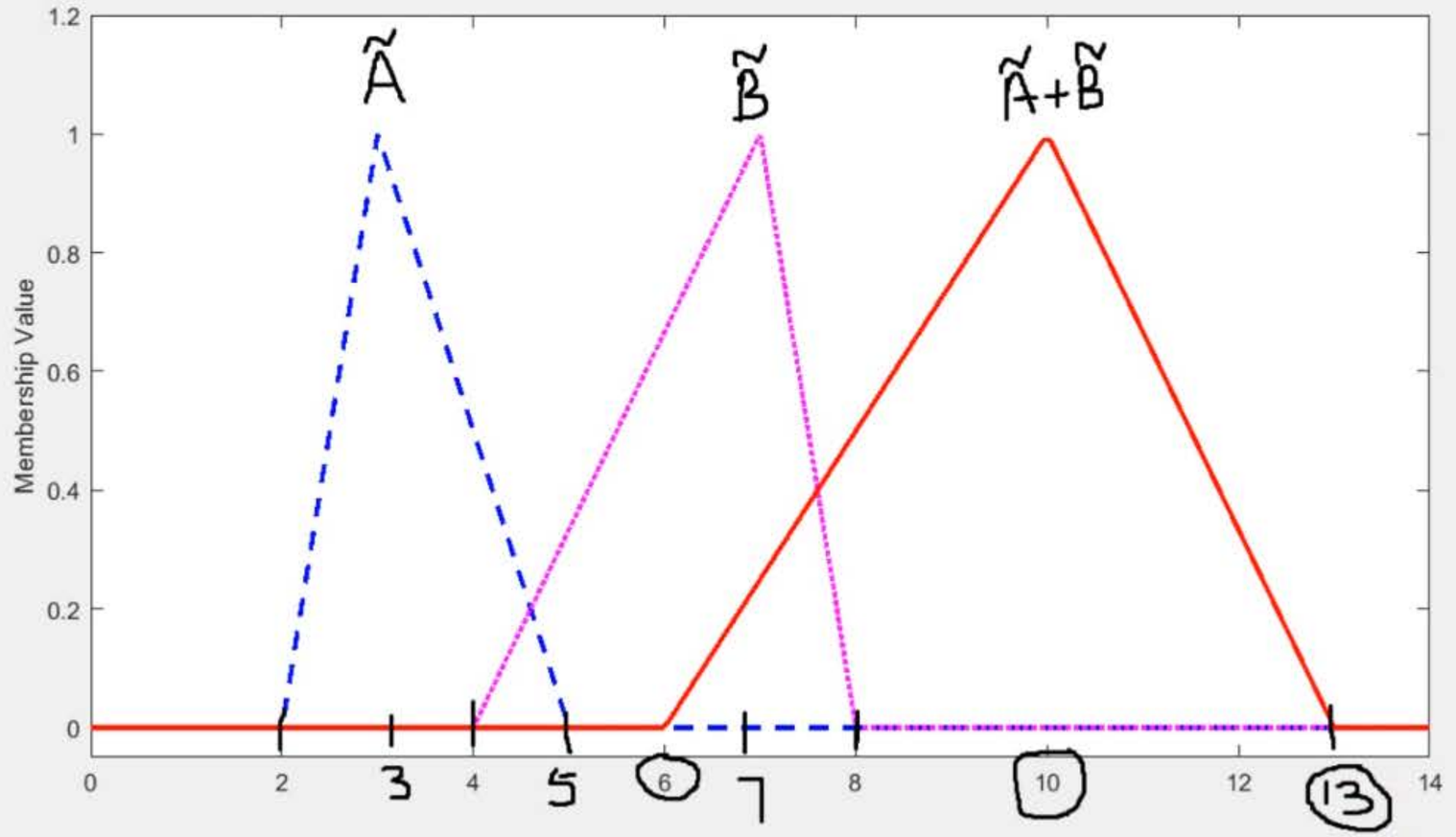
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FILE

```
1 -  x=linsp
2 -  A=trimf
3 -  B=trimf
4 -  TriSum=
5
6 -  plot(x,
7 -  hold on
8 -  plot(x,
9 -  plot(x,
10
11 -  ylim([-
12 -  ylabel(
```

Figure 1

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script Ln 12 Col 27

Negative of a triangular fuzzy number

Definition

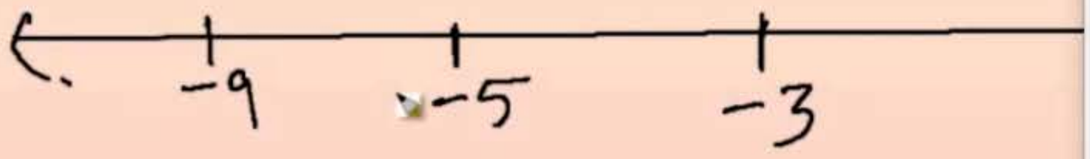
Let $\tilde{A} = (a, b, c)$ be a triangular fuzzy number, then $-\tilde{A}$ is defined as

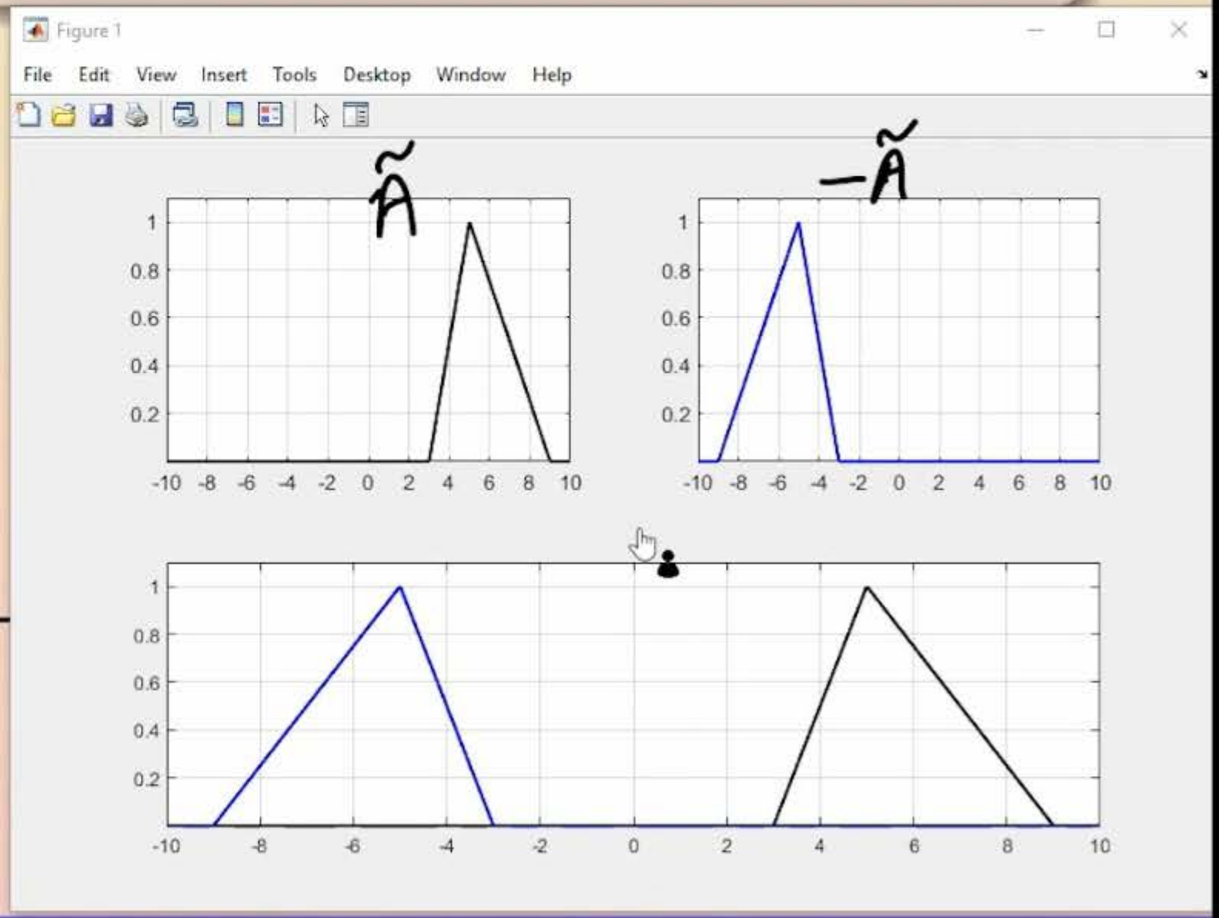
$$-\tilde{A} = (-c, -b, -a)$$

$\tilde{A} = (3, 5, 9)$ $\quad a \leq b \leq c$

$-\tilde{A} = (-9, -5, -3)$ ✓ $\quad -9 \leq -5 \leq -3$

$-\tilde{A} = (-3, -5, -9)$ $\quad -3 \not\leq -5 \not\leq -9$





LECTURE # 10

Subtraction

Definition

Let $\tilde{A} = (a_1, b_1, c_1)$ and $\tilde{B} = (a_2, b_2, c_2)$ be two triangular fuzzy numbers, then their subtraction is defined as

$$\tilde{A} - \tilde{B} = (a_1, b_1, c_1) - (a_2, b_2, c_2) = (a_1 - c_2, b_1 - b_2, c_1 - a_2)$$

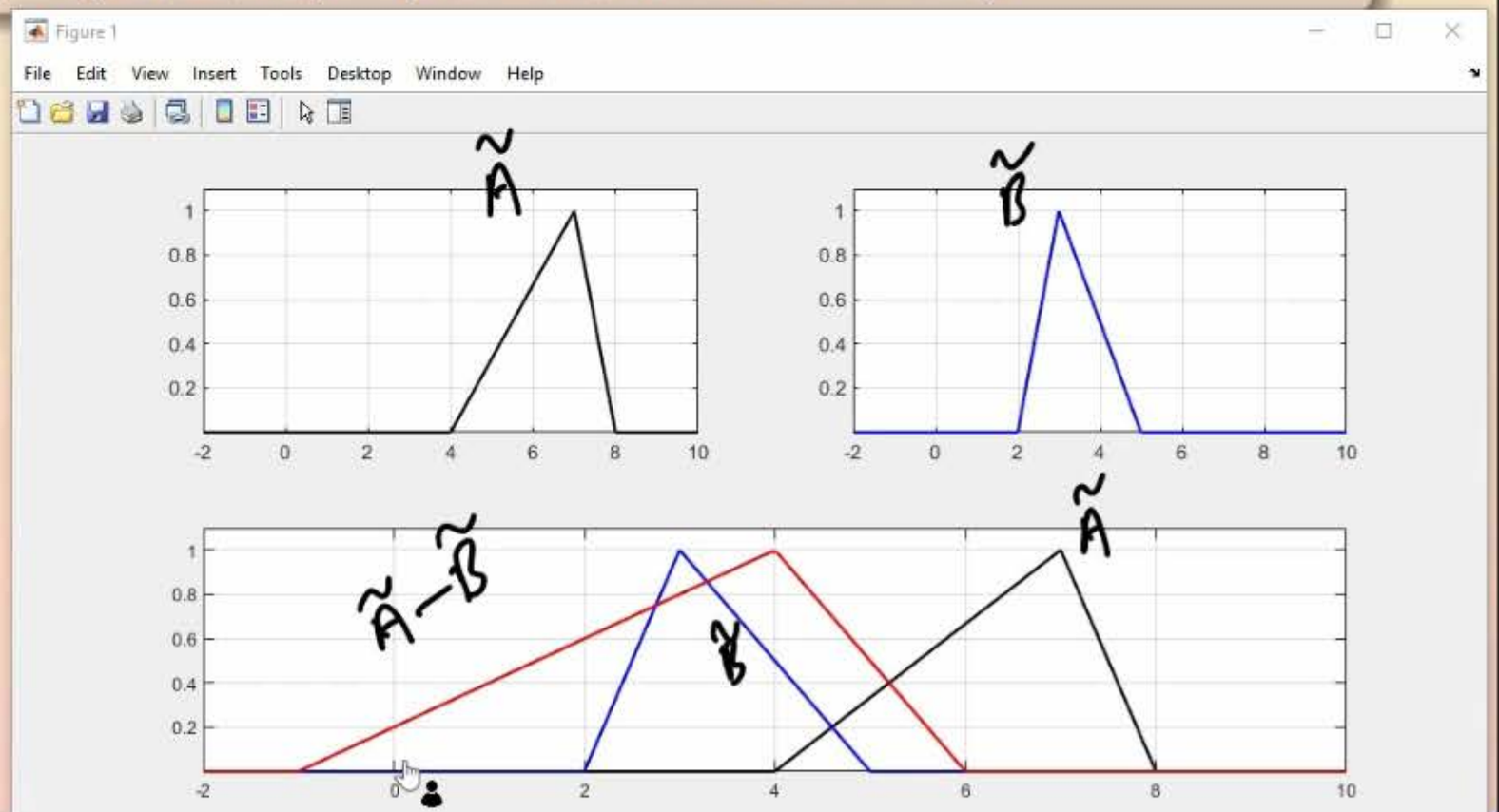
$\tilde{A} = (4, 7, 8)$; $\tilde{B} = (2, 3, 5)$

$\tilde{A} - \tilde{B} = (4, 7, 8) - (2, 3, 5)$

$\tilde{A} + (-\tilde{B}) = (4, 7, 8) + (-5, -3, -2)$

$= (4-5, 7-3, 8-2)$

$= (-1, 4, 6)$



Multiplication

Definition

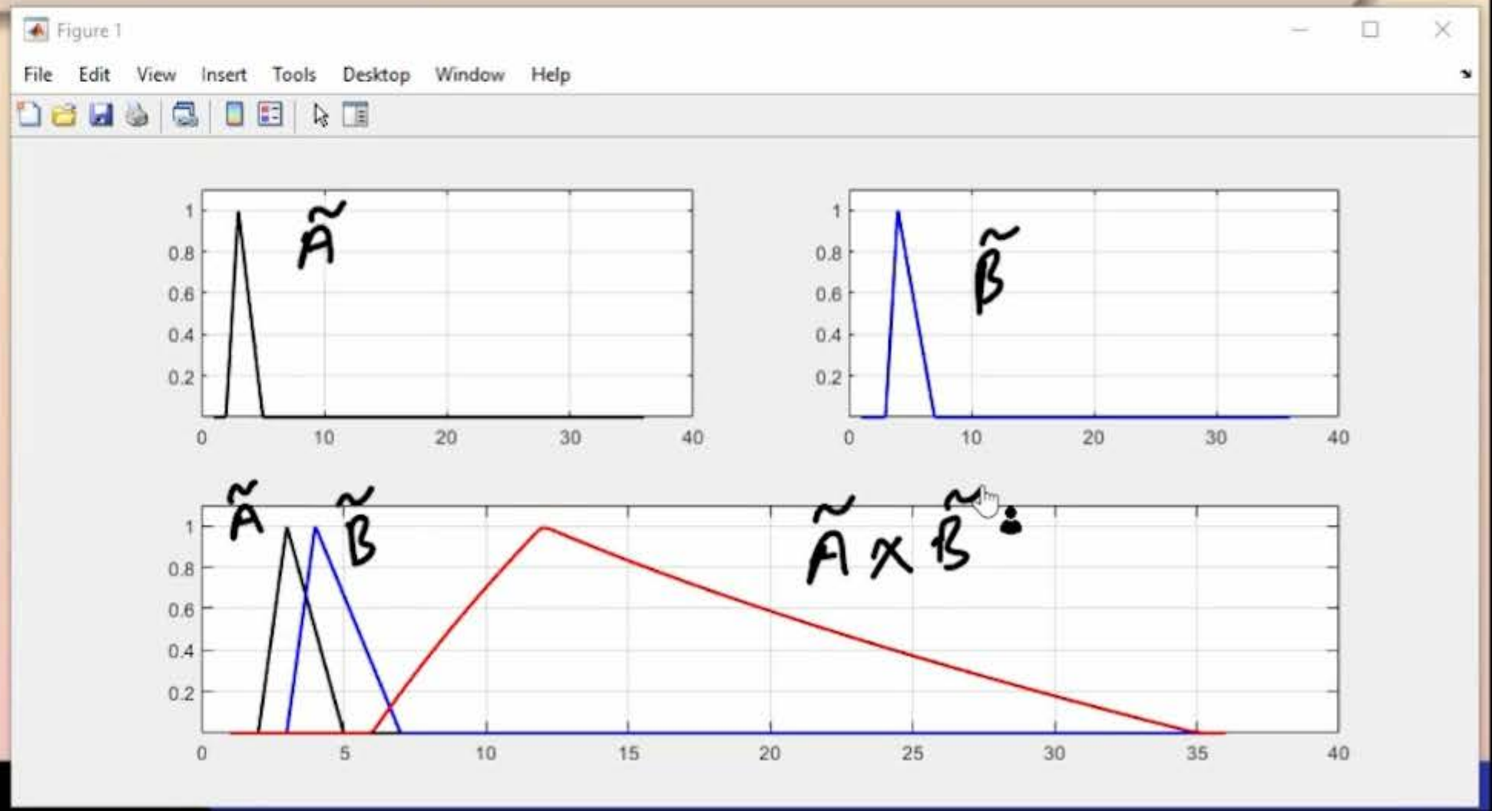
Let $\tilde{A} = (a_1, b_1, c_1)$ and $\tilde{B} = (a_2, b_2, c_2)$ be two triangular fuzzy numbers, then their multiplication is defined as

$$\begin{aligned}\tilde{A} \times \tilde{B} &= (a_1, b_1, c_1) \times (a_2, b_2, c_2) \\ &= (a_1 a_2, b_1 b_2, c_1 c_2)\end{aligned}$$

$\tilde{A} = (2, 3, 5)$; $\tilde{B} = (3, 4, 7)$

$\tilde{A} \times \tilde{B} = (2, 3, 5) \times (3, 4, 7)$

$= (6, 12, 35)$

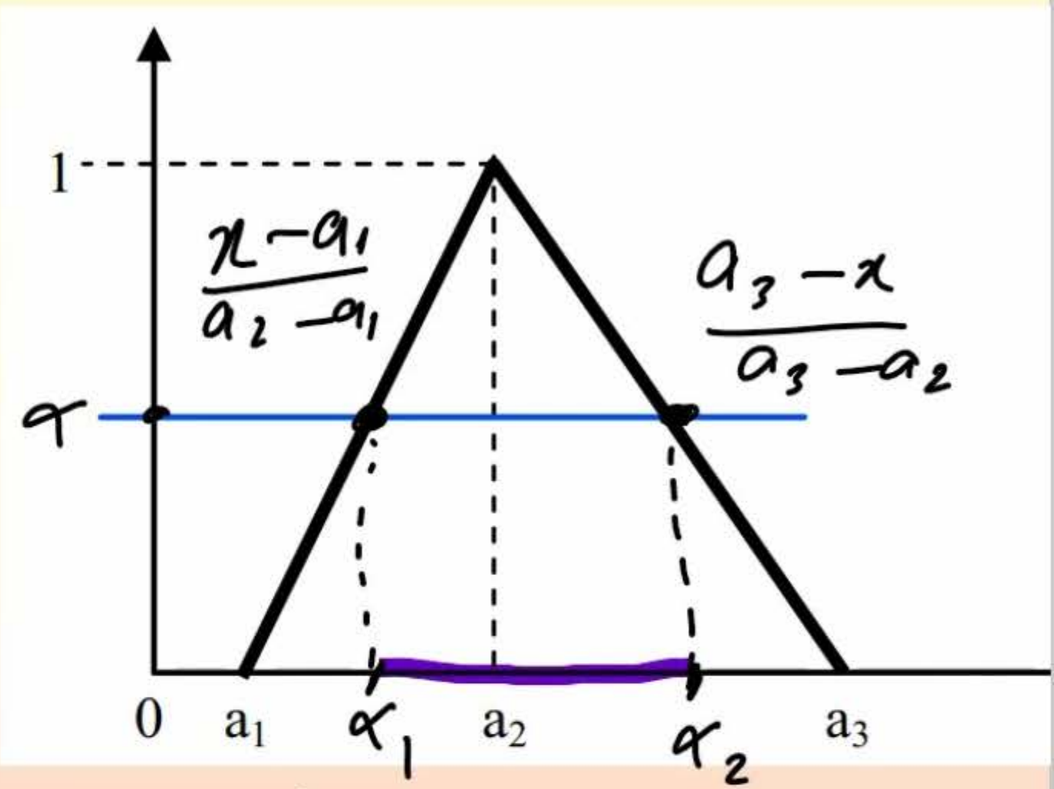


α cut of triangular fuzzy numbers

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \frac{x-a_1}{a_2-a_1} & a_1 \le x \le a_2 \\ \frac{a_3-x}{a_3-a_2} & a_2 \le x \le a_3 \\ 0 & x > a_3, \end{cases}$$

$\alpha \in [0,1]$

$A_\alpha = [\alpha_1, \alpha_2]$



$$\alpha = \frac{x - a_1}{a_2 - a_1}$$

$$\alpha(a_2 - a) = x - a_1$$

$$\boxed{a_1 + \alpha(a_2 - a_1) = x} = \alpha_1$$

$$\alpha = \frac{a_3 - x}{a_3 - a_2}$$

$$\alpha(a_3 - a_2) = a_3 - x$$

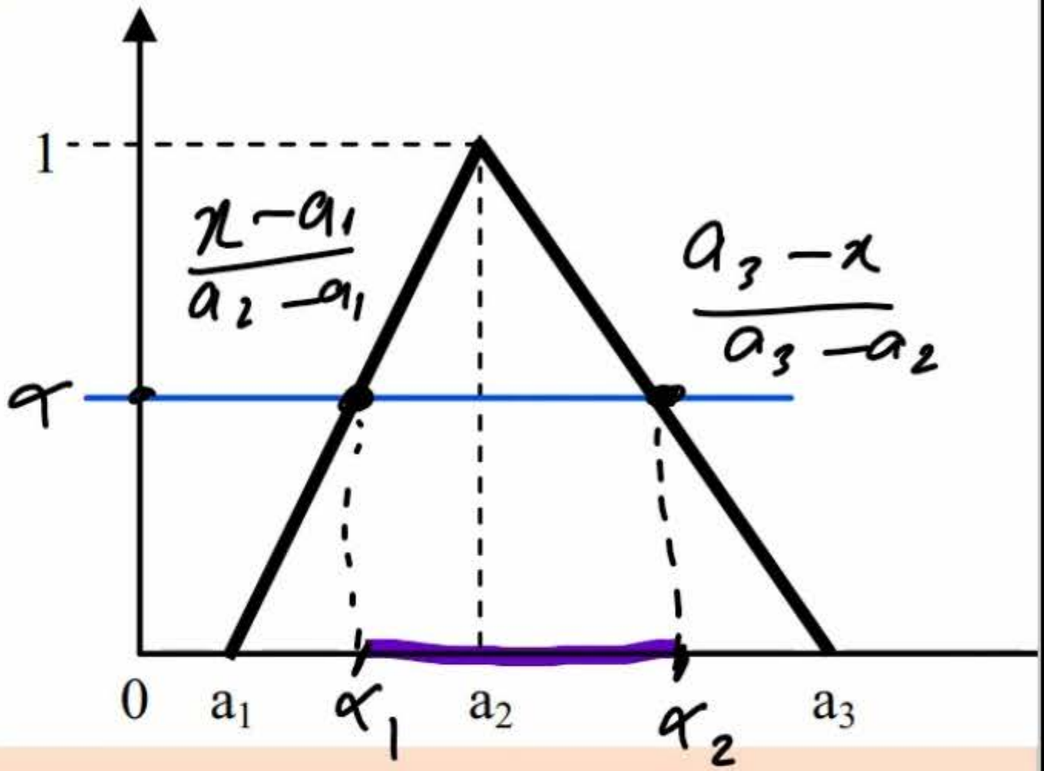
$$\alpha_2 = \boxed{x = a_3 - \alpha(a_3 - a_2)}$$

α cut of triangular fuzzy numbers

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \frac{x-a_1}{a_2-a_1} & a_1 \le x \le a_2 \\ \frac{a_3-x}{a_3-a_2} & a_2 \le x \le a_3 \\ 0 & x > a_3, \end{cases}$$

$\alpha \in [0,1]$

$A_\alpha = [\alpha_1, \alpha_2]$



$A_\alpha = [\alpha_1, \alpha_2]$

$= [a_1 + \alpha(a_2 - a_1),\ a_3 - \alpha(a_3 - a_2)]$

$a_1 + \alpha(a_2 - a_1) = x \Rightarrow \alpha_1$

$\alpha_2 = x = a_3 - \alpha(a_3 - a_2)$

α cut of triangular fuzzy numbers

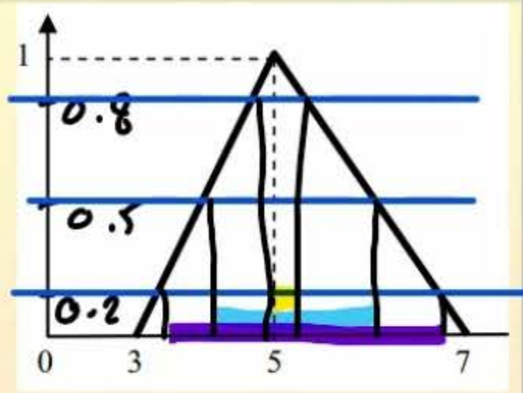
$$A_\alpha = [\alpha_1, \alpha_2]$$

$$= [a_1 + \alpha(a_2 - a_1),\ a_3 - \alpha(a_3 - a_2)]$$

$$\alpha = 0.2,\ 0.5,\ 0.8$$

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{5-3} & 3 \le x \le 5 \\ \frac{7-x}{7-5} & 5 \le x \le 7 \\ 0 & x > 7. \end{cases}$$

$$\tilde{A} = (3, 5, 7)$$



$$A_{0.2} = [3 + 0.2(5-3),\ 7 - 0.2(7-5)] = [3.4, 6.6]$$

$$A_{0.5} = [3 + 0.5(5-3),\ 7 - 0.5(7-5)] = [4, 6]$$

$$A_{0.8} = [3 + 0.8(5-3),\ 7 - 0.8(7-5)] = [4.6, 5.4]$$

$$A_{0.2} \supseteq A_{0.5} \supseteq A_{0.8}$$

A_α, $A_{\alpha'}$

$\alpha \le \alpha'$

$\Rightarrow A_\alpha \supseteq A_{\alpha'}$

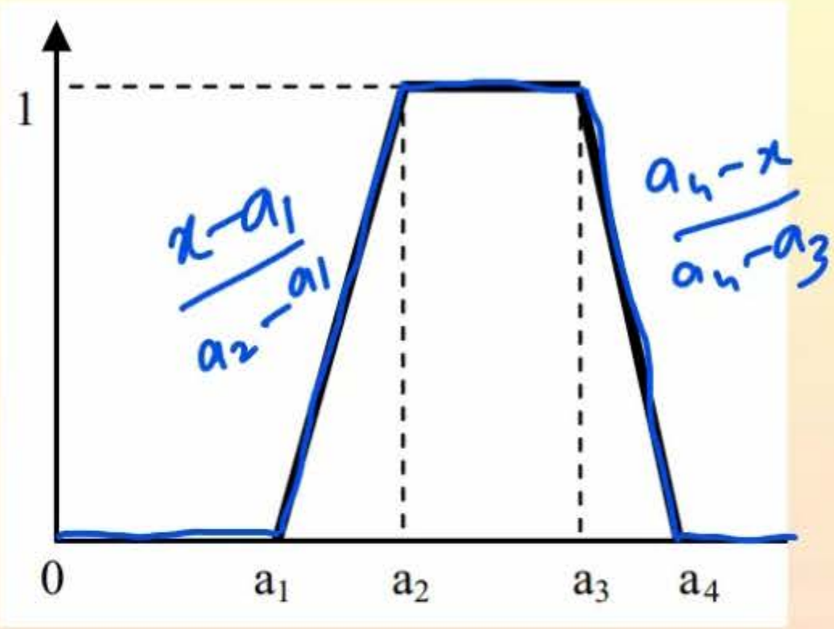
Trapezoidal Fuzzy Numbers

A fuzzy number represented with four points is called trapezoidal fuzzy number. It is defined as follows:

$$\tilde{A} = (a_1, a_2, a_3, a_4).$$

This fuzzy number is interpreted as fuzzy numbers

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \frac{x-a_1}{a_2-a_1} & a_1 \leq x \leq a_2 \\ 1 & a_2 \leq x \leq a_3 \\ \frac{a_4-x}{a_4-a_3} & a_3 \leq x \leq a_4, \\ 0 & x > a_4 \end{cases}$$



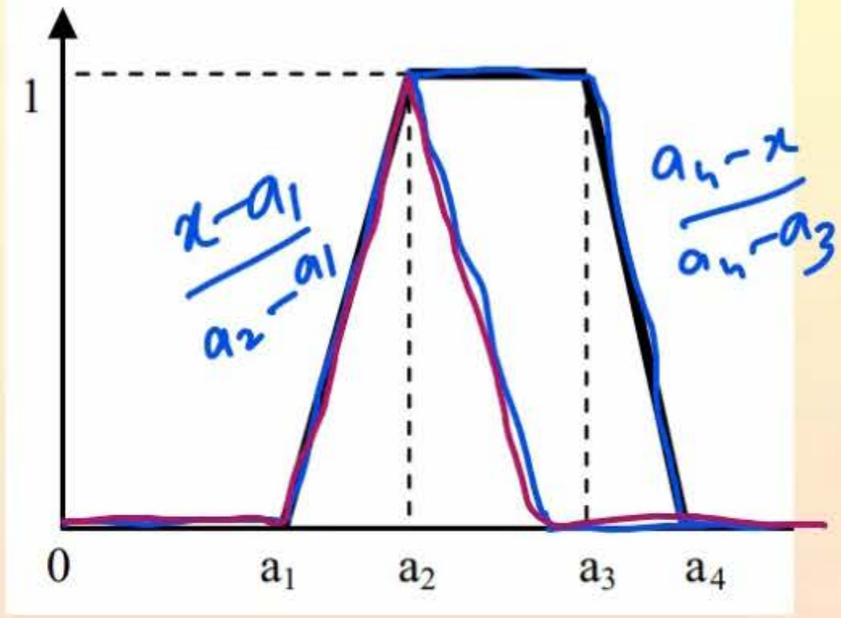
Trapezoidal Fuzzy Numbers

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$$\tilde{A} = (a_1, a_2, a_3, a_4).$$

This fuzzy number is interpreted as fuzzy numbers

$$\mu_{\tilde{A}}(x) = \begin{cases} 0 & x < a_1 \\ \frac{x-a_1}{a_2-a_1} & a_1 \le x \le a_2 \\ 1 & a_2 \le x \le a_3 \\ \frac{a_4-x}{a_4-a_3} & a_3 \le x \le a_4, \\ 0 & x > a_4 \end{cases}$$



$a_2 = a_3$

Note: For $a_2 = a_3$ Traperoidal fuzzy number becomes Triangular fuzzy number.

Addition of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their addition is defined as

$$\tilde{A} + \tilde{B} = (a_1, b_1, c_1, d_1) + (a_2, b_2, c_2, d_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2, d_1 + d_2)$$

$\tilde{A} = (2, 4, 5, 9)$, $\tilde{B} = (3, 5, 7, 8)$

$\tilde{A} + \tilde{B} = (2, 4, 5, 9) + (3, 5, 7, 8) = (2+3, 4+5, 5+7, 9+8)$

$= (7, 9, 12, 17)$

Addition of Trapezoidal Fuzzy Numbers

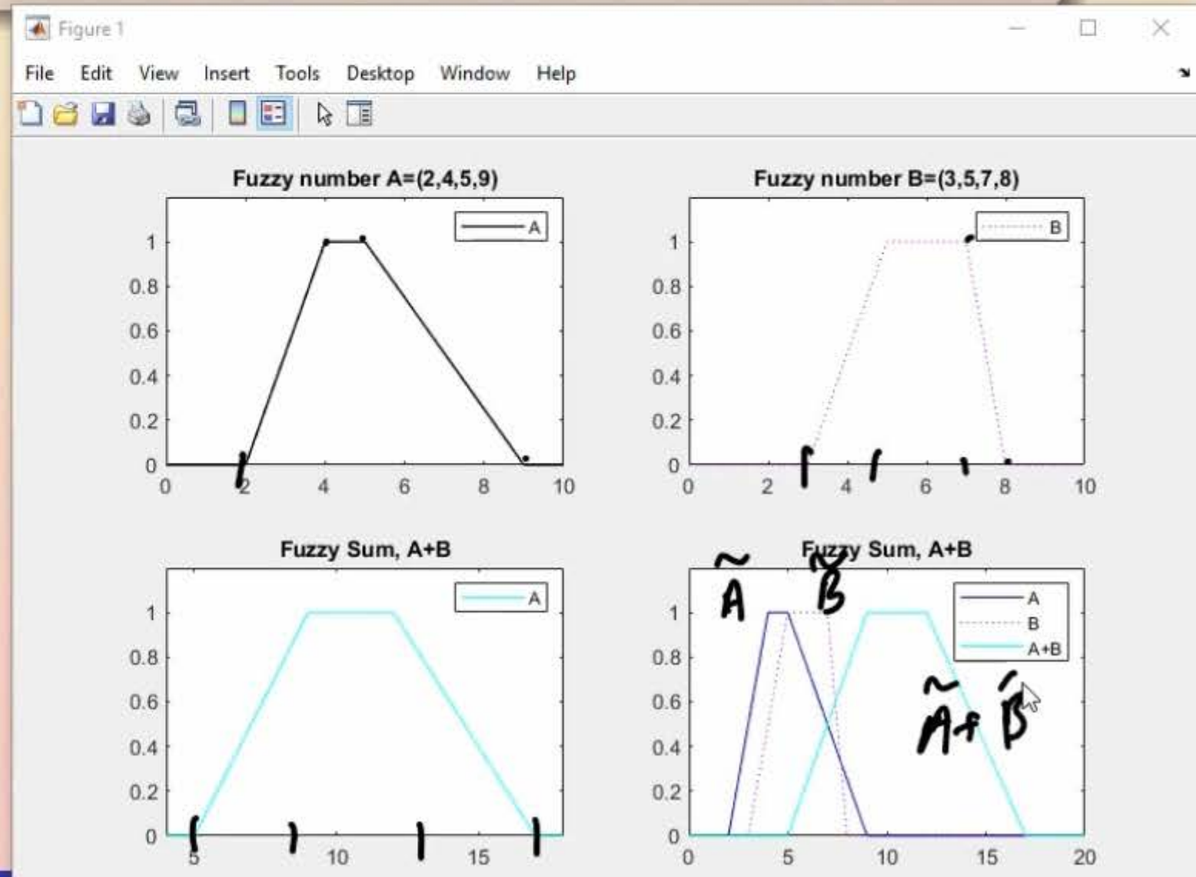
Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their addition is defined as

$$\tilde{A} + \tilde{B} = (a_1, b_1, c_1, d_1) + (a_2, b_2, c_2, d_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2, d_1 + d_2)$$

$\tilde{A} = (2,4,5,9)$, $\tilde{B} = (3,5,7,8)$

$\tilde{A} + \tilde{B} = (5,9,12,17)$



02:20

Addition of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their addition is defined as

$$\tilde{A} + \tilde{B} = (a_1, b_1, c_1, d_1) + (a_2, b_2, c_2, d_2) = (a_1 + a_2, b_1 + b_2, c_1 + c_2, d_1 + d_2)$$

$\tilde{A} = (2, 4, 5, 9)$, $\tilde{B} = (3, 5, 7, 8)$, $\tilde{A} + \tilde{B} = (5, 9, 12, 17)$

$$\tilde{A}(x) = \begin{cases} 0 & x < 2 \\ \frac{x-2}{4-2} & 2 \le x \le 4 \\ 1 & 4 \le x \le 5 \\ \frac{9-x}{9-5} & 5 \le x \le 9 \\ 0 & x > 9 \end{cases} \qquad \tilde{B}(x) = \begin{cases} 0 & x < 3 \\ \frac{x-3}{5-3} & 3 \le x \le 5 \\ 1 & 5 \le x \le 7 \\ \frac{8-x}{8-7} & 7 \le x \le 8 \\ 0 & x > 8 \end{cases}$$

$$\tilde{A} + \tilde{B} = \begin{cases} 0 & x < 5 \\ \frac{x-5}{9-5} & 5 \le x \le 9 \\ 1 & 9 \le x \le 12 \\ \frac{17-x}{17-12} & 12 \le x \le 17 \\ 0 & x > 17 \end{cases}$$

LECTURE # 11

Scalar Addition of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ be a trapezoidal fuzzy number, then its addition with a scalar k is defined by

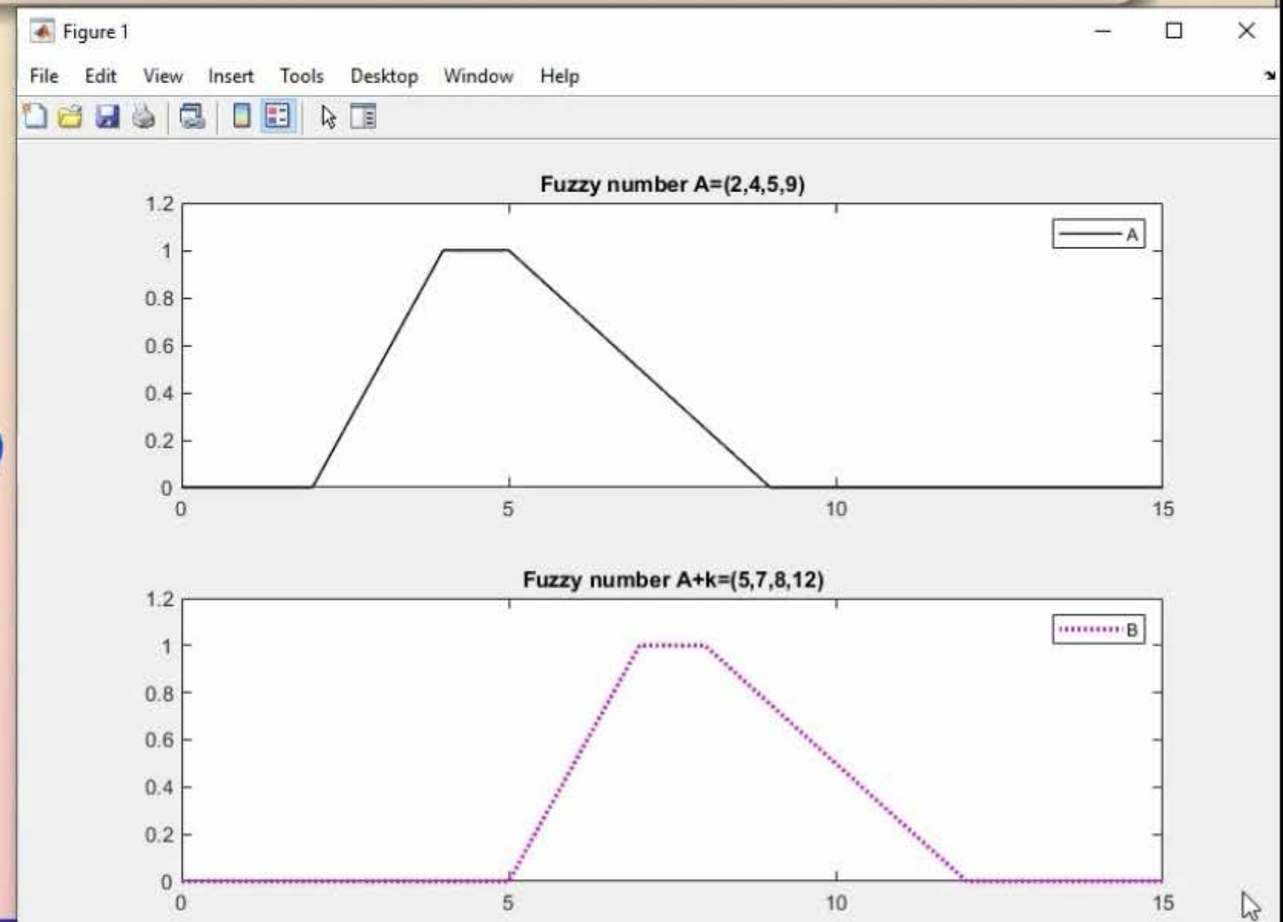
$$\tilde{A} + k = (a_1 + k, b_1 + k, c_1 + k, d_1 + k)$$

$\tilde{A} = (2, 4, 5, 9)$, $3 = k$

$\tilde{A} + k = \tilde{A} + 3$

$= (2+3, 4+3, 5+3, 9+3)$

$= (5, 7, 8, 12)$



Negative of a Trapezoidal Fuzzy Number

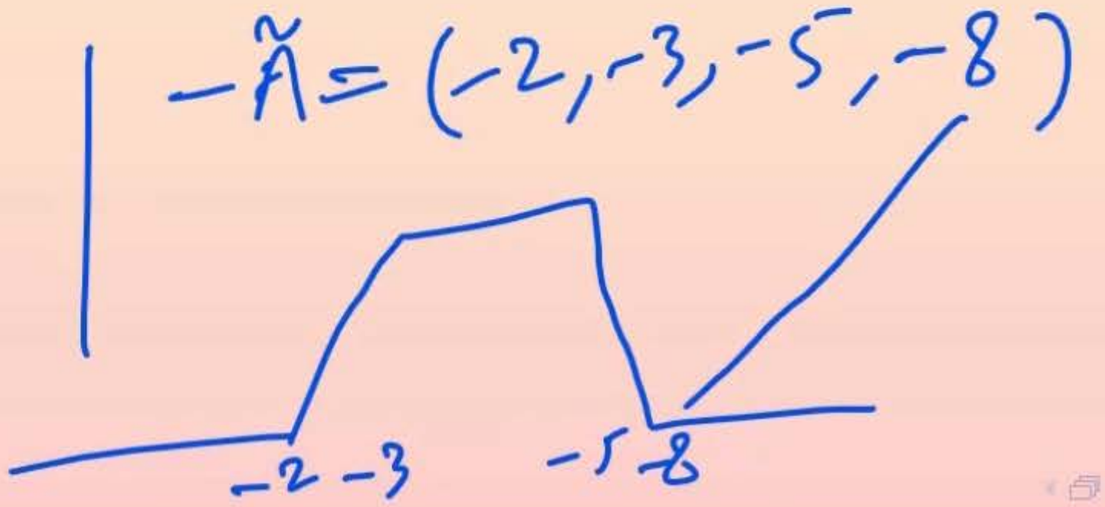
Definition

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then $-\tilde{A}$ is defined as

$$-\tilde{A} = (-d, -c, -b, -a)$$

$\tilde{A} = (2, 3, 5, 8)$

$-\tilde{A} = (-8, -5, -3, -2)$



Negative of a Trapezoidal Fuzzy Number

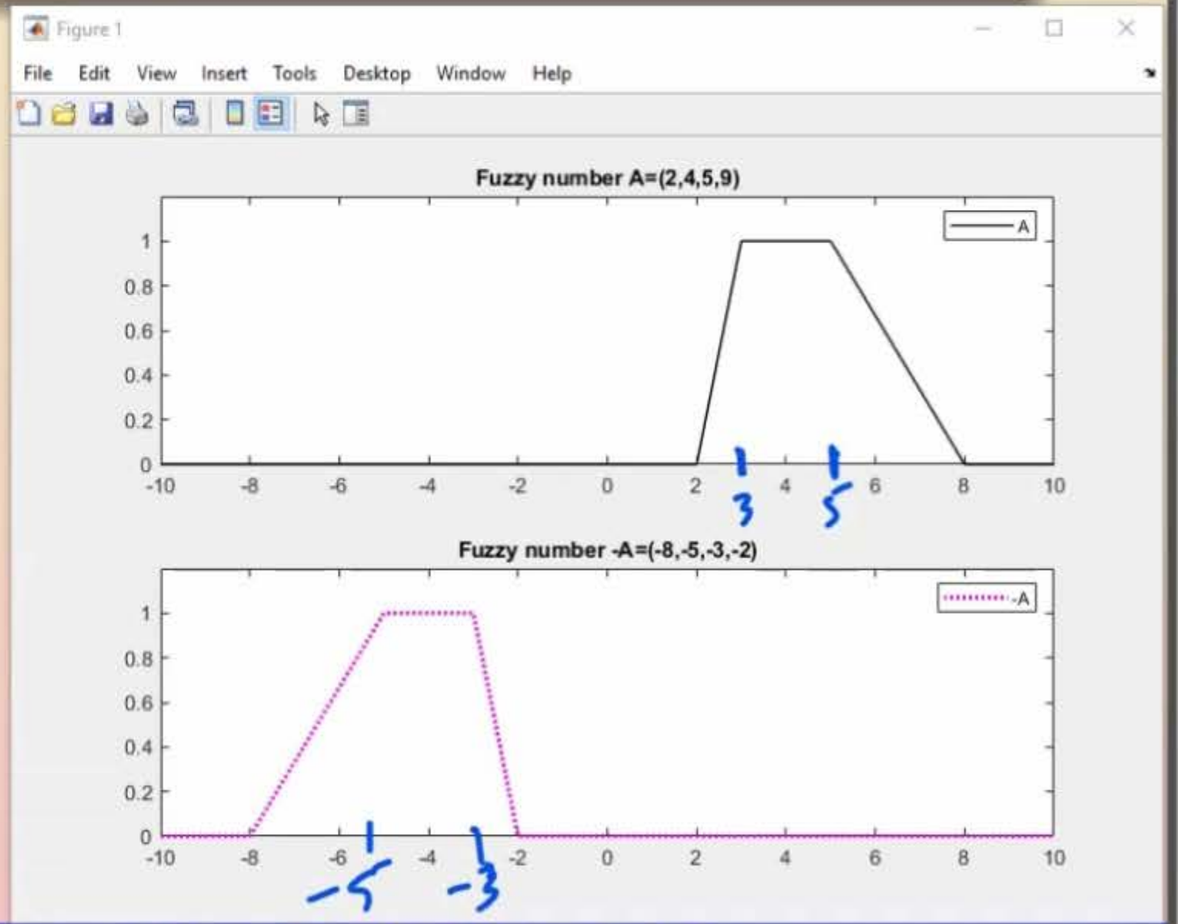
Definition

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then $-\tilde{A}$ is defined as

$$-\tilde{A} = (-d, -c, -b, -a)$$

$\tilde{A} = (2, 3, 5, 8)$

$-\tilde{A} = (-8, -5, -3, -2)$

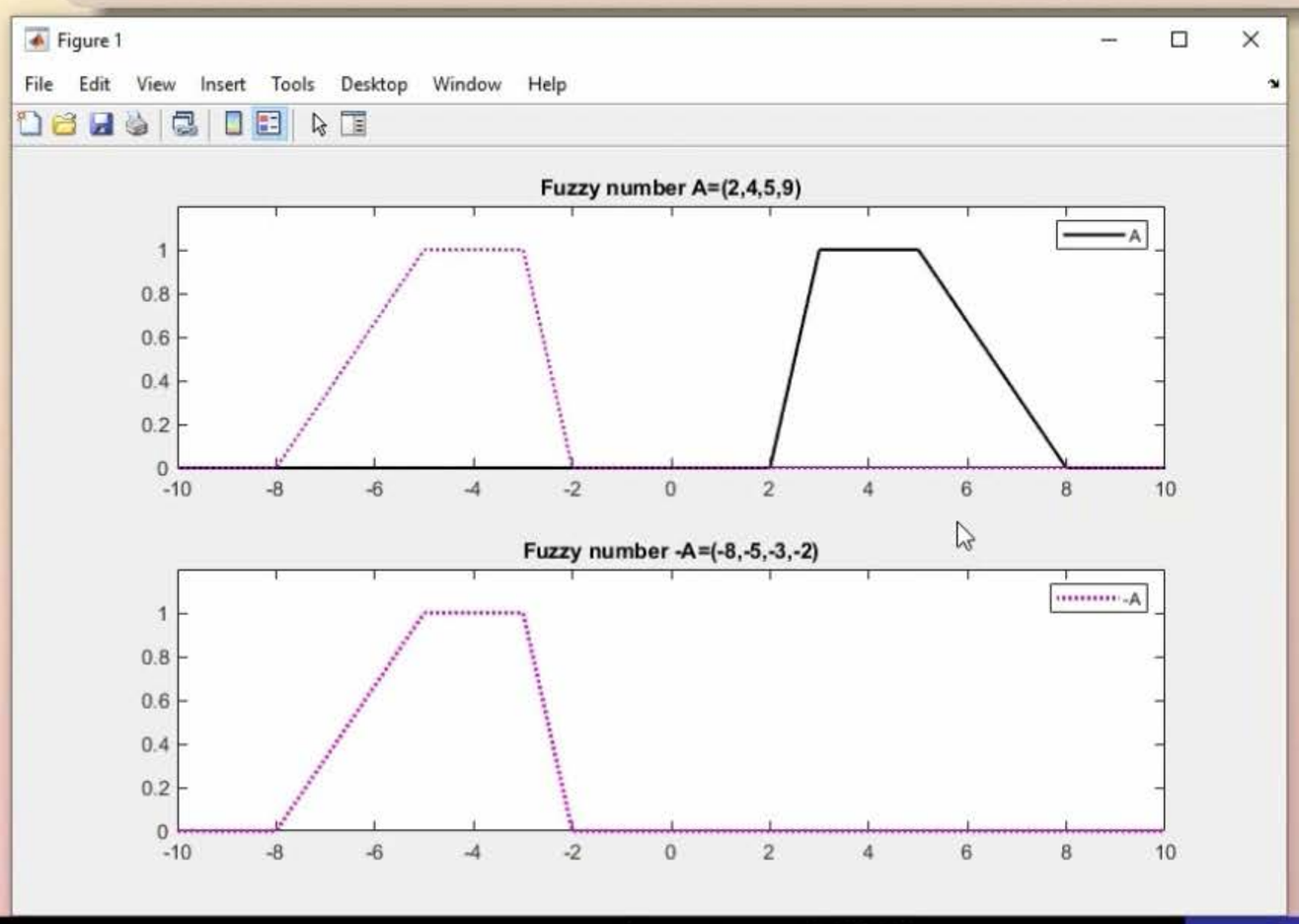


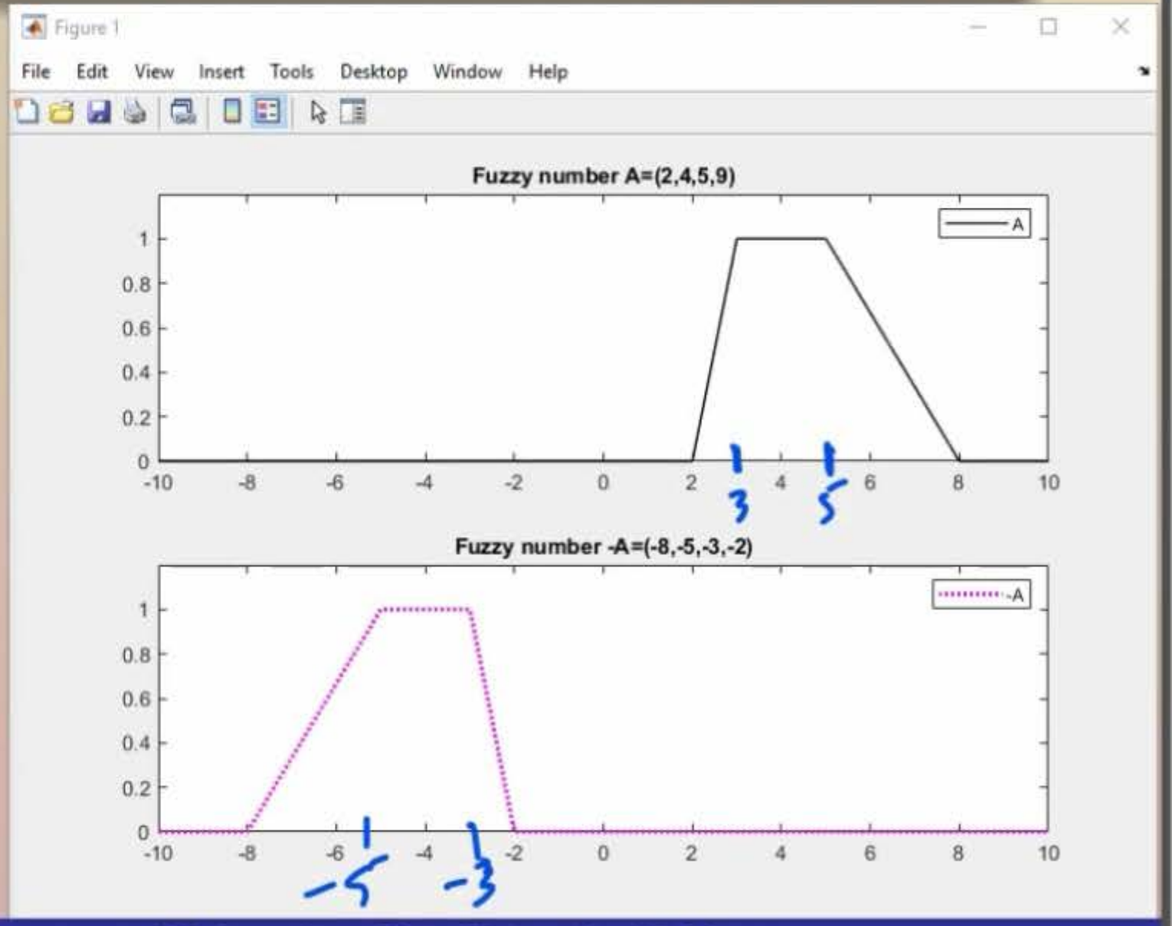
Negative of a Trapezoidal Fuzzy Number

Definition

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then $-\tilde{A}$ is defined as

$$-\tilde{A} = (-d, -c, -b, -a)$$





Subtraction of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their subtraction is defined as

$$\tilde{A} - \tilde{B} = (a_1, b_1, c_1, d_1) - (a_2, b_2, c_2, d_2) = (a_1 - d_2, b_1 - c_2, c_1 - b_2, d_1 - a_2)$$

$$-\tilde{B} = (-d_2, -c_2, -b_2, -a_2)$$

$$\tilde{A} + (-\tilde{B}) = \tilde{A} - \tilde{B}$$

Subtraction of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their subtraction is defined as

$$\tilde{A} - \tilde{B} = (a_1, b_1, c_1, d_1) - (a_2, b_2, c_2, d_2) = (a_1 - d_2, b_1 - c_2, c_1 - b_2, d_1 - a_2)$$

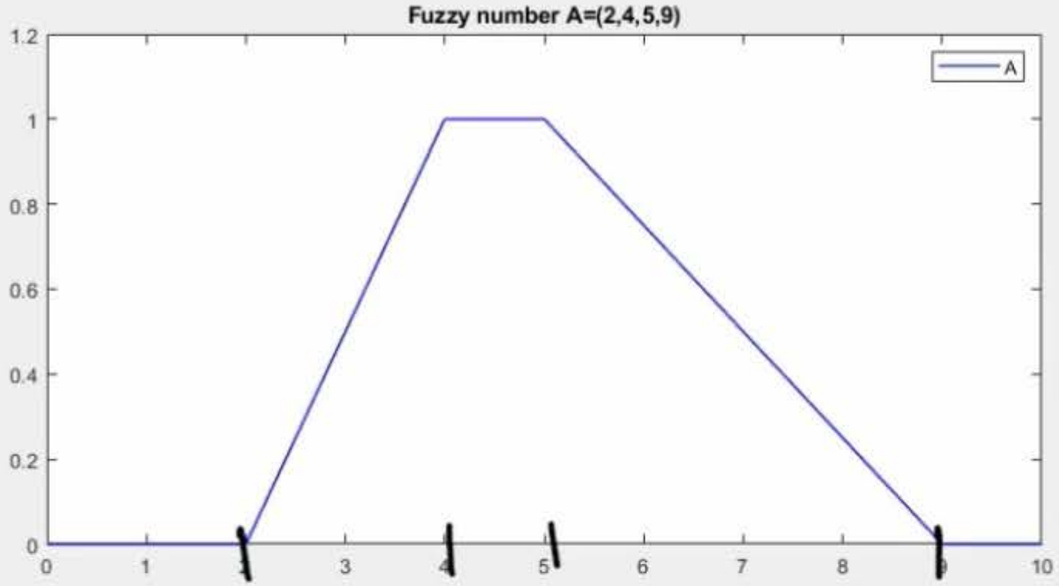
$\tilde{A} = (2, 4, 5, 9)$, $\tilde{B} = (3, 5, 7, 8)$

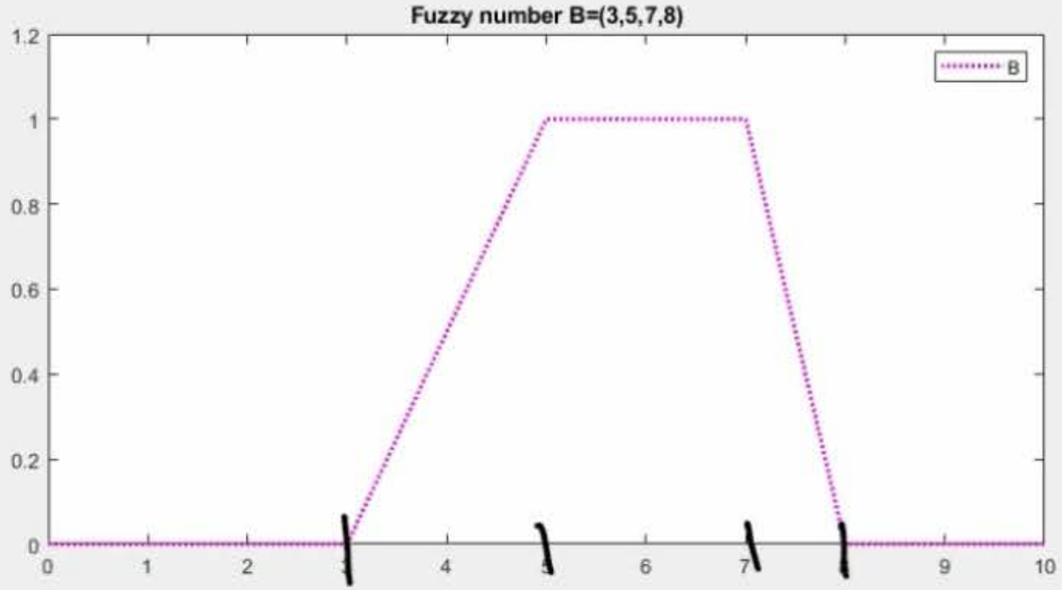
$\tilde{A} - \tilde{B} = (2, 4, 5, 9) - (3, 5, 7, 8)$

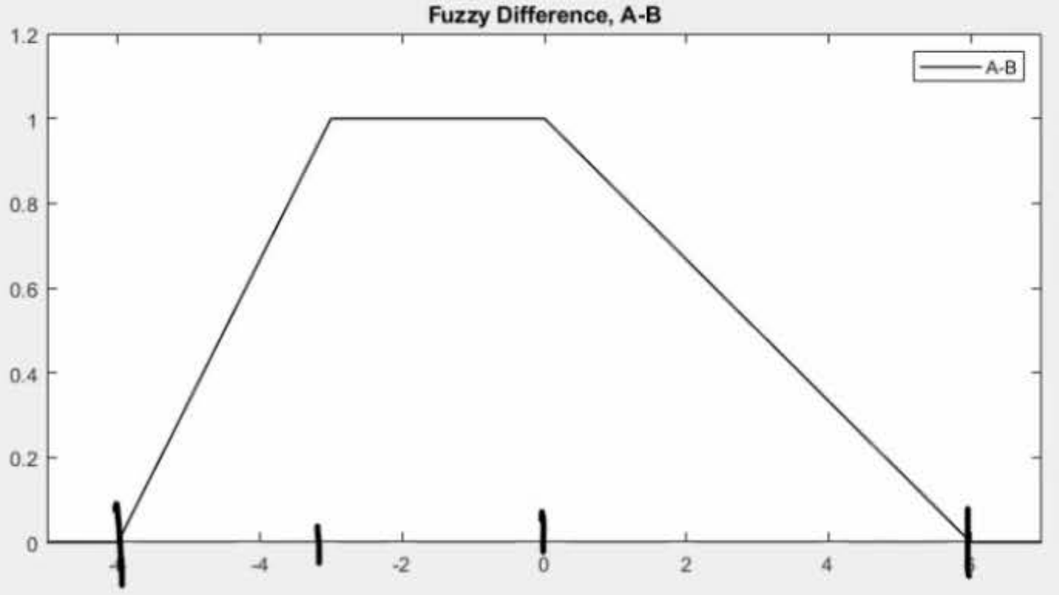
$= (2, 4, 5, 9) + (-8, -7, -5, -3)$

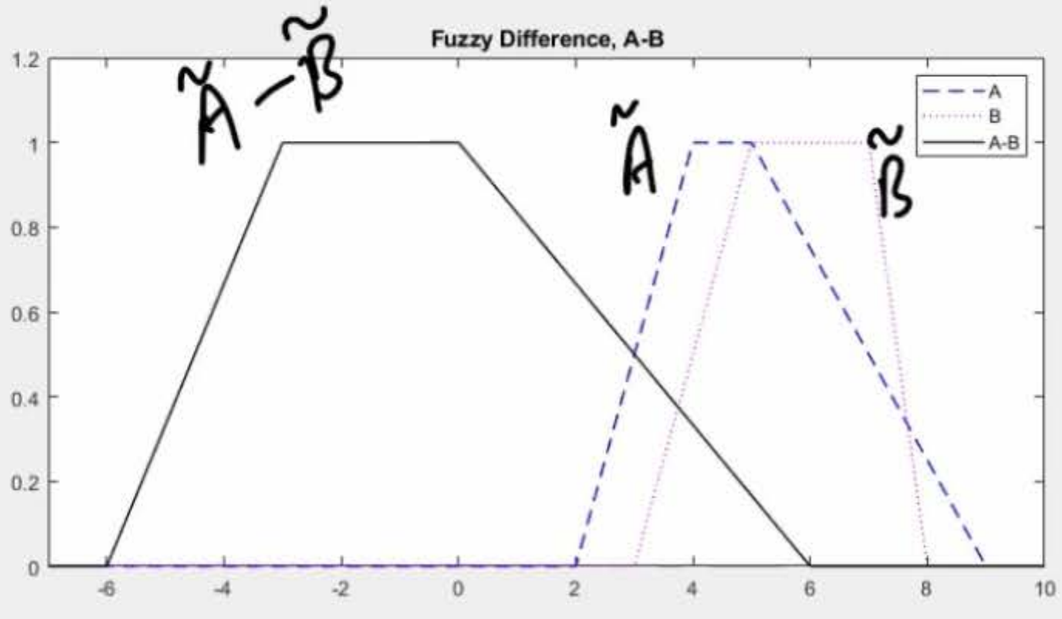
$= (2-8, 4-7, 5-5, 9-3)$

$= (-6, -3, 0, 6)$


Fuzzy number A=(2,4,5,9)
A
1.2
1
0.8
0.6
0.4
0.2
0
0 1 2 3 4 5 6 7 8 9 10


Fuzzy number B=(3,5,7,8)
B
1.2
1
0.8
0.6
0.4
0.2
0
0 1 2 3 4 5 6 7 8 9 10


Fuzzy Difference, A-B
A-B
1.2
1
0.8
0.6
0.4
0.2
0
-6 -4 -2 0 2 4 6


Fuzzy Difference, A-B
A
B
A-B
1.2
1
0.8
0.6
0.4
0.2
0
-6 -4 -2 0 2 4 6 8 10

Scalar Subtraction of Trapezoidal Fuzzy Numbers

Definition

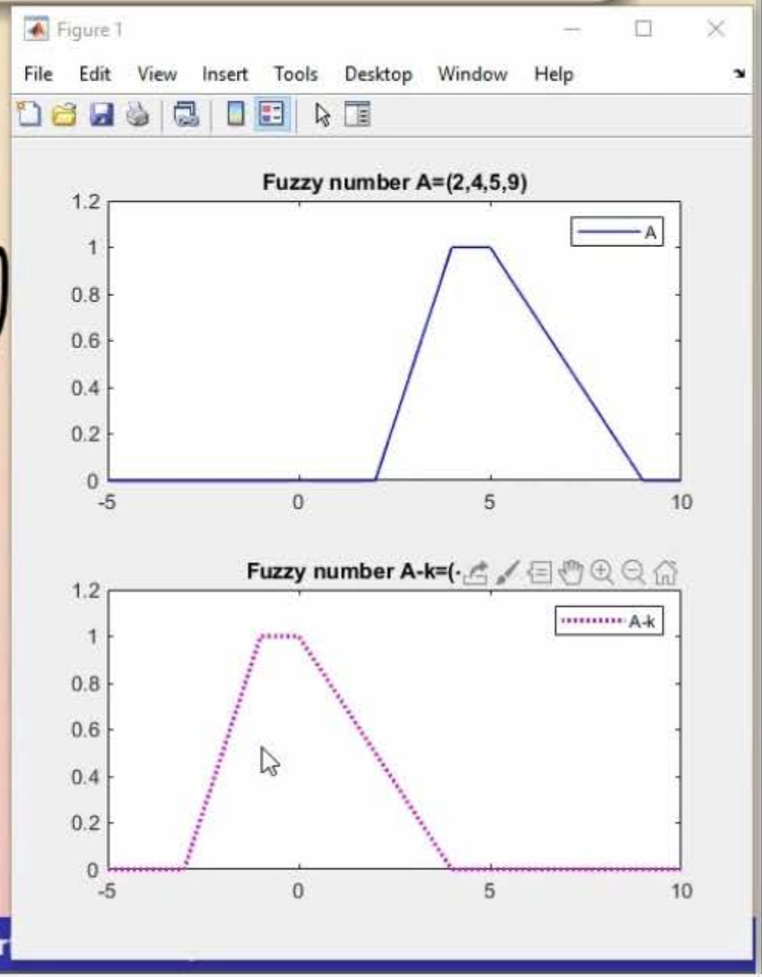
Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ be a trapezoidal fuzzy number, then its subtraction with a scalar k is defined as

$$\tilde{A} - k = (a_1 - k, b_1 - k, c_1 - k, d_1 - k)$$

$\tilde{A} = (2,4,5,9)$, $k = 5$

$\tilde{A} - k = \tilde{A} - 5 = (2-5, 4-5, 5-5, 9-5)$

$= (-3, -1, 0, 4)$



Scalar Subtraction of Trapezoidal Fuzzy Numbers

Definition

Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ be a trapezoidal fuzzy number, then its subtraction with a scalar k is defined as

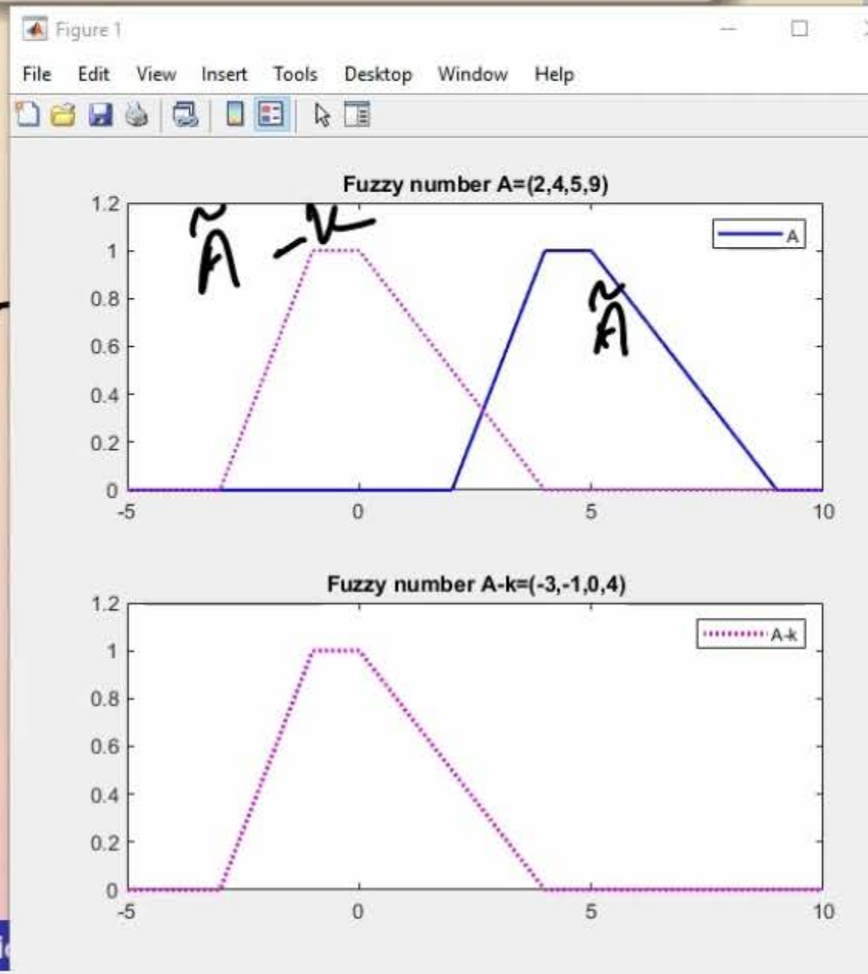
$$\tilde{A} - k = (a_1 - k, b_1 - k, c_1 - k, d_1 - k)$$

$k \in \mathbb{R}$

$\tilde{A} = (2,4,5,9)$, $k = 5$

$\tilde{A} - k = \tilde{A} - 5 = (2-5, 4-5, 5-5, 9-$

$= (-3, -1, 0, 4)$



Multiplication of Trapezoidal Fuzzy Numbers

Definition

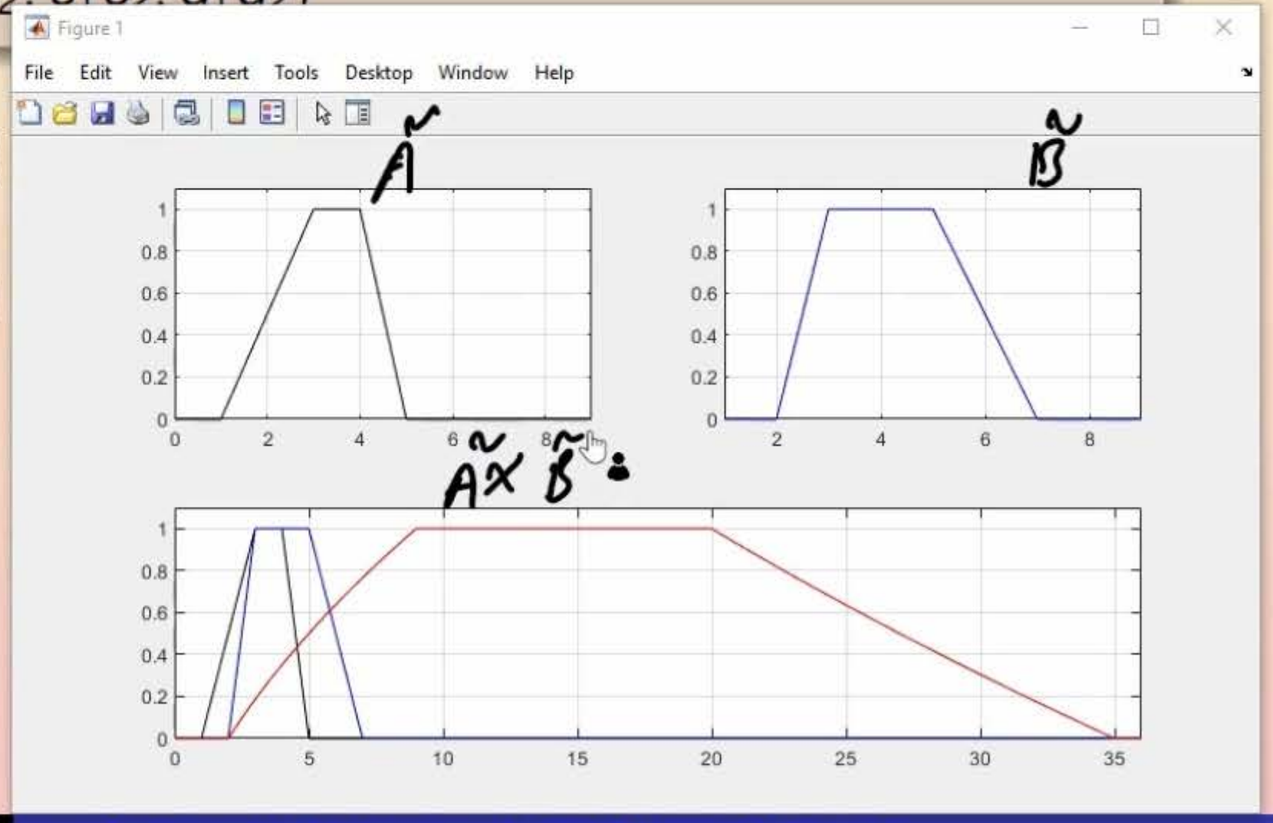
Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their multiplication is defined as

$$\tilde{A} \times \tilde{B} = (a_1, b_1, c_1, d_1) \times (a_2, b_2, c_2, d_2)$$
$$= (a_1 a_2, b_1 b_2, c_1 c_2, d_1 d_2)$$

$\tilde{A} = (1, 3, 4, 5)$, $\tilde{B} = (2, 3, 5, 7)$

$\tilde{A} \times \tilde{B} = (1, 3, 4, 5) \times (2, 3, 5, 7)$

$= (2, 9, 20, 35)$



Scalar multiplication of Trapezoidal Fuzzy Numbers

Definition

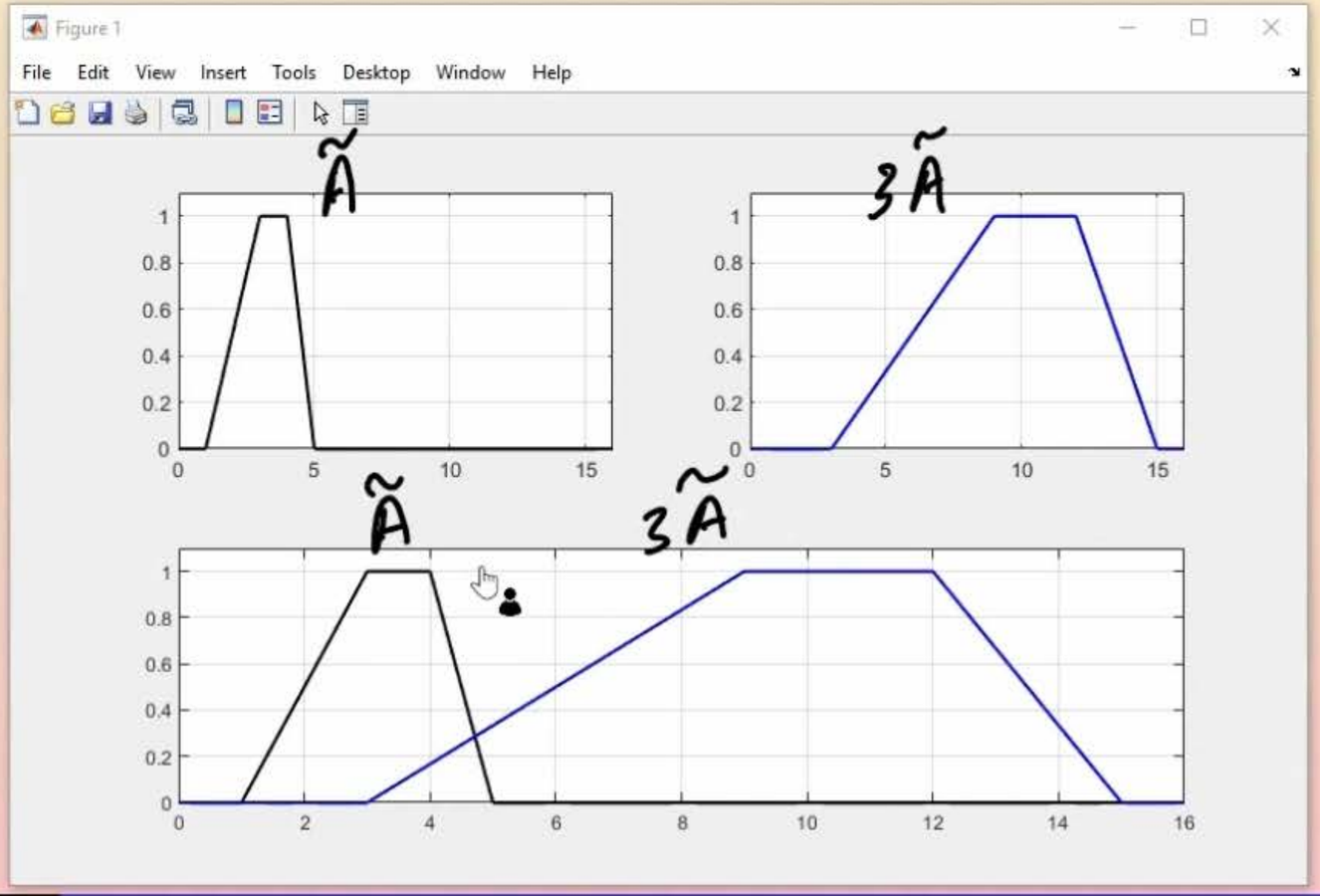
Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number and $\alpha \in R^+$, then

$$\alpha\tilde{A} = (\alpha a, \alpha b, \alpha c, \alpha d)$$

$\tilde{A} = (1,3,4,5)$ $\quad 3 \in R^+$

$3\tilde{A} = 3(1,3,4,5)$

$= (3, 9, 12, 15)$



Scalar multiplication of Trapezoidal Fuzzy Numbers

Definition

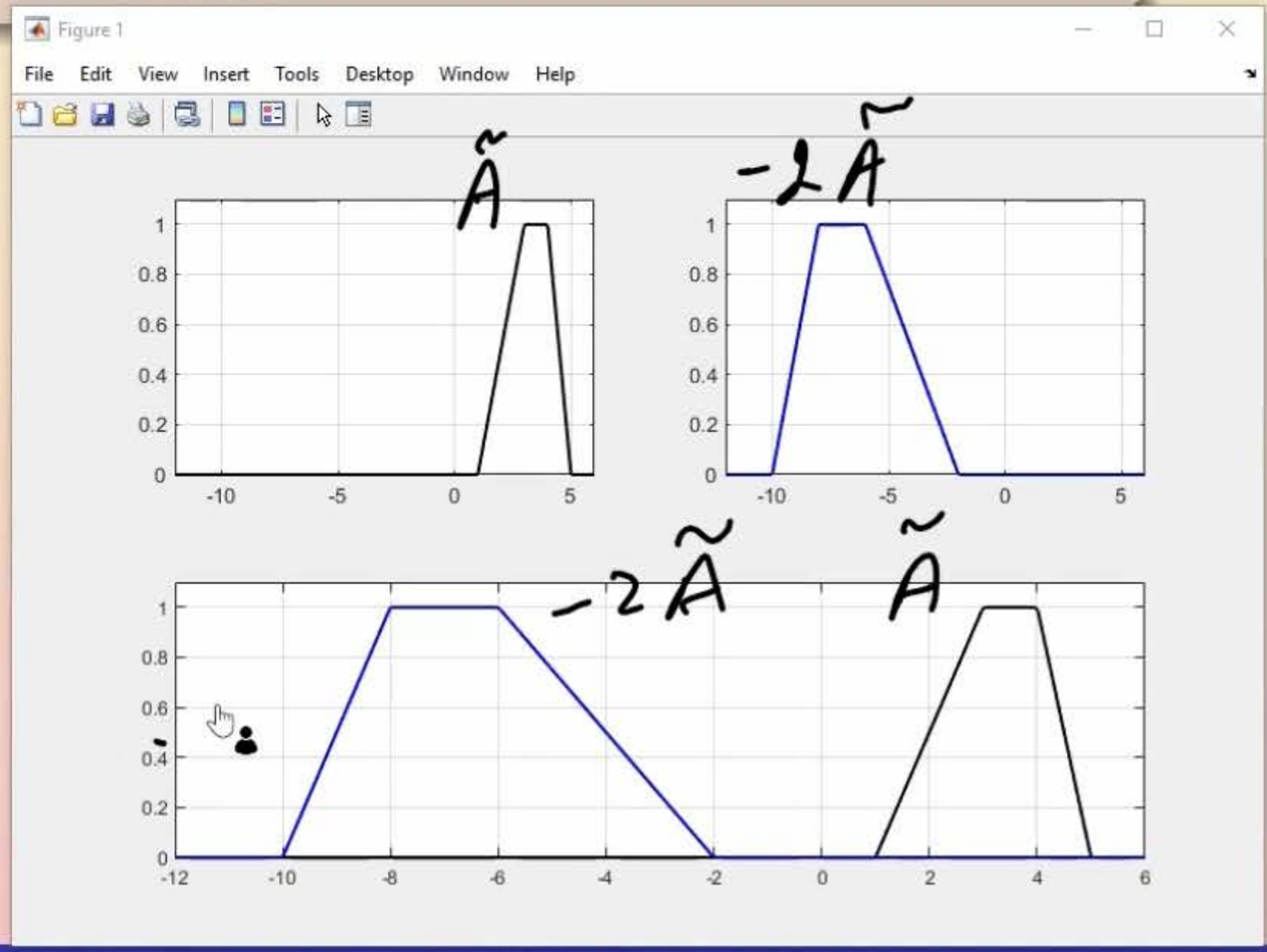
Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number and $\alpha \in R^-$, then

$$\alpha\tilde{A} = (\alpha d, \alpha c, \alpha b, \alpha a)$$

$\tilde{A} = (1, 3, 4, 5) \quad -2 \in R^-$

$-2\tilde{A} = -2(1, 3, 4, 5)$

$= (-10, -8, -6, -2)$



Inverse of a Trapezoidal Fuzzy Number

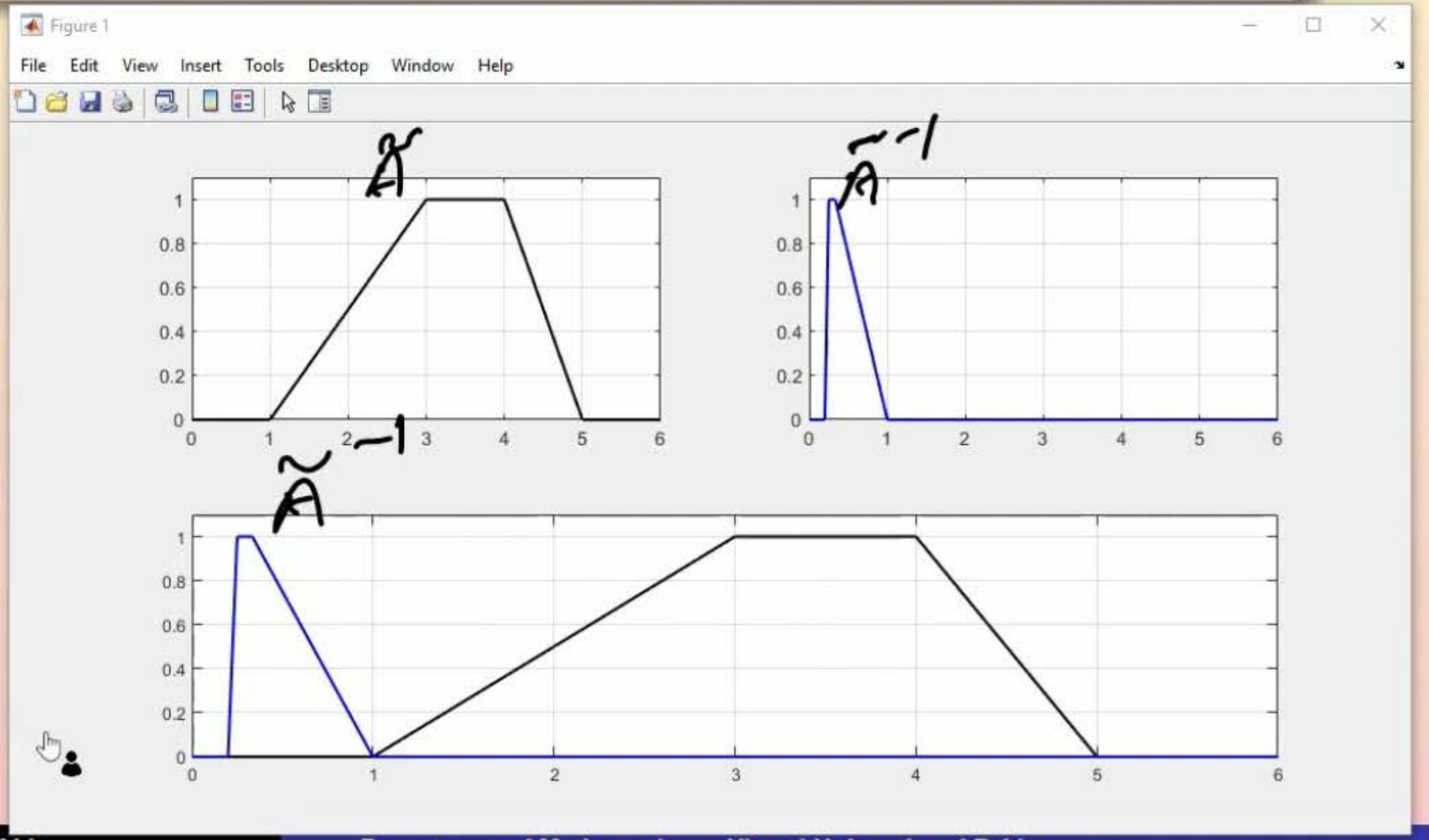
Definition

Let $\tilde{A} = (a, b, c, d)$ be a trapezoidal fuzzy number, then its inverse is defined as

$$\tilde{A}^{-1} = (1/d, 1/c, 1/b, 1/a)$$

$\tilde{A} = (1, 3, 4, 5)$,

$\tilde{A}^{-1} = \left(\frac{1}{5}, \frac{1}{4}, \frac{1}{3}, 1\right)$



Division of Trapezoidal Fuzzy Numbers

Definition

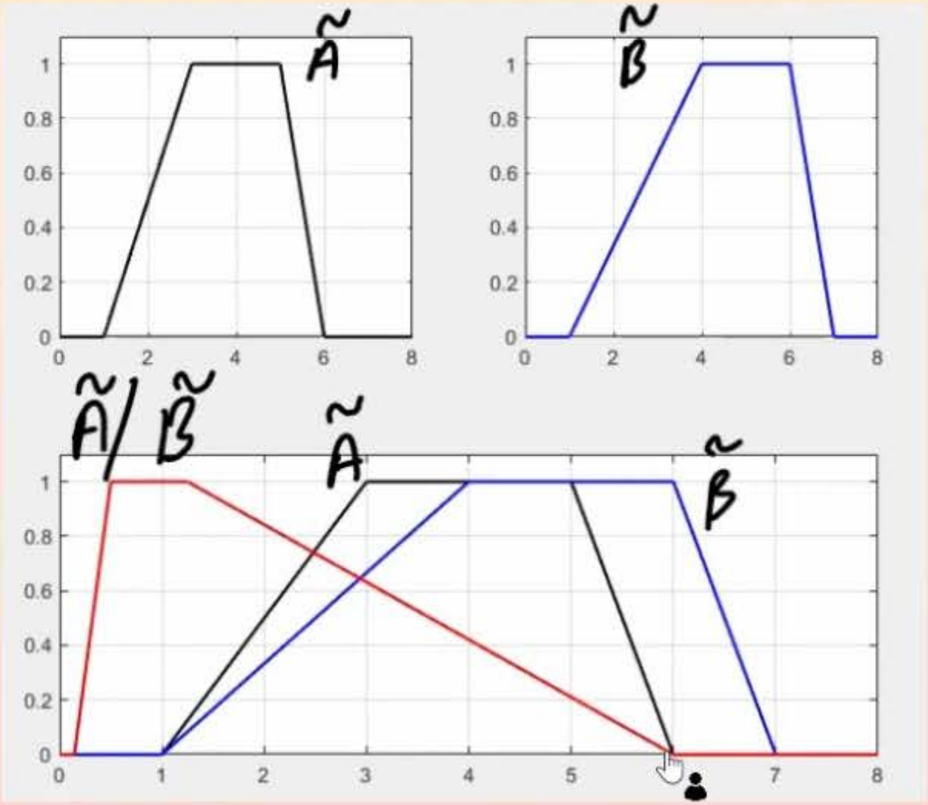
Let $\tilde{A} = (a_1, b_1, c_1, d_1)$ and $\tilde{B} = (a_2, b_2, c_2, d_2)$ be two trapezoidal fuzzy numbers, then their division is defined as

$$\tilde{A}/\tilde{B} = (a_1, b_1, c_1, d_1)/(a_2, b_2, c_2, d_2) = (a_1/d_2, b_1/c_2, c_1/b_2, d_1/a_2)$$

$\tilde{A} = (1, 3, 5, 6)$, $B = (1, 4, 6, 7)$

$\tilde{A}/\tilde{B} = (1, 3, 5, 6)/(1, 4, 6, 7)$

$= \left(\frac{1}{7}, \frac{3}{6}, \frac{5}{4}, \frac{6}{1}\right)$



LECTURE # 12

Fuzzy Relations

Product set

Let A and B be two non-empty sets, the product set or Cartesian product $A \times B$ is defined as follows,

$$A \times B = \{(a, b) \,|\, a \in A, b \in B\}$$

$A_1, A_2, A_3, A_4, \cdots, A_m$

$A_1 \times A_2 \times A \times \cdots \times A_n = \{(a_1, a_2, \cdots, a_n) \mid a_1 \in A_1, a_2 \in A_2, \cdots, a_n \in A_n\}$

$\prod_{i=1}^{n} A_i$

A^n

$A \times A = A^2$

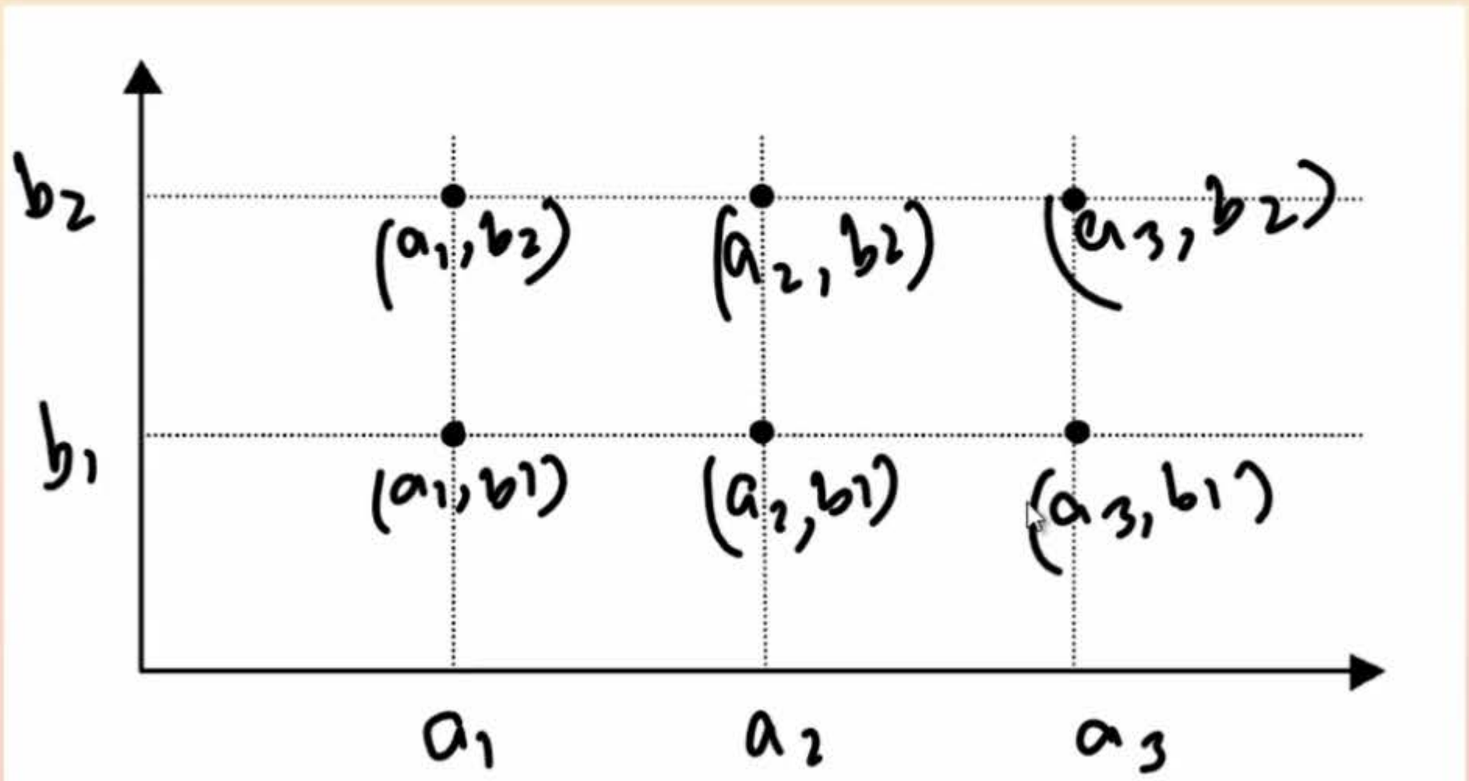
R^2, R^3

$R \times R \times R$

Example

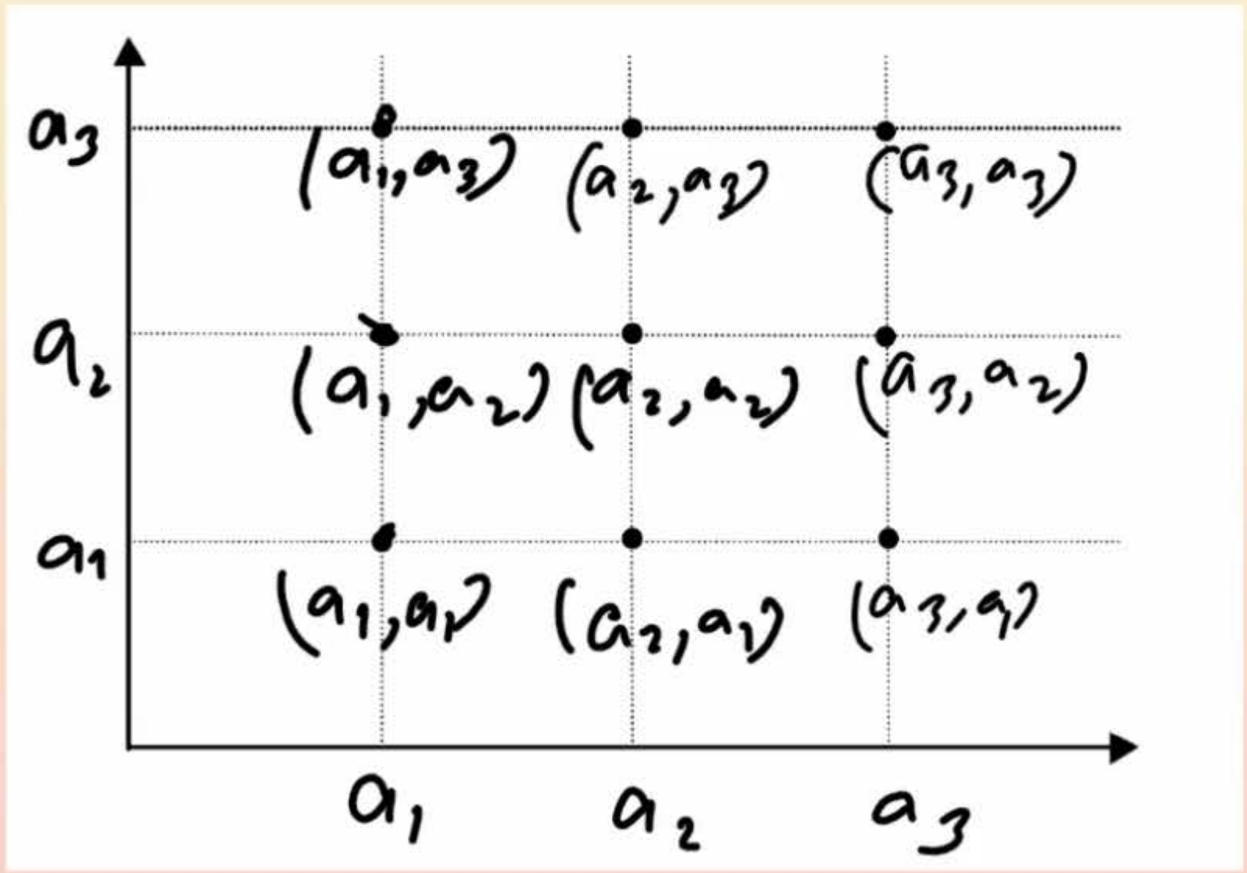
When $A = \{a_1, a_2, a_3\}, B = \{b_1, b_2\}$ the Cartesian product yields

$$A \times B = \{(a_1, b_1), (a_1, b_2), (a_2, b_1), (a_2, b_2), (a_3, b_1), (a_3, b_2)\}$$



Example

$$A \times A = \{(a_1, a_1), (a_1, a_2), (a_1, a_3), (a_2, a_1), (a_2, a_2), (a_2, a_3), (a_3, a_1), (a_3, a_2), (a_3, a_3)\}$$



Binary Relation

Definition

If A and B are two sets and there is a specific property between elements x of A and y of B, this property can be described using the ordered pair (x, y). A set of such (x, y) pairs, $x \in A$ and $y \in B$, is called a relation R.

$$R = \{(x, y) | x \in A, y \in B\}$$

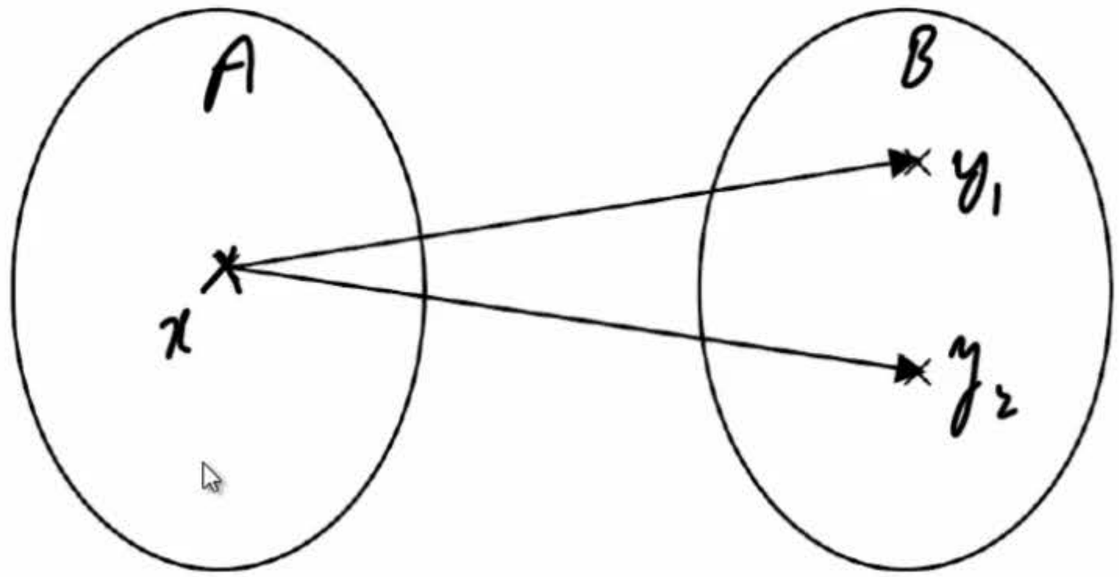
R is a binary relation and a subset of $A \times B$.

One to Many

One-to-many

R is said to be one-to-many if

$$\exists\, x \in A, \quad y_1, \quad y_2 \in B \quad (x, y_1) \in R, (x, y_2) \in R$$



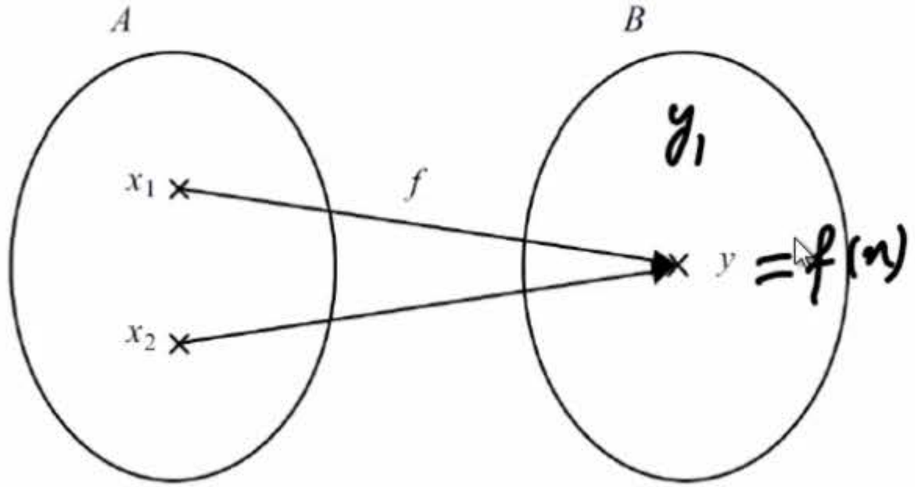
One-to-many relation (*not a function*)

Many to One

Surjection *(may-to-one)*

R is said to be a surjection if $f(A) = B$ or $ran(R) = B$.

$$\forall\, y \in B, \exists\, x \in A, y = f(x)$$



Surjection

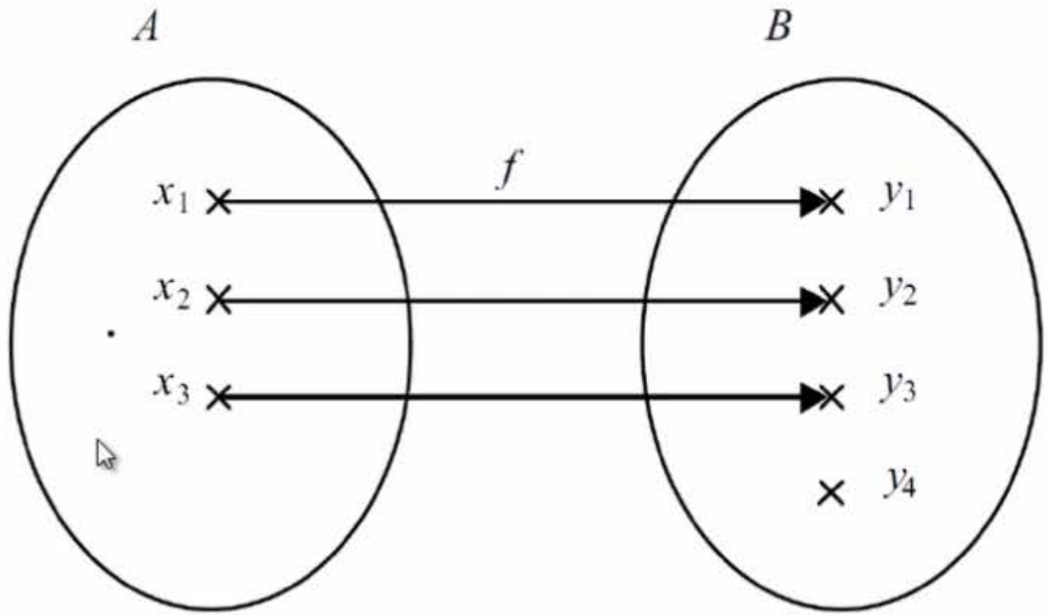
One to One

Injection (into, one-to-one)

R is said to be an injection if for all $x_1, x_2 \in A$,

$$\text{if} \quad x_1 \neq x_2, \quad f(x_1) \neq f(x_2).$$

Therefore, if R is an injection, $(x_1, y) \in R$ and $(x_2, y) \in R$ then $x_1 = x_2$.

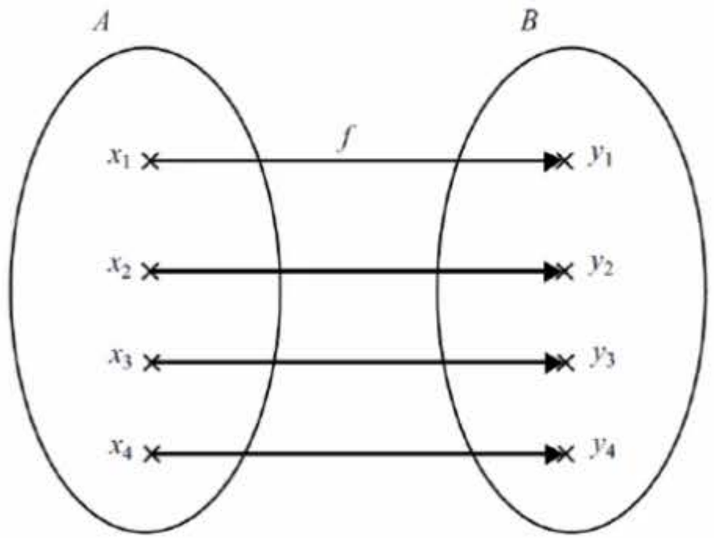


Injection

Bijective

Bijection (one-to-one correspondence)

R is said to be a bijection if it is both a surjection and an injection. Assuming A and B are in bijection, this is an equivalence relation holds, i.e. the elements and the number of the elements correspond.

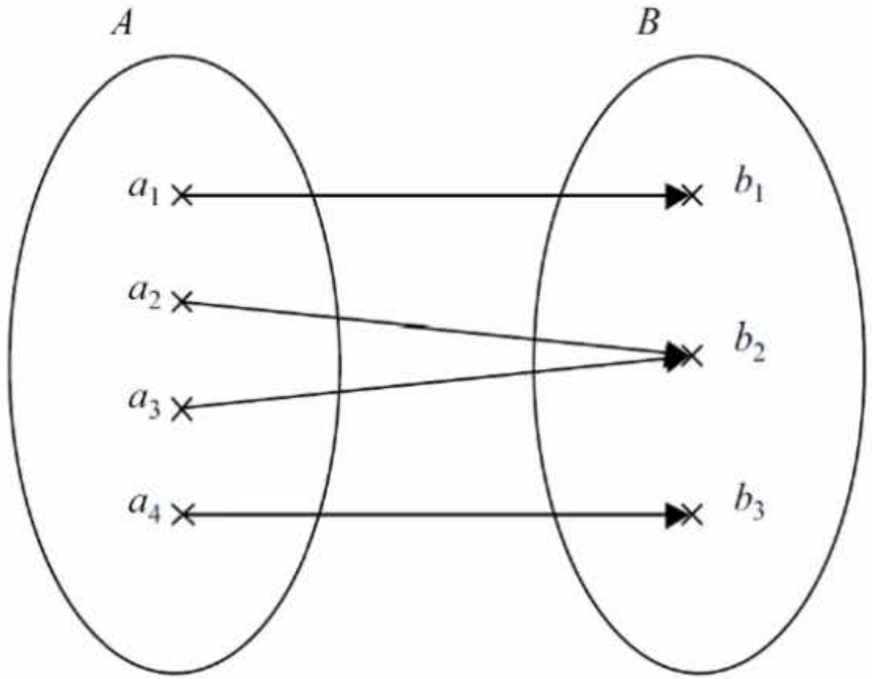


Bijection

Representation Methods of Relations

Bipartite graph

The first is by illustrating A and B in a figure and representing the relation by drawing arcs or edges.

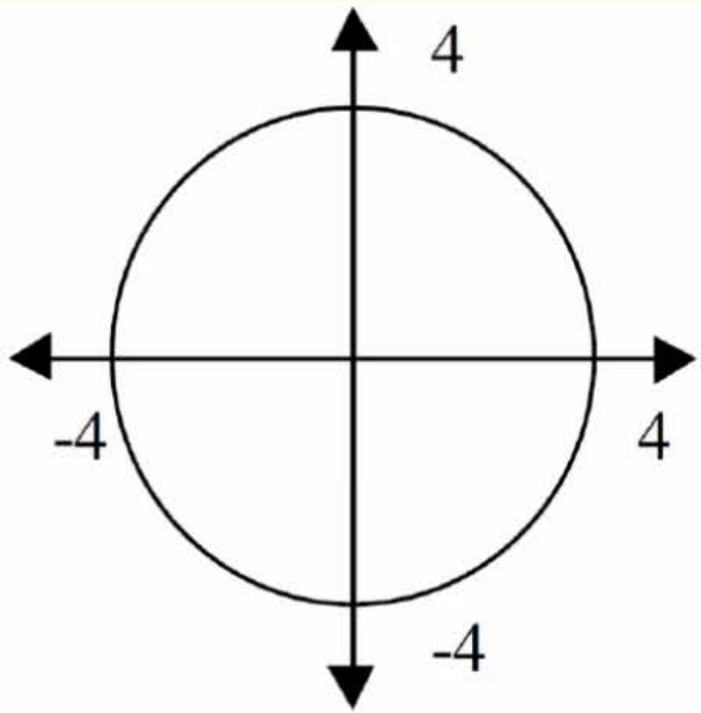


Binary relation from A to B

Representation Methods of Relations

Coordinate Diagram

The second is to use a coordinate diagram by plotting members of A on $x-axis$ and that of B on $y-axis$, and then the members of $A \times B$ lie on the space.



Relation of $x^2 + y^2 = 4$

Representation Methods of Relations

Matrix

The third method is by manipulating relation matrix. Let A and B be finite sets having m and n elements respectively. Assuming R is a relation between A and B, we may represent the relation by matrix $M_R = (m_{ij})$ which is defined as follows

$$M_R = (m_{ij})$$

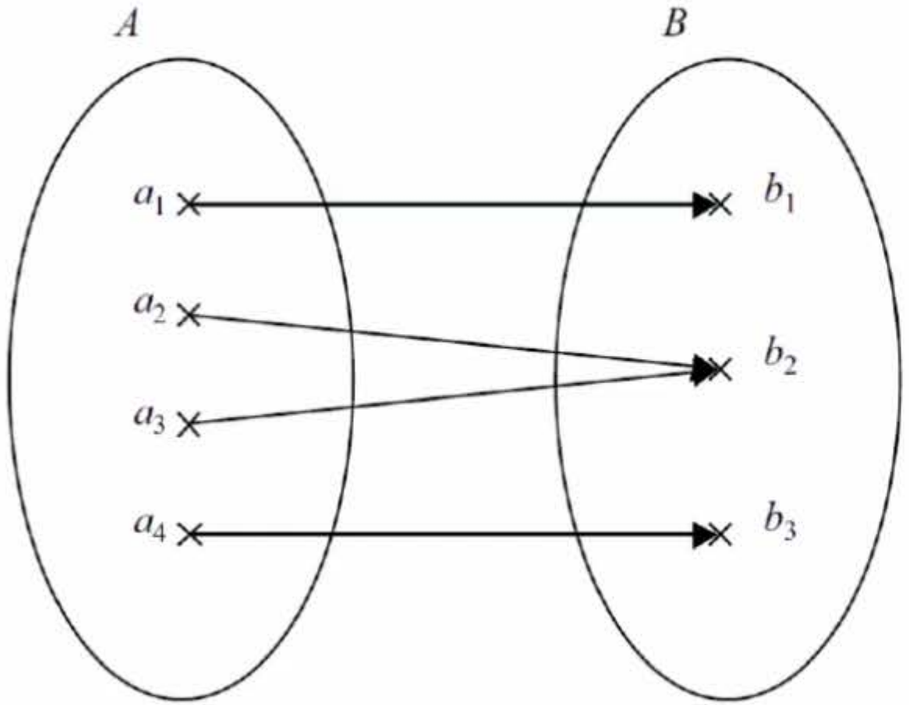
$$m_{ij} = \begin{cases} 1, & (a_i, b_j) \in R \\ 0, & (a_i, b_j) \notin R \end{cases}$$

$$i = 1, 2, 3, \ldots, m$$

$$j = 1, 2, 3, \ldots, n$$

Representation Methods of Relations

$R = \{(a_1, b_1), (a_2, b_2), (a_3, b_2), (a_4, b_3)\}$



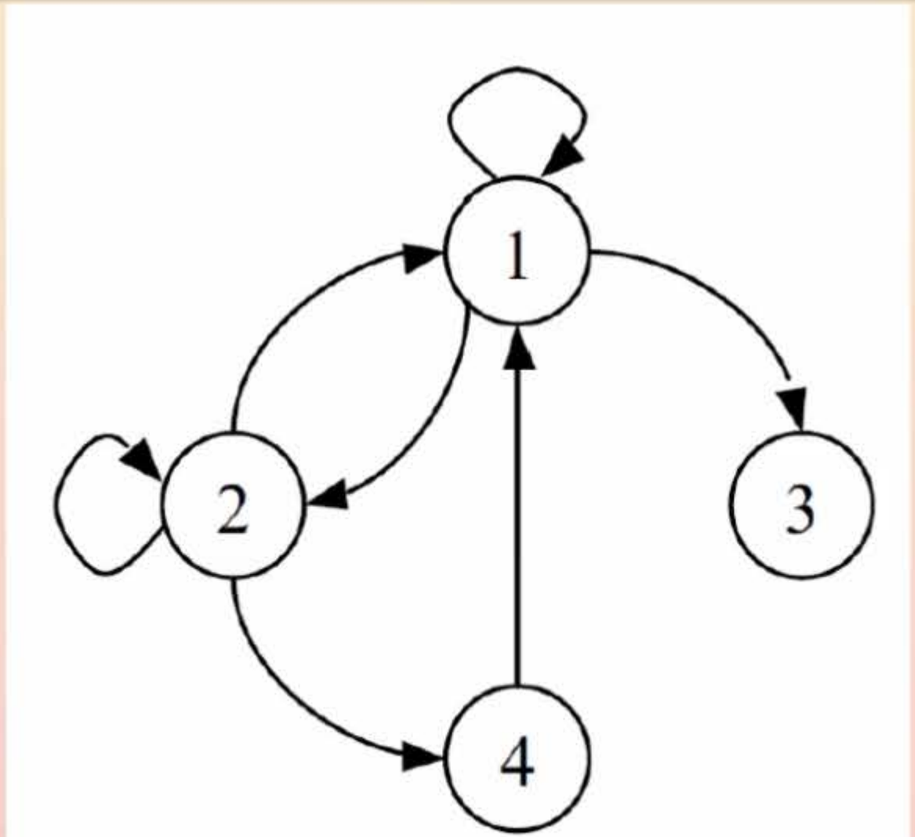
R	b_1	b_2	b_3
a_1	1	0	0
a_2	0	1	0
a_3	0	1	0
a_4	0	0	1

Representation Methods of Relations

Digraph

The fourth method is the directed graph or digraph method. Elements are represented as nodes, and relations between elements as directed edges.

$$A = \{1,2,3,4\} \ \textit{and} \ R = \{(1,1),(1,2),(2,1),(2,2),(1,3),(2,4),(4,1)\}$$



Fuzzy Relation

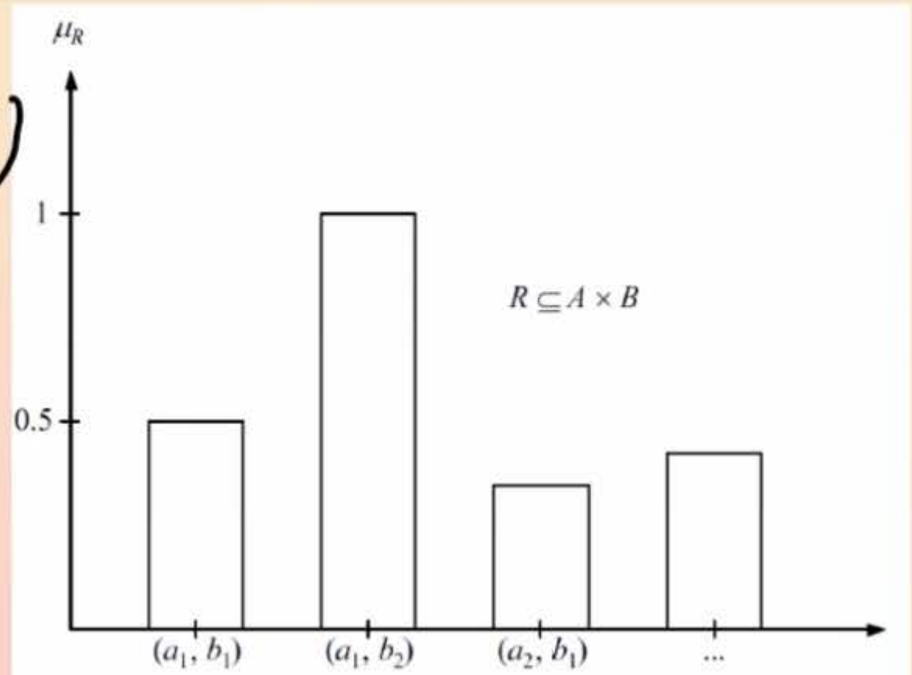
Definition

Fuzzy relation has degree of membership whose value lies in [0, 1].

$$\mu_R : A \times B \to [0, 1]$$
$$R = \{((x, y), \mu_R(x, y)) \,|\, \mu_R(x, y) \geqslant 0, x \in A, y \in B\}$$

$\mu_R(x,y) \geq \mu_R(x',y')$

$(x,y) \quad (x',y')$



Fuzzy relation as a fuzzy set A×B

$A \times B$

$A_1, A_2, \ldots, A_n$

LECTURE # 13

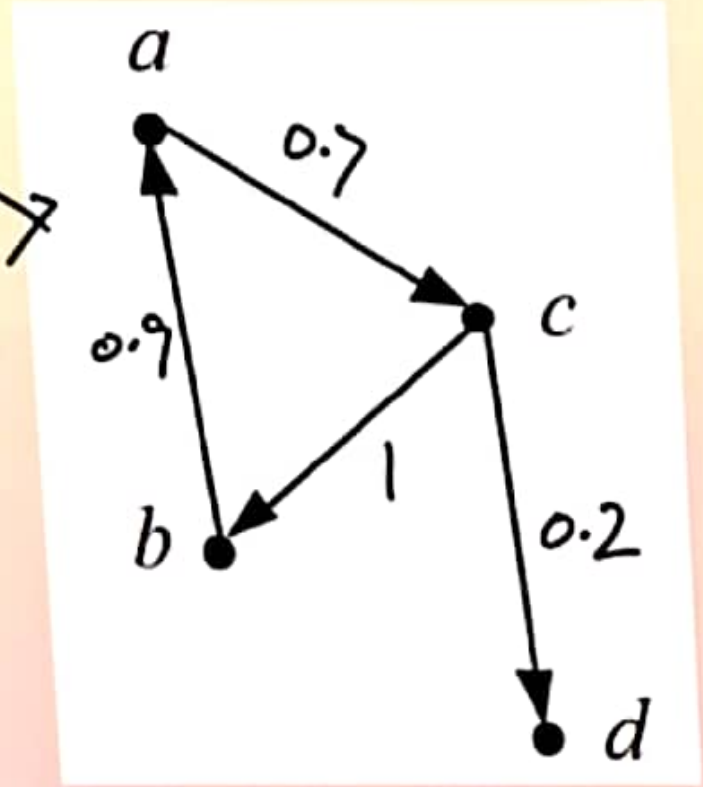
Example

$R = \{(a,c),(b,a),(c,b),(c,d)\}$

$R = \{((a,c),0.7),((b,a),0.9),((c,b),1),((c,d),0.2)\}$

$A \times A$

$A = \{a, b, c, d\}$



A \ A	a	b	c	d
a	0.0	0.0	0.7	0.0
b	0.9	0.0	0.0	0.0
c	0.0	1.0	0.0	0.2
d	0.0	0.0	0.0	0.0

Scanned with CamScanner

$$dom(R) = \{x \mid x \in A, y \in A, \mu_R(x, y) = 1\}$$

$$ran(R) = \{y \mid x \in A, y \in A, \mu_R(x, y) = 1\}$$

$R = \{(a, c), (b, a), (c, b), (c, d)\}$ $A = \{a, b, c, d\}$

(x, y)

$dom(R) = \{a, b, c\}$

$ran(R) = \{c, a, b, d\}$

Scanned with CamScanner

$$\mu_{dom(R)}(x) = \max_{y \in B} \mu_R(x, y)$$

$$\mu_{ran(R)}(y) = \max_{x \in A} \mu_R(x, y)$$

$(x_1, y_1), (x_1, y_2)$

$(x_1, y), (x_2, y)$

$X = \{x_1, x_2, x_3, x_4\}$

$$R = \begin{array}{c} \\ x_1 \\ x_2 \\ x_3 \\ x_4 \end{array} \begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & x_4 \end{array} \\ \begin{bmatrix} 0.2 & 0 & 0.5 & 0 \\ 0 & 0.3 & 0.7 & 0.8 \\ 0.1 & 0 & 0.4 & 0 \\ 0 & 0.6 & 0 & 0.1 \end{bmatrix} \end{array}$$

$Dom(R) = \{0.5, 0.8, 0.4, 0.6\}$

$Ran(R) = \{0.2, 0.6, 0.7, 0.8\}$

$A \times B$

$dom(R) \subseteq A$ ✓

$ran(R) \subseteq B$ ✓

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Fuzzy Relation

Given a fuzzy matrix $A = (a_{ij})$ and $B = (b_{ij})$, we can perform following operations on these fuzzy matrices.

i) Sum $A + B = \text{Max}[a_{ij}, b_{ij}]$

ii) Max Product $A \cdot B = AB = \max_k [\text{Min}(a_{ik}, b_{kj})]$

iii) Scalar product λA where $0 \leq \lambda \leq 1$

Scanned with CamScanner

$A =$

	a	b	c
a	0.2	0.5	0.0
b	0.4	1.0	0.1
c	0.0	1.0	0.0

$B =$

	a	b	c
a	1.0	0.1	0.0
b	0.0	0.0	0.5
c	0.0	1.0	0.1

$A + B =$

	a	b	c
a	1.0	0.5	0.0
b	0.4	1.0	0.5
c	0.0	1.0	0.1

$A \bullet B =$

	a	b	c
a	0.2	0.1	0.5
b	0.4	0.1	0.5
c	0.0	0.0	0.5

$a_{ij} \leq b_{ij} \iff A \leq B \iff SA \leq SB, AT \leq BT$

Scanned with CamScanner

Definition (Fuzzy relation matrix) If a fuzzy relation R is given in the form of fuzzy matrix, its elements represent the membership values of this relation. That is, if the matrix is denoted by M_R, and membership values by $\mu_R(i, j)$, then $M_R = (\mu_R(i, j))$

Scanned with CamScanner

LECTURE # 14

Union of Fuzzy Relations

Union of two relations R and S is defined as follows

$$\mu_{R \cup S}(x, y) = \text{Max}\,[\mu_R(x, y), \mu_S(x, y)] \qquad \forall\,(x, y) \in A \times B$$
$$= \mu_R(x, y) \vee \mu_S(x, y)$$

We generally use the sign $\vee$ for Max operation. For n relations, we extend it to the following.

$$\mu_{R_1 \cup R_2 \cup R_3 \cup \ldots \cup R_n}(x, y) = \bigvee_{R_i} \mu_{R_i}(x, y)$$

If expressing the fuzzy relation by fuzzy matrices, i.e. M_R and M_S, matrix $M_{R \cup S}$ concerning the union is obtained from the sum of two matrices $M_R + M_S$.

$$M_{R \cup S} = M_R + M_S$$

Scanned with CamScanner

Union of Fuzzy Relations

Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

Fuzzy relation matrices $M_{R \cup S}$ corresponding $R \cup S$

$M_{R \cup S}$	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.6	1.0	0.3

$R = \{((1,a), 0.3), ((1,b), 0.2), ((1,c), 1.0), ((2,a), 0.8), ((2,b), 1.0), ((2,c), 1.0), ((3,a), 0.0), ((3,b), 1.0), ((3,c), 0.0)\}$

$RUS = \{((1,a), 0.3), ((1,b), 0.2), ((1,c), 1.0), ((2,a), 0.8), ((2,b), 1.0), ((2,c), 1.0), ((3,a), 0.6), ((3,b), 1.0), ((3,c), 0.3)\}$

Scanned with CamScanner

Intersection of Fuzzy Relations

The intersection relation $R \cap S$ of set A and B is defined by the following membership function.

$$\mu_{R \cap S}(x) = \text{Min}\,[\mu_R(x, y), \mu_S(x, y)]$$
$$= \mu_R(x, y) \wedge \mu_S(x, y)$$

$\wedge$ wedge

min

The symbol $\wedge$ is for the Min operation. In the same manner, the intersection relation for n relations is defined by

$$\mu_{R_1 \cap R_2 \cap R_3 \cap \ldots \cap R_n}(x, y) = \bigwedge_{R_i} \mu_{Ri}(x, y)$$

Scanned with CamScanner

Intersection of Fuzzy Relations

Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

Fuzzy relation matrices $M_{R \cap S}$ corresponding $R \cap S$

$M_{R \cap S}$	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.0	0.9	0.0

$R \cap S = \{((1,a), 0.3), ((1,b), 0.0), ((1,c), 0.1), ((2,a), 0.1), ((2,b), 0.8), ((2,c), 1.0), ((3,a), 0.0), ((3,b), 0.9), ((3,c), 0.0)\}$

Scanned with CamScanner

Complement relation $\overline{R}$ for fuzzy relation R shall be defined by the following membership function.

$$\mu_{\overline{R}}(x, y) = 1 - \mu_R(x, y) \quad \forall (x, y) \in A \times B$$

Scanned with CamScanner

Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

$M_{\bar{R}}$	a	b	c
1	0.7	0.8	0.0
2	0.2	0.0	0.0
3	1.0	0.0	1.0

$M_{\bar{S}}$	a	b	c
1	0.7	1.0	0.1
2	0.9	0.2	0.0
3	0.4	0.1	0.7

R S

$R \cup S, R \cap S, \bar{R}, \bar{S}$

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Inverse of Fuzzy Relations

When a fuzzy relation $R \subseteq A \times B$ is given, the inverse relation of R^{-1} is defined by the following membership function.

$$\mu_{R^{-1}}(y, x) = \mu_R(x, y) \quad \text{For all } (x, y) \in A \times B,$$

Scanned with CamScanner

Two fuzzy relation matrices M_R and M_S are given.

M_R	a	b	c
1	0.3	0.2	1.0
2	0.8	1.0	1.0
3	0.0	1.0	0.0

M_S	a	b	c
1	0.3	0.0	0.1
2	0.1	0.8	1.0
3	0.6	0.9	0.3

$\bar{R}^{-1}$, $\bar{S}^{-1}$

$M_{\bar{R}^{-1}}$, $M_{\bar{S}^{-1}}$

$\mu_R^{-1}(y,x) = M_R(x,y)$

$M_{\bar{R}^{-1}}$	1	2	3
a	0.3	0.2	1.0
b	0.8	1.0	1.0
c	0.0	1.0	0.0

$M_{\bar{S}^{-1}}$	1	2	3
a	0.3	0.0	0.1
b	0.1	0.8	1.0
c	0.6	0.9	0.3

$R = \{((1,a), 0.3), ((1,b), 0.2), ((1,c), 1.0), ((2,a), 0.8), ((2,b), 1.0), ((2,c), 1.0), ((3,a), 0.0), ((3,b), 1.0), ((3,c), 0.0)\}$

$\bar{R}^{-1} = \{((a,1), 0.3), ((b,1), 0.2), ((c,1), 1.0), ((b,a), 0.8), ((b,2), 1.0), ((b,3), 1.0), ((c,1), 1.0), ((c,2), 1.0), ((c,3), 0.0)\}$.

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LECTURE # 15

Definition (Composition of fuzzy relation) Two fuzzy relations R and S are defined on sets A, B and C. That is, $R \subseteq A \times B$, $S \subseteq B \times C$. The composition $S \bullet R = SR$ of two relations R and S is expressed by the relation from A to C, and this composition is defined by the following.

For $(x, y) \in A \times B$, $(y, z) \in B \times C$,

$$\mu_{S \bullet R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))] = \vee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

$S \bullet R$ from this elaboration is a subset of $A \times C$. That is, $S \bullet R \subseteq A \times C$.

If the relations R and S are represented by matrices M_R and M_S, the matrix $M_{S \bullet R}$ corresponding to $S \bullet R$ is obtained from the product of M_R and M_S.

$$M_{S \bullet R} = M_R \bullet M_S$$

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Example Consider fuzzy relations $R \subseteq A \times B$, $S \subseteq B \times C$. The sets A, B and C shall be the sets of events. By the relation R, we can see the possibility of occurrence of B after A, and by S, that of C after B. For example, by M_R, the possibility of $a \in B$ after $1 \in A$ is 0.1. By M_S, the possibility of occurrence of α after a is 0.9.

R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

Here, we can not guess the possibility of C when A is occurred. So our main job now will be the obtaining the composition $S \bullet R \subseteq A \times C$.

Presuming that the relations R and S are the expressions of rules that guide the occurrence of event or fact. Then the possibility of occurrence of event B when event A is happened is guided by the rule R. And rule S indicates the possibility of C when B is existing. For further cases, the possibility of C when A has occurred can be induced from the composition rule $S \bullet R$. This manner is named as an "inference" which is a process producing new information.

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_y [\min(\mu_R(x, y), \mu_S(y, z))] \quad \checkmark$$

$$= \vee_y [\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4		
2			
3			

Min

(1, a), (a, α) 0.1, 0.9 = 0.1

(1, b), (b, α) 0.2, 0.2 = 0.2

(1, c), (c, α) 0.0, 0.8 = 0.0

(1, d), (d, α) 1.0, 0.4 = 0.4

$= \max[0.1, 0.2, 0.0, 0.4]$

$= 0.4$

0.4

$\max[\min[(1, a), (a, \alpha)], \min[(1, b), (b, \alpha)], \min[(1, c), (c, \alpha)], \min[(1, d), (d, \alpha)]]$

$= \max[\min(0.1, 0.9), \min(0.2, 0.2), \min(0.0, 0.8), \min(1.0, 0.4)]$

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4		
2	0.3		
3			

(2, a), (a, α) 0.3, 0.9 Min 0.3

(2, b), (b, α) 0.3, 0.2 0.2

(2, c), (c, α) 0.0, 0.8 0.0

(2, d), (d, α) 0.2, 0.4 0.2

Max 0.3

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4		
2	0.3		
3	0.8		

(3, a), (a, α) 0.8, 0.9 Min 0.8

(3, b), (b, α) 0.9, 0.2 0.2

(3, c), (c, α) 1.0, 0.8 0.8

(3, d), (d, α) 0.4, 0.4 0.4

Max 0.8

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	
2	0.3		
3	0.8		

(1, a), (a, B) 0.1, 0.0 → Min 0.0

(1, b), (b, B) 0.2, 1.0 → 0.2

(1, c), (c, B) 0.0, 0.0 → 0.0

(1, d), (d, B) 1.0, 0.2 → 0.2

Max 0.2

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \vee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	
2	0.3	0.3	
3	0.8		

(2, a), (a, B) 0.3, 0.0 → 0.0

(2, b), (b, B) 0.3, 1.0 → 0.3

(2, c), (c, B) 0.0, 0.6 → 0.0

(2, d), (d, B) 0.2, 0.2 → 0.2

Min

Max 0.3

Scanned with CamScanner

R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	
2	0.3	0.3	
3	0.8	0.9	

(3, a), (a, B) 0.8, 0.0 → Min 0.0

(3, b), (b, B) 0.9, 1.0 → 0.9

(3, c), (c, B) 1.0, 0.0 → 0.0

(3, d), (d, B) 0.4, 0.2 → 0.2

Max 0.9

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	
3	0.8	0.9	

(1, a), (a, γ) 0.1, 0.3 → Min 0.1

(1, b), (b, γ) 0.2, 0.8 → 0.2

(1, c), (c, γ) 0.0, 0.7 → 0.0

(1, d), (d, γ) 1.0, 0.3 → 0.3

Max 0.3

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \bigvee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	0.3
3	0.8	0.9	

(2, a), (a, γ) 0.3, 0.3 → Min 0.3

(2, b), (b, γ) 0.3, 0.8 → 0.3

(2, c), (c, γ) 0.0, 0.7 → 0.0

(2, d), (d, γ) 0.2, 0.3 → 0.2

Max 0.3

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R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \vee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	0.3
3	0.8	0.9	0.8

(3, a), (a, γ) 0.8, 0.3

(3, b), (b, γ) 0.9, 0.8

(3, c), (c, γ) 1.0, 0.7

(3, d), (d, γ) 0.4, 0.3

Min

0.3

0.8

0.7

0.3

Max 0.8

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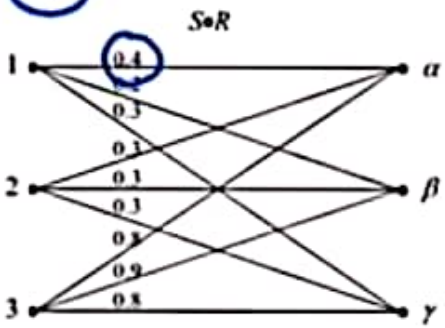
R	a	b	c	d
1	0.1	0.2	0.0	1.0
2	0.3	0.3	0.0	0.2
3	0.8	0.9	1.0	0.4

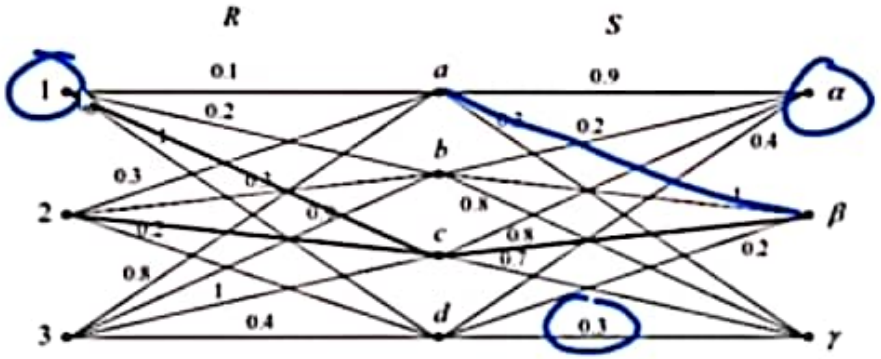
S	α	β	γ
a	0.9	0.0	0.3
b	0.2	1.0	0.8
c	0.8	0.0	0.7
d	0.4	0.2	0.3

$$\mu_{S \circ R}(x, z) = \max_{y}[\min(\mu_R(x, y), \mu_S(y, z))]$$

$$= \vee_{y}[\mu_R(x, y) \wedge \mu_S(y, z)]$$

S•R	α	β	γ
1	0.4	0.2	0.3
2	0.3	0.3	0.3
3	0.8	0.9	0.8





Composition of fuzzy relation

0.1, 0.2, 0.0, 0.4

S•R

LECTURE # 16

FUZZY SETS AND LOGIC WITH APPLICATIONS

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TRIANGULAR NORMS AND CONORMS:

Triangular norms and conorms generalize the basic connectives between fuzzy sets. They were first introduced in the theory of probabilistic metric spaces. Later these were found to be very suitable to be used with fuzzy sets.



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TRIANGULAR NORMS:

A binary operation $\Delta : [0, 1]^2 \rightarrow [0, 1]$ is a triangular norm (t-norm) if it satisfies the following conditions

1. $1 \Delta x = x$
2. $x \Delta y = y \Delta x$
3. $(x \Delta y) \Delta z = x \Delta (y \Delta z)$
4. If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$.

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TRIANGULAR CONORM

A binary operation $\nabla : [0, 1]^2 \rightarrow [0, 1]$ is a triangular conorm (t-conorm) if it satisfies the following conditions

1. $1 \nabla x = x$
2. $x \nabla y = y \nabla x$
3. $(x \nabla y) \nabla z = x \nabla (y \nabla z)$
4. If $w \leq x$ and $y \leq z$, then $w \nabla y \leq x \nabla z$.

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Theorem: If Δ is a t-norm, then prove that $0 \Delta x = 0$.

Proof:

We use 4th property of t-norm.

If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$.

Or

If $x \leq 1$ and $0 \leq 0$, then $x \Delta 0 \leq 1 \Delta 0$.

$x \Delta 0 \leq 1 \Delta 0 = 0$ Using 1st property of t-norm i.e. $1 \Delta x = x$

So $x \Delta 0 \leq 0$ since 0 is least number.

Hence $x \Delta 0 = 0$ which complete the proof.

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SOME EXAMPLES OF T-NORMS:

- $x \Delta_0 y = \begin{cases} x \wedge y & ; if\ x \vee y = 1 \\ 0 & ; otherwise \end{cases}$ (where $\wedge$ = min and $\vee$ = max)
- $x \Delta_1 y = 0 \vee (x+y-1)$
- $x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$
- $x \Delta_3 y = xy$
- $x \Delta_4 y = \frac{xy}{x+y-xy}$
- $x \Delta_5 y = x \wedge y = \min(x,y)$

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LECTURE # 17

SOME EXAMPLES OF T-NORMS:

- $x \Delta_0 y = \begin{cases} x \wedge y & ; if\ x \vee y = 1 \\ 0 & ; otherwise \end{cases}$ (where $\wedge$ = min and $\vee$ = max)
- $x \Delta_1 y = 0 \vee (x+y-1)$
- $x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$
- $x \Delta_3 y = xy$
- $x \Delta_4 y = \frac{xy}{x+y-xy}$
- $x \Delta_5 y = x \wedge y = \min(x,y)$

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Example 1:

Check *min* is T-norm or not?

Solution:

This T-norm is aso called Δ_5.

We check the four properties of the T-norm.

1. $1 \Delta x = \min(1,x) = x$
2. $x \Delta y = \min(x,y) = \min(y,x) = y \Delta x$
3. $(x \Delta y) \Delta z = (\min(x,y)) \Delta z$

$= \min((x,y),z)$

$= \min(x,y,z)$

$= x \Delta \min(y,z)$

$= x \Delta (y \Delta z)$

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4. If $w \leq x$ and $y \leq z$, then $w \Delta y \leq x \Delta z$.

Since $w \leq x$ and $y \leq z$

$\Longrightarrow \min(w,y) \leq \min(x,z)$

$\Longrightarrow w \Delta y \leq x \Delta z$.

Since all the four properties hold so min is a T-norm.



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Example 2:

Check $x \Delta_2 y = \frac{xy}{2-(x+y-xy)}$ is T-norm or not?

Solution:

This T-norm is aso called Δ_2.

We check the four properties of the T-norm.

1. $1 \Delta_2 x = \frac{1 \cdot x}{2-(1+x-1x)} = \frac{x}{2-(1+x-x)} = \frac{1 \cdot x}{2-1} = x$

2. $x \Delta_2 y = \frac{xy}{2-(x+y-xy)} = \frac{yx}{2-(y+x-yx)} = y \Delta_2 x$

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3. L.H.S. $= x \Delta_2 (y \Delta_2 z) = x \Delta_2 \left(\frac{yz}{2-(y+z-yz)}\right)$

$$= \frac{x\left(\frac{yz}{2-(y+z-yz)}\right)}{2-\left(x+\frac{yz}{2-(y+z-yz)}-x\frac{yz}{2-(y+z-yz)}\right)}$$

$$= \frac{xyz}{2(2-(y+z-yz))-[x(2-(y+z-yz))+yz-xyz}$$

$$= \frac{xyz}{4-2x-2y-2z+yz+xy+xz}$$

$$= \frac{xyz}{4-2(x+y+z)+xy+yz+zx}$$

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$$\text{R.H.S.} = (x \,\Delta_2\, y)\,\Delta_2\, z = \frac{xy}{2-(x+y-xy)}\,\Delta_2\, z$$

$$= \frac{\left(\frac{xy}{2-(x+y-xy)}\right)z}{2-\left(\frac{xy}{2-(x+y-xy)}+z-\frac{xy}{2-(x+y-xy)}z\right)}$$

$$= \frac{\left(\frac{xyz}{2-(x+y-xy)}\right)}{2-\left(\frac{xy}{2-(x+y-xy)}+z-\frac{xyz}{2-(x+y-xy)}z\right)}$$

$$= \frac{xyz}{2\big(2-(x+y-xy)\big)-xy-z\big(2-(x+y-xy)\big)+xyz}$$

$$= \frac{xyz}{2(2-x-y+xy)-xy-z(2-x-y+xy)+xyz}$$

$$= \frac{xyz}{4-2x-2y-2z+xy+yz+zx} = \frac{xyz}{4-2(x+y+z)+xy+yz+zx} = \text{L.H.S.}$$

So $x\,\Delta_2\,(y\,\Delta_2\,z) = (x\,\Delta_2\,y)\,\Delta_2\,z$

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4. If $x \leq y$ and $z \leq w$, then $x \Delta_2 z \leq y \Delta_2 w$.

So we show that

$$\frac{xz}{2-(x+z-xz)} \leq \frac{yw}{2-(y+w-yw)}$$

Or $\frac{xz}{2-x-z+xz)} \leq \frac{yw}{2-y-w+yw)}$

Since $x \leq y$ *and* $z \leq w$

$\Rightarrow$ $-x \geq -y$ and $-z \geq -w$

$\Rightarrow$ $1-x \geq 1-y$ and $1-z \geq 1-w$

$\Rightarrow$ $(1-x)(1-z) \geq (1-y)(1-w)$

$\Rightarrow$ $1-x-z+xz \geq 1-y-w+yw$

$\Rightarrow$ $2-x-z+xz \geq 2-y-w+yw$

$$\Rightarrow \frac{1}{2-x-z+xz} \leq \frac{1}{2-y-w+yw} \quad \text{(1)}$$

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Also $xz \leq yw$ _____(2)

Multiplying equation (1) and (2).

$$\frac{xz}{2\text{-}x\text{-}z+xz} \leq \frac{yw}{2\text{-}y\text{-}w+yw}$$

$$\Rightarrow \frac{xz}{2\text{-}(x+z\text{-}xz)} \leq \frac{yw}{2\text{-}(y+w\text{-}yw)}$$

$$\Rightarrow x\Delta_2 z \leq y\Delta_2 w$$

Since all the four properties hold so Δ_2 is a T-norm.

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LECTURE # 18

Example 3:

Check $x \Delta_3 y = xy$ is T-norm or not?

Solution:

We check the four properties of the T-norm.

1. $1 \Delta_3 x = 1 * x = x$
2. $x \Delta_3 y = x * y = y * x = x$
3. L.H.S. $= x \Delta_3 (y \Delta_3 z) = x \Delta_3 (y * z)$

$= x * (y * z) = (x * y) * z = (x \Delta_3 y) \Delta_3 z =$ R.H.S.

So $x \Delta_3 (y \Delta_3 z) = (x \Delta_3 y) \Delta_3 z$

Scanned with CamScanner

4. If $x \leq y$ and $z \leq w$, then $x \Delta_3 z \leq y \Delta_3 w$.

So we show that

$$xz \leq yw$$

$$Since\ x \leq y\ and\ z \leq w$$

$$\Rightarrow xz \leq yw$$

$$\Rightarrow x \Delta_3 z \leq y \Delta_3 w$$

Since all the four properties hold so Δ_3 is a T-norm.

Scanned with CamScanner

Example 4:

Check $x \Delta_4 y = \frac{xy}{x+y-xy}$ is T-norm or not?

Solution:

We check the four properties of the T-norm.

1. $1 \Delta_4 x = \frac{1x}{1+x-1.x} = \frac{x}{1} = x$

2. $x \Delta_4 y = \frac{xy}{x+y-xy} = \frac{yx}{y+x-yx} = y\Delta_4 x$

3. L.H.S. $= x\Delta_4 (y \Delta_4 z) = x\Delta_4 \left(\frac{yz}{y+z-yz}\right)$

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$$= \frac{x\frac{yz}{y+z-yz}}{x+\frac{yz}{y+z-yz}-x\frac{yz}{y+z-yz}} = \frac{\frac{xyz}{y+z-yz}}{x+\frac{yz}{y+z-yz}-\frac{xyz}{y+z-yz}}$$

$$= \frac{xyz}{xy+yz+zx-2xyz}$$

R.H.S. $= (x\Delta_4 y)\,\Delta_4 z = \left(\frac{xy}{x+y-xy}\right)\Delta_4 z$

$$= \frac{\frac{xy}{x+y-xy}z}{\frac{xy}{x+y-xy}+z-\frac{xy}{x+y-xy}z} = \frac{\frac{xyz}{x+y-xy}}{\frac{xy}{x+y-xy}+z-\frac{xyz}{x+y-xy}}$$

$$= \frac{xyz}{xy+yz+zx-2xyz}$$

So $x\,\Delta_4\,(y\,\Delta_4\,z) = (x\,\Delta_4\,y)\,\Delta_4\,z$

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If $x \le y$ and $z \le w$, then $x \Delta_4 z \le y \Delta_4 w$.
So we show that

$$\frac{xz}{x+z-xz} \le \frac{yw}{y+w-yw}$$

Since $x \le y$ *and* $z \le w$

$\Rightarrow xz \le yw$ _____(1)

$x-1 \le y-1 \le 0$ and $z-1 \le w-1 \le 0$

$\Rightarrow (x-1)(z-1) \ge (y-1)(w-1)$

$xz-x-z+1 \ge yw-y-w+1$

$0 \ge xz-x-z \ge yw-y-w$

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$$\Rightarrow \frac{1}{xz-x-z} \leq \frac{1}{yw-y-w} \leq 0 \quad \text{(2)}$$

Multiplying equation (1) by equation (2).

$$\frac{xz}{xz-x-z} \geq \frac{yw}{yw-y-w}$$

on (2).

$$\frac{xz}{-(x+z-xz)} \geq \frac{yw}{-(y+w-yw)}$$

Multiplying by -1 on both sides.

$$\frac{xz}{(x+z-xz)} \leq \frac{yw}{(y+w-yw)}$$

$$\Rightarrow x\Delta_4 z \leq y\Delta_4 w$$

Since all the four properties hold so Δ_4 is a T-norm.

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LECTURE # 19

Idempotent T-norm:

Δ is idempotent if $x \Delta x = x$, $\forall\ x \in [0,1]$.

Converse T-norm:

If $x \Delta y \leq c \leq x_1 \Delta y_1$

so there exist r, s such that $x \leq r \leq x_1$ and $y \leq s \leq y_1$

such that $c = r \Delta s$.

$\Rightarrow$ Δ is converse t-norm.

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Theorem: If a t-norm Δ is convex and $a < b$, then there is $c \in [a,1]$ such that $a = b \Delta c$.

Proof:

We are given that Δ is t-norm.

If $a \leq a$ and $b \leq 1$,

then $a \Delta b \leq a \Delta 1 = a$

$\Rightarrow$ $a \Delta b \leq a \leq b = 1 \Delta b$

$\Rightarrow$ $a \Delta b \leq a \leq 1 \Delta b$

So by convexity there exist $a \leq c \leq 1$ and $b \leq b \leq b$.

$\Rightarrow$ $a = c \Delta b$ which completes the proof.

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Archimedean T-norm:

A t-norm Δ is archimedean if it is convex and for each a, b $\in$ (0, 1), there exist a positive integer n such that

$[a]^n = a \Delta a \Delta a \ldots \Delta a < b$.

Examples:

1- $[a]^3 = a \Delta a \Delta a < b$; for a, b $\in$ (0, 1).

2- Δ_1, Δ_2, Δ_3 and Δ_4 are all archimedean t-norms.

3- min is not and archimedean t-norm.

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Theorem: If Δ is archimedean t-norm, then $\forall\ a,b \in (0, 1)$

$a \Delta b < b$.

Proof:

Δ is archimedean t-norm

so $[a]^n < b$; $\forall\ a, b \in (0,1)$.

Case I:

If $a \leq b$

Take $a \leq b$ and $b \leq b$

then by t-norm $a \Delta b \leq b \Delta b < b$.

$\Rightarrow$ $a \Delta b < b$.

Case II:

If $a > b$

Take $a < 1$ and $b \leq b$

$\Rightarrow$ $a \Delta b < 1 \Delta b = b$

$\Rightarrow$ $a \Delta b < b$.

which completes the proof.

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LECTURE # 20

Exercise : $x \bigtriangleup y = 0 \vee \left(\frac{x+y-1+a xy}{1+a} \right)$; for $a > -1$

Show that $\bigtriangleup$ is a t-norm.

Sol = (1) $1 \bigtriangleup x = x$

L.H.S. = $1 \bigtriangleup x = 0 \vee \left(\frac{1+x-1+a(1)(x)}{1+a} \right)$

$= 0 \vee \left(\frac{x+ax}{1+a} \right) = 0 \vee \left(\frac{x(1+a)}{(1+a)} \right)$

$1 \bigtriangleup x = x$

$= 0 \vee x = x = R.H.S.$

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(i) $x \triangle y = y \triangle x$

$$L.H.S. = x \triangle y = o v \left(\frac{x+y-1+axy}{1+a} \right)$$

$$= o v \left(\frac{y+x-1+ayx}{1+a} \right)$$

$$= y \triangle x$$

$$\Rightarrow x \triangle y = y \triangle x$$

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(ii) $x \Delta (y \nabla z) = (x \nabla y) \Delta z$

L.H.S. $= x \Delta (y \nabla z) = x \Delta \left[\nabla \left(\frac{y + z - 1 + ayz}{1 + a} \right) \right]$

$= \nabla \left[x + \left(\frac{y + z - 1 + ayz}{1 + a} \right) - 1 + a x \left(\frac{y + z - 1 + ayz}{1 + a} \right) \right]$

$= \nabla \left[\frac{x(1+a) + (y + z - 1 + ayz) - 1(1+a) + ax(y + z - 1 + ayz)}{(1+a)^2} \right]$

$= \nabla \left[\frac{x + ax + y + z - 1 + ayz - 1 - a + axy + axz - ax + axyz}{(1+a)^2} \right]$

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$$\text{L.H.S.} = \nabla\left(\frac{x+y+z+a(xy+yz+zx)-2-a+a^2xyz}{(1+a)^2}\right)$$

$$\text{R.H.S.} = (x \triangle y) \triangle z$$

$$= \left[\nabla\left(\frac{x+y-1+axy}{1+a}\right)\right] \triangle z$$

$$= \nabla\left[\frac{\left(\frac{x+y-1+axy}{1+a}\right)+z-1+a\left(\frac{x+y-1+axy}{1+a}\right)z}{1+a}\right]$$

$$= \nabla\left[\frac{x+y-1+axy+z(1+a)-1(1+a)+az(x+y-1+axy)}{1+a}\right]$$

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$$= a \circ \left[\frac{x+y-1+axy+z+az-1-a+a^2x+ayz-az+a^2xyz}{(1+a)^2} \right]$$

$$= a \circ \left[\frac{x+y+z+a(xy+yz+zx)-(2-a+a^2xyz)}{(1+a)^2} \right] = \text{L.H.S.}$$

$$x \circ (y \circ z) = (x \circ y) \circ z$$

(3) is satisfied.

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(iv) If $x \le y$ and $z \le w$ then

$$x \triangle z \le y \triangle w.$$

we have to show that

$$\sigma v\left(\frac{x+z-1+a x z}{1+a}\right) \le \sigma v\left(\frac{y+w-1+a y w}{1+a}\right)$$

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Since $x \leq y$ and $z \leq w$ $\quad x, y, z, w \in F_{\{0,1\}}$

$\Rightarrow x + z \leq y + w$

$x + z - 1 \leq y + w - 1$ ——— (1)

Moreover, $xz \leq yw$

$\Rightarrow \underline{a \vee xz} \leq \underline{a \vee yw} \quad \because a > -1$

$a \vee a x z \leq a \vee a y w$ ——— (2)

$\because \vee$ with a cannot be $-\infty$.

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$$\text{GV}(x+z-1+axz) \leq \text{GV}(y+w-1+ayw)$$

$$a > -1 \Rightarrow 1+a > 0 \Rightarrow \frac{1}{1+a} > 0$$

$$\text{GV}\left(\frac{x+z-1+axz}{1+a}\right) \leq \text{GV}\left(\frac{y+w-1+ayw}{1+a}\right)$$

$$\Rightarrow x \triangle z \leq y \triangle w$$

(4) satisfied.

$\Rightarrow \triangle$ is a t-norm.

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LECTURE # 21

Theorem = Show that

$$x \triangle_0 y \leq x \triangle y \leq x \triangle_5 y$$

Proof = $x \triangle_0 y \leq x \triangle y$

$$x \triangle_0 y = \begin{cases} \min(x, y) & \text{if } \max(x, y) = 1 \\ 0 & \text{, otherwise} \end{cases}$$

Case I - If $\max(x, y) \neq 1$

$$\Rightarrow x \triangle_0 y = 0 \leq x \triangle y$$

$$\Rightarrow x \triangle_0 y \leq x \triangle y \quad \text{——} (1)$$

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Case (iii) :- If $\max(x,y) = 1$, Let $\max(x,y) = y = 1$

$\Rightarrow \min(x,y) = x$

$\Rightarrow \min(x,y) = x \Delta_{0.y} = x \Delta_{0.1} = x = x \Delta y$

$\Rightarrow x \Delta_{0.y} \leq x \Delta y$ —— (2)

(1), (2) $\Rightarrow x \Delta_{0.y} \leq x \Delta y$

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$x \Delta y \leq x \Delta_5 y$

$x \Delta_5 y = \min(x, y)$

Let $\min(x, y) = x$

Take $x \leq x$ and $y \leq 1$

using (iv) of t-norm

$\Rightarrow x \Delta y \leq x \Delta 1 = x = \min(x, y)$

$\Rightarrow x \Delta y \leq x \Delta_5 y$

Hence

$x \Delta_6 y \leq x \Delta y \leq x \Delta_5 y$

Completes the proof

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Theorem : $\Delta_1 \leq \Delta_2 \leq \Delta_3$

Part : (i) $\Delta_1 \leq \Delta_2$

$$x \Delta_1 y = 0 \vee (x + y - 1)$$

$$x \Delta_2 y = \frac{xy}{2 - (x + y - xy)}$$

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$\because \quad 0 \leq x \leq 1 \quad \hookrightarrow \quad 0 \leq y \leq 1$

$\Rightarrow \quad 0 \leq x+y \leq 2$

$\Rightarrow \quad x+y-2 \leq 0 \quad \text{---(1)}$

Also $\quad 1 \geq 1-x \geq 0 \quad \hookrightarrow \quad 1 \geq 1-y \geq 0$

$\Rightarrow \quad 0 \leq (1-x)(1-y) \leq 1$

$1 \leq 1+(1-x)(1-y) \leq 2$



$$1 \leq 2 - x - y + xy \leq 2$$

$$1 \geq \frac{1}{2 - x - y + xy} \geq \frac{1}{2} \quad \text{---} (2)$$

$$x + y - 2 \leq \frac{x + y - 2}{2 - x - y + xy}$$

$$x + y - 2 + 1 \leq \frac{x + y - 2}{2 - x - y + xy} + 1 = \frac{x + y - 2 + 2 - x - y + xy}{2 - x - y + xy} = \frac{xy}{2 - x - y + xy}$$

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$$x+y-1 \leq \frac{xy}{2-x-y+xy}$$

$$\Rightarrow \text{ar}(x+y-1) \leq \frac{xy}{2-x-y+xy} = \Delta_2$$

$$\Delta_1$$

$$x\Delta_1 y \leq x\Delta_2 y$$

$$\Rightarrow \Delta_1 \leq \Delta_2$$

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$x \Delta_2 y \leq x \Delta_3 y$

$\because \quad 0 \leq x \leq 1 \hookrightarrow 0 \leq y \leq 1$

$0 \geq -x \geq -1 \hookrightarrow 0 \geq -y \geq -1$

$1 \geq 1-x \geq 0 \quad \hookrightarrow \quad 1 \geq 1-y \geq 0$

$\Rightarrow \quad 1 \geq (1-x)(1-y) \geq 0$

$\Rightarrow \quad 1 \geq 1-y-x+xy \geq 0$

$2 \geq 2-x-y+xy \geq 1$

i $\quad x \Delta_2 y = \dfrac{xy}{2-x-y+xy} = \dfrac{xy}{2-(x+y-xy)}$

$x \Delta_3 y = xy$

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$$2 \geq 2 - x - y + xy \geq 1$$

$$\frac{1}{2} \leq \frac{1}{2 - x - y + xy} \leq 1 \quad \text{---} (3)$$

$\because$ $0 < xy < 1$ multiply xy with eq (3) on b.s.

$$\frac{xy}{2} \leq \frac{xy}{2 - x - y + xy} \leq xy$$

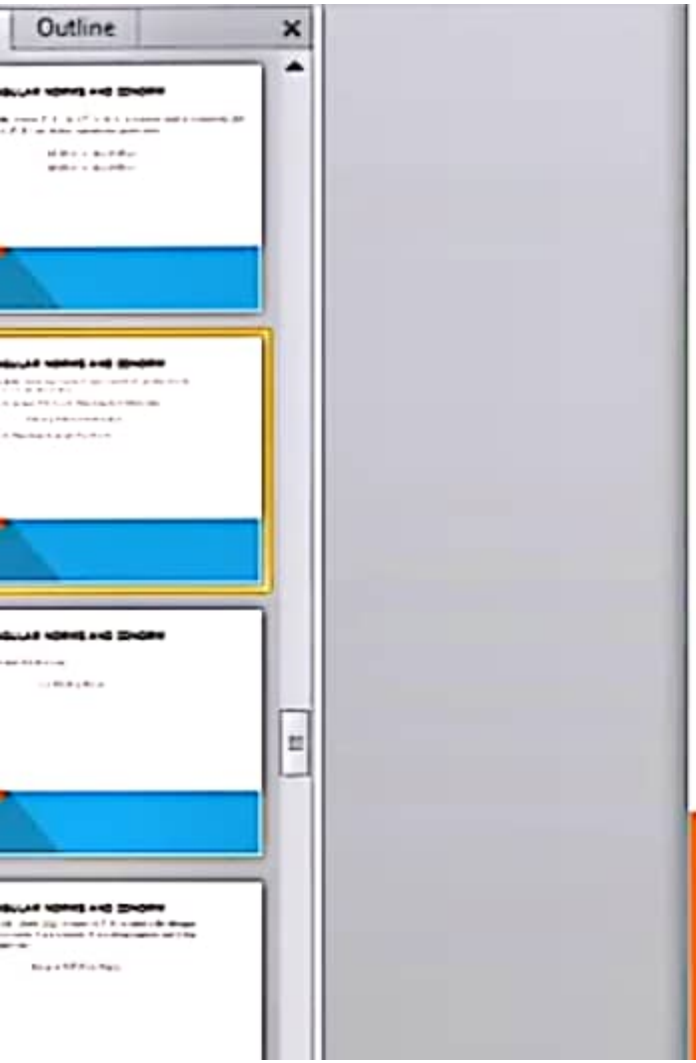
$$\Rightarrow \frac{xy}{2 - x - y + xy} \leq xy$$

$$x \mathbin{\square_2} y \leq x \mathbin{\square_3} y \Rightarrow \square_2 \leq \square_3$$

$$\Rightarrow \square_1 \leq \square_2 \leq \square_3$$

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LECTURE # 22



TRIANGULAR NORMS AND CONORM

Proposition 2.11. *Given any t-norm T and t-conorm S, we have $T(x,0) = 0$ and $S(x,1) = 1$, for all $x \in [0,1]$.*

Proof. From T_1 we have $T(0,1) = 0$. Then from T_4 it follows that

$$T(0,x) \leq T(0,1) = 0, \forall x \in [0,1],$$

i.e., $T(0,x) = 0$. Then from T_2 we get $T(x,0) = 0$.

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$T(x, 0) = 0 \quad ; \quad x \Delta 0 = 0$

$0 \leq 0 \quad \text{and} \quad x \leq 1$

T_4: $\quad 0 \Delta x \leq 0 \Delta 1 = 0$

$0 \Delta x \leq 0$

$0 \Delta x = 0$

T_2: $\quad x \Delta 0 = 0$

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$S(x,1) = 1 \Rightarrow x \nabla 1 = 1$

$0 \leq x$ and $1 \leq 1$

S4: $0 \nabla 1 \leq x \nabla 1$

S1: $1 = 0 \nabla 1$

$1 \leq x \nabla 1$

$x \nabla 1 = 1 \Rightarrow S(x,1) = 1$

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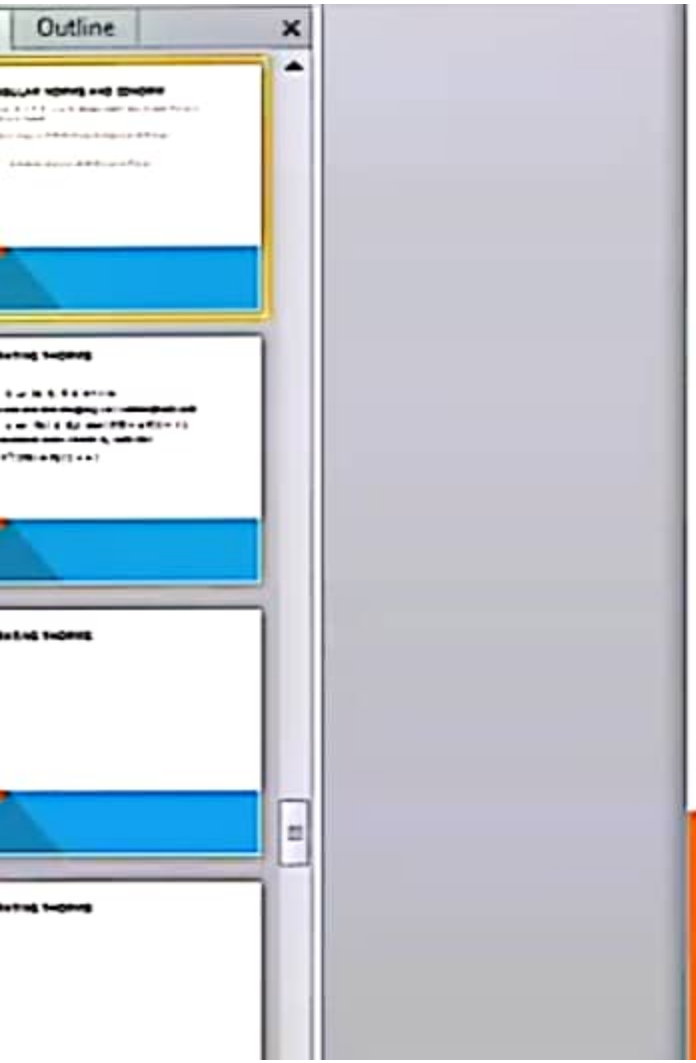


TRIANGULAR NORMS AND CONORM

Definition 2.13. *(Zadeh [154]) A triplet (S,T,N) is called a De Morgan triplet if T is a t-norm, S is a t-conorm, N is a strong negation, and if they fulfill De Morgan's law*

$$S(x,y) = N(T(N(x),N(y))).$$

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TRIANGULAR NORMS AND CONORM

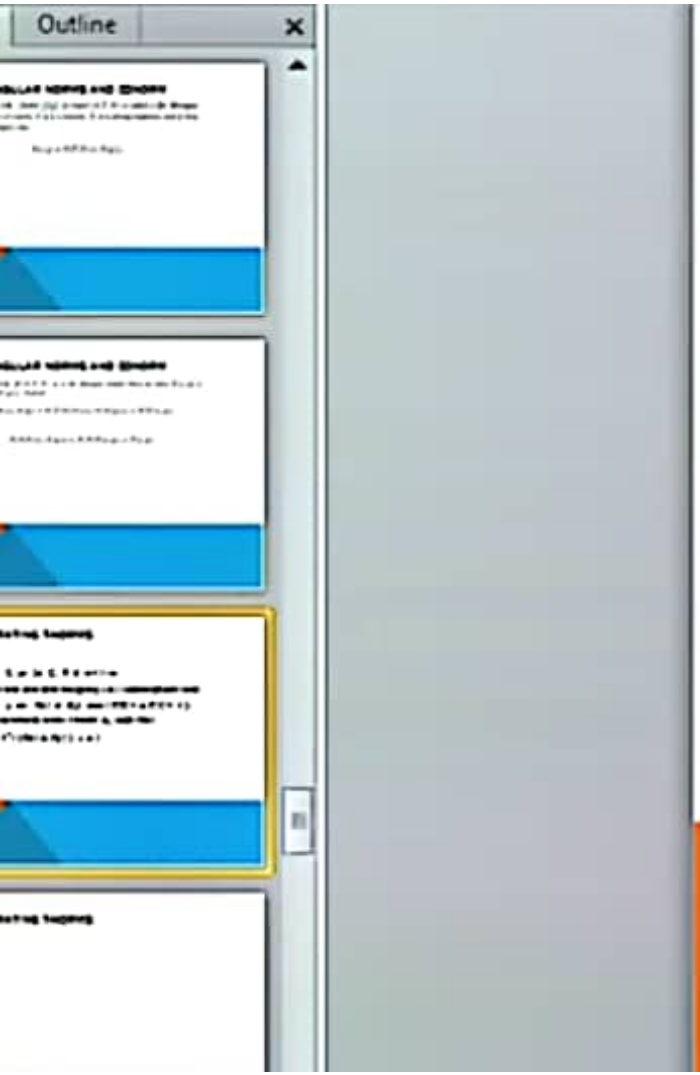
Remark 2.14. *If (S,T,N) is a De Morgan triplet then we have $T(x,y) = N(S(N(x),N(y)))$. Indeed,*

$$S(N(x),N(y)) = N(T(N(N(x)),N(N(y)))) = N(T(x,y)).$$

Then

$$N(S(N(x),N(y))) = N(N(T(x,y)) = T(x,y).$$

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GENERATING T-NORMS

Let $f : [0, 1] \rightarrow [a, 1]$; $0 \leq a < 1$ is

1- One to one and onto mapping i.e. (isomorphism) and

2- If $x \leq y \Rightarrow f(x) \leq f(y)$ also ($f(0) = a$, $f(1) = 1$).

Then we can make a new t-norm Δ_f such that

$x \ \Delta_f \ y = f^{-1}([f(x) \Delta f(y)] \vee a)$

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GENERATING T-NORMS

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$$x \ \Delta_f \ y = f^{-1}([f(x) \Delta f(y)] \vee a)$$

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Example 1:- $\Delta_f = ?$, $\Delta = xy$, $I(x) = x$,

Sol:-

$$x \Delta_f y = I^{-1}\left((f(x) \Delta f(y)) \vee a \right)$$

$$= f^{-1}\left([f(x) f(y)] \vee a \right)$$

$$= f^{-1}\left([f(x) f(y)] \vee 0 \right)$$

$$= f^{-1}\left(f(x) f(y) \right)$$

$$= f^{-1}(xy) = xy$$

$f(0) = 0$
$a = 0$

$I(x) = x$
$I^{-1}(x) = x$

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Example = $f(x) = x^n$; $n > 0$; $x \odot y = xy$

Sol =

Let $y = x^n$

$y^{1/n} = x$

$\Rightarrow f^{-1}(x) = x^{1/n}$

$I(a) = a^n = a \Rightarrow \overline{a = a}$

$$x \Delta_f y = I^{-1}\left(\left[I(x) \Delta I(y)\right] \vee a\right)$$

$$= I^{-1}\left(\left[I(x)\, I(y)\right] \vee a\right)$$

$$= I^{-1}\left(\left[I(x)\, I(y)\right] \vee a\right)$$

$$= f^{-1}\left(f(x)\, f(y)\right) \qquad \because I(x) = x^n$$

$$= f^{-1}\left(x^n \cdot y^n\right) \qquad f^{-1}(x) = x^{1/n}$$

$$= \left(x^n y^n\right)^{1/n} = xy \implies x \Delta_f y = xy$$

Best of luck to all

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